Partial Differential Equations with Applications

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These notes contain materials for one period (seven weeks) of the course *BM20A4703-PDEs with Applications* for a master's degree in Applied Mathematics at the Department of Computational Engineering, School of Engineering Science, LUT University. The course covers the basic principles of modelling with PDEs, certain classes of PDEs and their applications. Some topics that will be treated in more detail include the heat transfer problem, shock waves and nonlinear conservation laws, wave equations, and Navier–Stokes equations. Main references of the course include [**Cra18, Eva10, Rob01, Sal16b, Smo94, RRS16**].

Disclaimer!!! These notes are by no means original or complete. Please send any typos you have found to my email address given on the front page.

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Notations

General notations

- $\mathbb{R}^n = n$-dimensional real Euclidean space, $\mathbb{R} = \mathbb{R}^1$.
- A typical point in $\mathbb{R}^n$ is $x = (x_1, \ldots, x_n)$.
- For $x, y \in \mathbb{R}^n$, $x \cdot y \in \mathbb{R}$ is called the *dot product* of x and y and is defined as
$$x \cdot y = \sum_{i=1}^{n} x_i y_i.$$
- Ω, U, V and W usually denote open subsets of $\mathbb{R}^n$. We write
$$V \subset\subset U \quad \text{or} \quad V \Subset U$$
if $V \subset \bar{V} \subset U$ and $\bar{V}$ is compact (i.e., closed and bounded), and say V is *compactly contained* in U.
- ∂U = boundary of U, $\bar{U} = U \cup \partial U$ = closure of U.
- $U_T = U \times (0, T]$, $\Gamma_T = \overline{U_T} \setminus U_T$, *parabolic cylinder* and its boundary. See illustration on page 11.
- $|U| = \text{Vol}(U) = \int_U \mathrm{d}\mu$ = volume of the set U for a given measure μ on $\mathbb{R}^n$ (typically Lebesgue measure).
- We say $U \subset \mathbb{R}^n$ is a *domain* (or *region*) if it is a non-empty connected open set.
- $B(x, r) = B_r(x) = \{y \in \mathbb{R}^n \mid |y - x| < r\}$ = open ball center x, radius r.
- $B^0(x, r) = B_r^0(x) = \{y \in \mathbb{R}^n \mid |y - x| \leq r\}$ = closed ball center x, radius r.

Functions and derivatives

- If $u : U \to \mathbb{R}$, we denote the *support* of u by
$$\text{supp } u := \overline{\{x \in U \mid u(x) \neq 0\}}.$$
u is said to have *compact support* if supp u is a compact set.
- A function $u : U \to \mathbb{R}$ is called *Lipschitz continuous* if
$$|u(x) - u(y)| \leq C|x - y|,$$

for some constant C and all $x, y \in U$. We write

$$\mathrm{Lip}[u] := \sup_{\substack{x,y\in U \\ x\neq y}} \frac{|u(x)-u(y)|}{|x-y|}.$$

- We usually use subscript to denote partial derivative $u_t = \frac{\partial u}{\partial t}$. Similarly, $u_{x_i x_j} = \frac{\partial^2 u}{\partial x_i \partial x_j}$, etc.
- Multi-index: $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$ is called a *multi-index* of order $|\alpha| = \alpha_1 + \cdots + \alpha_n$.
- Given a multi-index α, define

$$D^\alpha u(x) = \frac{\partial^{|\alpha|} u(x)}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}} = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n} u(x).$$

- $Du := (u_{x_1}, \ldots, u_{x_n}) = \nabla u =$ *gradient vector*.
- Vector-valued function (mapping) is usually **boldface**.
- If $\mathbf{u} = (u^1, \ldots, u^n) : U \to \mathbb{R}^n$, $\operatorname{div} \mathbf{u} := \sum_{i=1}^n u^i_{x_i} = \nabla \cdot \mathbf{u} =$ *divergence* of $\mathbf{u}$.
- $\Delta u = \sum_{i=1}^n u_{x_i x_i} = \operatorname{div} Du = \nabla \cdot \nabla u =$ *Laplacian* of u.
- $\mathbf{n}$ and ν are usually used to denote the *outward pointing unit normal* of some surface or boundary.
- $\frac{\partial u}{\partial \nu} = \nabla u \cdot \boldsymbol{\nu}$ is the (outward) *normal derivative* of u.
- $\hat{u} = \mathcal{F}u =$ *Fourier transform* of u (see definition on page 8).
- $\check{u} = \mathcal{F}^{-1}u =$ *inverse Fourier transform* of u (see page 8).

Function spaces

- $C(U) = \{u : U \to \mathbb{R} \mid u \text{ is continuous}\}$.
- $C^k(U) = \{u : U \to \mathbb{R} \mid u \text{ is } k\text{-times continuously differentiable}\}$.
- $C^\infty(U) = \{u : U \to \mathbb{R} \mid u \text{ is infinitely differentiable (smooth)}\}$.
- $C_c^k(U), C_c^\infty(U)$ (or $C_0^k(U), C_0^\infty(U)$) denotes the spaces of functions in $C^k(U), C^\infty(U)$ with *compact support*.
- $C^{k,\alpha}(U)$ denotes the *Hölder space* of functions on U having continuous derivatives up to order k such that the k-th partial derivatives are Hölder continuous with exponent α.
- $L^p(U) = \{u : U \to \mathbb{R} \mid u \text{ is Lebesgue measurable}, \|u\|_{L^p(U)} < \infty\}, (1 \le p < \infty)$, where

$$\|u\|_{L^p(U)} := \left(\int_U |u|^p \mathrm{d}x \right)^{\frac{1}{p}}.$$

Note that two L^p functions that differ only on a set of measure zero are considered identical.

- $L^\infty(U) = \{u : U \to \mathbb{R} \mid u \text{ is Lebesgue measurable}, \|u\|_{L^\infty(U)} < \infty\}$, where

$$\begin{aligned}\|u\|_{L^\infty(U)} &:= \operatorname*{ess\,sup}_U |u| \\ &:= \inf\{C \geq 0 : |u(x)| \leq C \text{ for almost every } x\}.\end{aligned}$$

- $L^p_{\text{loc}}(U) = \{u : U \to \mathbb{R} \mid u \in L^p(V) \text{ for each } V \subset\subset U\} =$ locally p-integrable functions, for $1 \leq p \leq \infty$.
- $TV =$ total variation, $BV =$ space of functions of bounded variation. See Definition 2.11.
- $W^{k,p}(U), H^k(U)$, etc. denote Sobolev spaces. See page 56.
- $C(I;X) = \{u(t) : I \to X \mid u(t) \text{ is continuous in } I \text{ with } X\text{-norm}\}$, I is an interval $\in \mathbb{R}$ and X is a Banach space.
- $L^p(I;X) = \{u(t) : I \to X \mid \|u\|_{L^p(I;X)} < \infty\}$, where

$$\|u\|_{L^p(I;X)} := \left(\int_I \|u\|_X^p \mathrm{d}t\right)^{\frac{1}{p}}.$$

When $p = \infty$, the norm is defined as

$$\|u\|_{L^\infty(I;X)} := \sup_{t \in I} \|u\|_X.$$

CHAPTER 1

Introduction to Partial Differential Equations

1.1. Mathematical modelling

> *All models are wrong, but some are useful.*
>
> — George Box

Mathematical modelling is the process of describing real-world problems in mathematical terms, usually in the form of so-called *governing equations*, and then using these equations both to help understand the original problem and also to discover new features about the problem. Mathematical modelling plays a crucial role in science, engineering and several aspects of social and industrial activity. It is not an overstatement to say that, mathematical modelling not only lies at the heart of much of our understanding of the world, it moreover allows engineers to design the technology of the future. With modelling, one may be able to travel to the edge of the universe, peer into the heart of the atom, understand the future of our climate, and much more.

In many modelling problems, the governing equations are usually given in the form of some *partial differential equations* (PDEs), that is, equations involving unknown functions of two or more variables and their partial derivatives. The vast majority of equations describing physical, chemical, biological and even social phenomena are PDEs. Examples of these are the vibrations of solids, the flow of fluids, the spread of heat, the diffusion of chemical elements, the structure of molecules, the moving of cells, the dynamics of pedestrians, the pricing of stocks, the forming of opinions in a presidential election, and so on. Moreover, disciplines of pure mathematics such as geometry and topology also give rise to PDEs.

In a nutshell, a mathematical model is constructed based on two main ingredients:

`general physical laws` *and* `constitutive relations.`

Examples of physical laws include the general laws of conservation of mass, energy, momentum, etc. The constitutive relations are of an experimental nature and depend strongly on the features of the phenomena under examination, such as Newton's laws of motion, Fourier's law of heat conduction, Fick's law for the diffusion of a substance, Darcy's law for the fluid flow in porous media or the way the speed of a driver depends on the car density downstream.

1.2. Starting example: Heat transfer

Heat, like gravity, penetrates every substance of the universe, its rays occupy all parts of space. The object of our work is to set forth the mathematical laws which this element obeys.

— Joseph Fourier

Let us consider the problem of heat transfer in some domain $U \subset \mathbb{R}^n$, where $n = 1, 2$ or 3 (straight line, plan or usual three-dimensional space). Let $u(x,t)$ denote the density of the *thermal energy* at time t and position $x \in U$, thus $\int_U u(x,t)\mathrm{d}x$ represents the total thermal energy in U. Let $V \subset U$ be any subdomain. For the following to hold, we will assume that V is an open, simply connected (i.e., without holes) set with a piecewise smooth boundary ∂V (also denoted by S). The law of conservation of energy tells us that the rate of change of the thermal energy in V is equal to the flux of the energy going through the boundary ∂V plus the amount of energy generated in V. This statement can be written mathematically as[1]

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V u(x,t)\mathrm{d}x = -\int_{\partial V} \mathbf{F}(x,t)\cdot\mathbf{n}\mathrm{d}S + \int_V f(x,t)\mathrm{d}x.$$

Here $\mathbf{F}$ is the flux density, $\mathbf{n}$ is the *outward pointing unit normal* at each point on the boundary ∂V, and f is the source of energy generated inside V. The first integral on the right-hand side takes a minus sign since $\mathbf{n}$ is pointing outward (see Figure 1.1). The *divergence theorem*

[1] In $\mathbb{R}^3$, this is typically written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\iiint_V u(x,t)\mathrm{d}x = -\oiint_{\partial V} \mathbf{F}(x,t)\cdot\mathbf{n}\mathrm{d}S + \iiint_V f(x,t)\mathrm{d}x.$$

Here we adopt a simplified notation for integrals in any dimension for convenience.

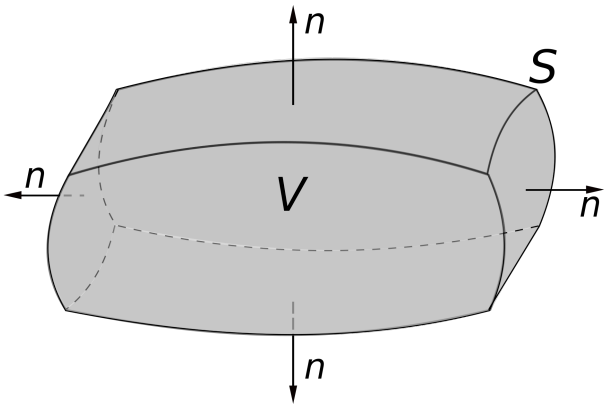


FIGURE 1.1. A domain V with the surface normal $\mathbf{n}$. (*WikiCommons*)

(or *Gauss's theorem*) says that (see Notations on page 1)

$$\int_{\partial V} \mathbf{F}(x,t)\cdot\mathbf{n}\mathrm{d}S = \int_V \operatorname{div}\mathbf{F}(x,t)\mathrm{d}x = \int_V \nabla\cdot\mathbf{F}(x,t)\mathrm{d}x.$$

Putting these together and assuming that u is relatively "nice" (in order to interchange the order of derivative and integral[2]), we have

$$\int_V \left(u_t(x,t) + \nabla\cdot\mathbf{F}(x,t) - f(x,t)\right)\mathrm{d}x = 0.$$

Since V is arbitrary, this leads to[3]

$$u_t(x,t) + \nabla\cdot\mathbf{F}(x,t) = f(x,t). \tag{1.1}$$

Now, the law of thermal conduction, as known as *Fourier's*[4] *law*, states that the rate of heat transfer through a material is proportional to the negative gradient in the temperature and to the area, at right angles to that gradient, through which the heat flows. That is

$$\mathbf{F} = -a\nabla u,$$

where a represents the *thermal conductivity*, which for simplicity is assumed to be a positive constant. The minus sign describes the intuitively expected fact that the heat energy flows from warmer to cooler

[2]when exactly can we do that? Exercise!

[3]why? Exercise!

[4]Joseph Fourier (1768–1830), a French mathematician and physicist who pioneered the study of the heat transfer problem by inventing Fourier transform and laid the foundation for Fourier analysis.

regions. We arrive at

$$u_t(x,t) - a\Delta u(x,t) = f(x,t),$$

which for $a = 1$ is the *nonhomogeneous heat equation*

$$u_t(x,t) - \Delta u(x,t) = f(x,t).$$

In the case where no source or sink is present, we have the *heat equation*

$$u_t - \Delta u = 0.$$

Depending on the interpretation, the heat equation is also called *diffusion equation* where u describes the density of some chemical concentration (with an analogue of Fourier's law called *Fick's laws*).

REMARK 1.1. The equation (1.1) is called *continuity equation*. In fact, this equation is the cornerstone for many PDE models. We will encounter (1.1) again in later chapters.

1.3. Solving heat equations: Fourier transform

> *Mathematics compares the most diverse phenomena and discovers the secret analogies that unite them.*
>
> — Joseph Fourier

This section will discuss how to solve the heat equation using Fourier transform. This gives a glimpse of one of the most powerful tools in the analysis of PDEs with important applications beyond the field.

DEFINITION 1.2. Let $u \in L^1(\mathbb{R}^n)$, we define its *Fourier transform* $\mathcal{F}u = \hat{u}$ by

$$\hat{u}(\xi) = \mathcal{F}u(\xi) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{-ix\cdot\xi} u(x)\mathrm{d}x, \quad \xi \in \mathbb{R}^n,$$

and its *inverse Fourier transform* $\mathcal{F}^{-1}u = \check{u}$ by

$$\check{u}(x) = \mathcal{F}^{-1}u(x) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{ix\cdot\xi} u(\xi)\mathrm{d}\xi, \quad x \in \mathbb{R}^n.$$

We recall a few useful properties of the Fourier transform.

THEOREM 1.3. *Under appropriate assumptions on u, v, the following hold*

(i) $\mathcal{F}(D^\alpha u) = (i\xi)^\alpha \mathcal{F}u$. *In particular,* $\mathcal{F}(\Delta u) = -|\xi|^2 \mathcal{F}u$.
(ii) $\mathcal{F}^{-1}(\mathcal{F}u) = u$.

(iii) $\mathcal{F}(u * v) = (2\pi)^{n/2}\mathcal{F}u\mathcal{F}v$, *where* $u * v$ *denotes the* convolution *of* u *and* v, *that is,*

$$(u * v)(x) := \int_{\mathbb{R}^n} u(y)v(x-y)\mathrm{d}y = \int_{\mathbb{R}^n} u(x-y)v(y)\mathrm{d}y.$$

Back to the heat equation, we usually couple it with some initial condition (which records the thermal energy at time $t = 0$) to form the *Cauchy problem*

$$\begin{cases} u_t - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0,\infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases} \tag{1.2}$$

By taking Fourier transforms with respect to x, we arrive at

$$\begin{cases} \hat{u}_t + |\xi|^2\hat{u} = 0 & \text{for } t > 0, \\ \hat{u} = \hat{g} & \text{for } t = 0. \end{cases}$$

This is a linear ordinary differential equation (ODE), with the solution given by

$$\hat{u}(\xi, t) = e^{-t|\xi|^2}\hat{g}(\xi).$$

Consequently, using Theorem 1.3 (ii-iii) we get

$$u = \mathcal{F}^{-1}\left(e^{-t|\xi|^2}\hat{g}\right) = H * g,$$

where $\hat{H} = \frac{1}{(2\pi)^{n/2}}e^{-t|\xi|^2}$. Hence, with standard computations (exercise!)

$$H = \frac{1}{(2\pi)^{n/2}}\mathcal{F}^{-1}\left(e^{-t|\xi|^2}\right) = \frac{1}{(4\pi t)^{n/2}}e^{\frac{-|x|^2}{4t}}. \tag{1.3}$$

The function H is called the *heat kernel* (or the *fundamental solution of the heat equation*). A graph of H taken at different snapshots in time is given in Figure 1.3.

Finally, by putting everything together,

$$u(x,t) = (H(\cdot,t) * g)(x) = \frac{1}{(4\pi t)^{n/2}}\int_{\mathbb{R}^n} e^{\frac{-|x-y|^2}{4t}}g(y)\mathrm{d}y. \tag{1.4}$$

We will review in the exercises that the formula (1.4) does indeed provide a solution to the heat equation (1.2) by a direct computation method.

1.3.1. Maximum principle. Assume that the initial data $g \in L^1(\mathbb{R}^n)$ and satisfies $g(x) \geq 0$ for all $x \in \mathbb{R}^n$. Observe from (1.4) that either $u(x,t) > 0$ for all $x \in \mathbb{R}^n, t > 0$, or else both $u(x,t) \equiv 0$ and $g(x) \equiv 0$ (exercise!). This is called the *maximum principle*, which asserts a so-called *infinite speed of propagation*. In other words, if the initial temperature is nonnegative and is positive somewhere, the

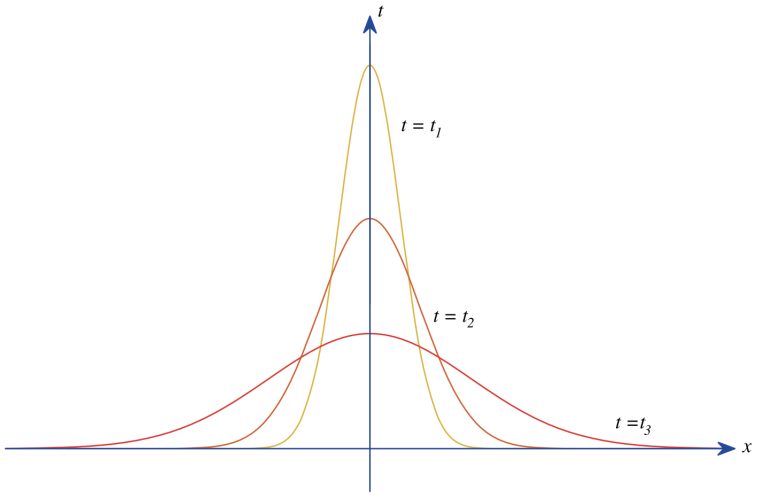


FIGURE 1.2. Graph of the heat kernel at times $t_1 < t_2 < t_3$.

temperature at any later time, no matter how small, is everywhere positive.[5] It derives the following result, in particular, the uniqueness theorem for the heat equation.

THEOREM 1.4. *Suppose that* $g_1, g_2 \in L^1(\mathbb{R}^n)$ *and* $g_1 \leq g_2$ *for all* $x \in \mathbb{R}^n$. *Then the corresponding solutions* u_1, u_2 (*given as the form* (1.4)) *either satisfy*

$$u_1(x,t) < u_2(x,t),$$

for all $t > 0$, *or else* $u_1 \equiv u_2$ *and* $g_1 \equiv g_2$.

PROOF. It follows directly from the maximum principle and the fact that $u_2 - u_1$ solves the heat equation with initial data $g_2 - g_1$. Uniqueness follows by putting $g_1 = g_2$. □

REMARK 1.5. Note carefully that the Fourier transform only provides one possible solution for the heat equation, and any solution obtained via this method (or satisfying all the conditions of this method) is unique. It does not rule out the possibility of another solution not given by this form. In fact, there are infinitely many solutions to the heat equation (first constructed by Tikhonov, see [**Joh82**]). Each of these solutions (besides $u \equiv 0$) grows rapidly as $|x| \to \infty$. To recover the uniqueness, either some appropriate growth conditions should be imposed, or some restriction on the boundary (such as a bounded domain) should be provided.

[5]In reality, the speed of heat transfer cannot be faster than the speed of light. In fact, the rate of thermal radiation is exactly the speed of light.

1.3.2. Energy method. We exhibit another method to prove the uniqueness of the heat equation, now with a boundary value problem. Assume that U is a bounded smooth domain in $\mathbb{R}^n$ and $T > 0$. Consider the *parabolic cylinder*

$$U_T := U \times (0, T] = \{(x, t) \,|\, x \in U, t \in (0, T]\},$$

and its *parabolic boundary*

$$\Gamma_T := \overline{U_T} \setminus U_T = (\partial U \times [0, T]) \cup (U \times \{t = 0\}).$$

Notice that U_T contains the top of the cylinder $U \times \{t = T\}$. Likewise, Γ_T contains the bottom and the vertical sides but not the top (see Figure 1.3).

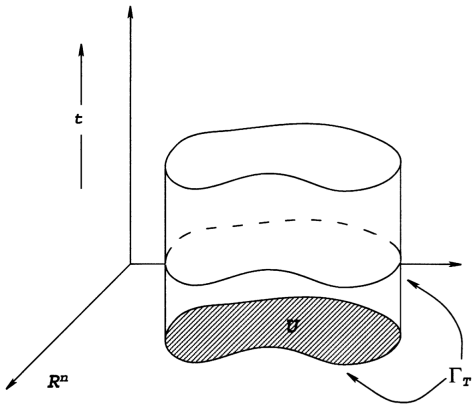


FIGURE 1.3. The parabolic cylinder and its boundary. [**Eva10**]

Let $u \in C^2(\overline{U_T})$ be a solution to the boundary value problem of the heat equation

$$u_t - \Delta u = 0, \text{ in } U, \quad u = 0 \text{ on } \Gamma_T.$$

Let $e(t)$ be the *energy functional* defined by

$$e(t) = \int_U u^2(x, t) \mathrm{d}x, \quad (0 \le t \le T).$$

Then $e(0) = 0, e(t) \geq 0$ and is differentiable in $(0, T)$ and, by *Green's identity*[6] (or integration by parts)

$$\begin{aligned} e'(t) &= 2 \int_U u u_t \mathrm{d}x \\ &= 2 \int_U u \Delta u \mathrm{d}x \\ &= -2 \int_U |Du|^2 \mathrm{d}x \leq 0, \end{aligned}$$

which implies that $e(t)$ is non-increasing in $(0, T)$. Hence $e(t) \leq e(0) = 0$. Therefore $e \equiv 0$. Consequently, $u \equiv 0$ in U_T.

REMARK 1.6. The uniqueness also follows from a so-called *strong maximum principle* which states that

$$\max_{\overline{U_T}} u = \max_{\Gamma_T} u,$$

that is, u only attains its maximum (or minimum) at the boundary. Furthermore, if U is connected and the maximum is attained at some interior point $(x_0, t_0) \in U_T$, then u is constant on $\overline{U_T}$ for all $t \leq t_0$.

1.3.3. Regularity. Remarkably, solutions to the heat equation will become smooth immediately after the initial time (see also the solution given by (1.4)). This property is called the *regularity* of the heat equation.

THEOREM 1.7. *Let $u \in C^2(\overline{U_T})$ solve the heat equation in U_T. Then $u \in C^\infty(U_T)$.*

We refer to [**Eva10**] for a proof of this theorem. We will see later in the course that, this desirable feature is not always the case for other PDEs.

1.4. PDEs and other important PDE models

> *Science is a differential equation. Religion is a boundary condition.*
>
> — Alan Turing

A *partial differential equation* (of *order* k) is an expression of the following type

$$F(x, u(x), Du(x), \ldots, D^k u(x)) = 0, \tag{1.5}$$

[6] `https://en.wikipedia.org/wiki/Green_identities`

where $u = u(x_1, \dots, x_n)$ is the *unknown*. Equation (1.5) is *linear* if F is linear with respect to u and all its derivatives, otherwise it is *nonlinear*. Moreover, we call (1.5)

- *semilinear* if F is nonlinear only with respect to u but is linear with respect to all its derivatives, with coefficients depending only on x;
- *quasilinear* if F is linear with respect to the highest order derivatives of u, with coefficients depending only on x, u and lower order derivatives;
- *fully nonlinear* if F is nonlinear with respect to the highest order derivatives of u.

To give the reader an idea of the wide range of applications of PDEs, apart from the heat equation, we present here a series of other examples, together with at least one of the possible interpretations for each example. We start with the most common linear models and continue with nonlinear ones, with systems of PDEs appearing at the end of the list.

(1) *Transport equation* (1st order):

$$u_t + \mathbf{v} \cdot \nabla u = 0,$$

which describes the transport of, say, a solid polluting substance along a channel. Here u is the concentration of the substance and $\mathbf{v}$ is the stream speed.

(2) *Wave equation* (2nd order):

$$u_{tt} - c^2 \Delta u = 0,$$

which describes for example the propagation of transversal waves of small amplitude in a perfectly elastic chord (e.g. of a violin) if $n = 1$, or membrane (e.g. of a drum) if $n = 2$. If $n = 3$ it governs the propagation of electromagnetic waves in vacuum or of small amplitude sound waves. Here u may represent the wave amplitude and c is the propagation speed.

(3) *Laplace's equation* (2nd order):

$$\Delta u = 0,$$

which describes the *steady state* of the heat or wave equation, that is, when their solution does not depend on time. The nonhomogeneous version

$$\Delta u = f$$

is called *Poisson's equation*. They also play an important role in electrostatics.

(4) *Helmholtz equation* (or *eigenvalue problem*) (2nd order):

$$-\Delta u = k^2 u,$$

which models wave travelling in seismology, electromagnetic radiation, and acoustics. Here the "eigenvalue" k is known as the wave number.

(5) *Black-Scholes equation* (2nd order):

$$u_t + \frac{1}{2}\sigma^2 x^2 u_{xx} + rxu_x - ru = 0,$$

where $r, \sigma \geq 0$. This equation is fundamental in mathematical finance. It models the evolution of the price u of a so-called derivative (e.g. a European option), based on an underlying asset (a stock, a currency, etc.) whose price is x.

(6) *Schrödinger equation* (2nd order):

$$iu_t + \Delta u + V(x)u = 0,$$

where i is the imaginary unit. This equation is fundamental in quantum mechanics and governs the evolution of a particle subject to a potential V (that models the environment in which the particle exists). Here u is the wave function and $|u|^2$ represents a probability density function for the system.

(7) *Klein-Gordon equation* (2nd order):

$$u_{tt} - \Delta u + m^2 u = 0,$$

is a relativistic wave equation, which describes the relation in relativity of the total energy and the invariant mass and momentum. This equation is related to the Schrödinger equation and is a quantized version of the relativistic energy-momentum relation (or relativistic dispersion relation)

$$E^2 = (pc)^2 + \left(m_0 c^2\right)^2.$$

(8) *Telegraph equation* (or *Telegrapher's equation*) (2nd order):

$$u_{tt} - c^2 u_{xx} + (\alpha + \beta)u_t + \alpha\beta u = 0,$$

which models the intensity of the current in a conductor, considered as a function of time t and distance x from any fixed point of the conductor. Here, c is the speed of light, α is a capacity coefficient and β is the induction coefficient.

(9) *Burgers equation* (quasilinear, 1st order):

$$u_t + uu_x = 0,$$

which governs a one-dimensional flux of a nonviscous fluid and is used to model traffic dynamics as well. Its viscous variant

$$u_t + uu_x = \varepsilon u_{xx}, \quad (\varepsilon > 0)$$

constitutes a basic example of competition between dissipation (due to the term εu_{xx}) and steepening (shock formation due to the term uu_x). A more general version of Burgers' equation

$$u_t + \nabla \cdot \mathbf{F}(u) = 0$$

is called the *scalar conservation law* and will be considered in more detail in the next chapter.

(10) *Porous medium equation* (quasilinear, 2nd order):

$$u_t = k\nabla \cdot (u^\gamma \nabla u),$$

where $k > 0, \gamma > 1$ are constant. This equation appears in the description of filtration phenomena, e.g. of the motion of water through the ground.

(11) *Minimal surface equation* (quasilinear, 2nd order):

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = 0.$$

The graph of a solution u minimizes the area among all surfaces $z = v(x_1, x_2)$ whose boundary is a given curve. One famous example is soap balls, which are minimal surfaces.

(12) *Eikonal equation* (fully nonlinear, 1st order):

$$|\nabla u| = c(x),$$

which appears in geometric optics (here c is the refractive index of the medium). If u is a solution, its level surfaces $u(x) = t$ describe the position of a light wavefront at time t.

(13) *Hamilton-Jacobi equation* (nonlinear, 1st order):

$$u_t + H(Du, x) = 0.$$

It has wide applications in optics, mechanics, and semi-classical quantum theory. Its solutions determine infinite families of solutions of Hamilton's ordinary differential equations, which are the equations of motion of a mechanical system or an optical system in the ray approximation.

(14) *Monge–Ampère equation* (fully nonlinear, 2nd order):

$$\det(D^2 u) = f.$$

This equation arises naturally in several problems in Riemannian geometry, conformal geometry, for which one simplest application is to the problem of prescribed Gauss curvature. It also relates to the optimal mass transport problem.

(15) *Korteweg-de Vries (KdV) equation* (nonlinear, 3rd order)

$$u_t + uu_x + u_{xxx} = 0,$$

which models shallow water waves. This equation also describes the development of the so-called solitons in wave travelling.

(16) *Maxwell's equations* (six scalar linear equations of first order):

$$\begin{cases} \mathbf{E}_t - \operatorname{curl} \mathbf{B} = 0, \\ \mathbf{B}_t + \operatorname{curl} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{E} = 0, \ \operatorname{div} \mathbf{B} = 0, \end{cases}$$

where $\mathbf{E}$ is the electric field and $\mathbf{B}$ is the magnetic induction field. The unit measures are the "natural" ones, i.e. the light speed is $c = 1$ and the magnetic permeability is $\mu_0 = 1$. Equations on the first line are called Ampère and Faraday laws, and Gauss laws on the second line.

(17) *Navier-Stokes equations* (three quasilinear scalar equations of second order and one linear equation of first order):

$$\begin{cases} \mathbf{u}_t - \nu \Delta \mathbf{u} + (\mathbf{u} \cdot \nabla) u + \frac{1}{\rho} \nabla p = 0, \\ \operatorname{div} \mathbf{u} = 0, \end{cases} \tag{1.6}$$

where $\mathbf{u} = (u_1, u_2, u_3)$ is the three-component fluid velocity field, p is the fluid pressure and ρ its density (constant). The Navier-Stokes equations govern the motion of a viscous, homogeneous and incompressible fluid. The term $(\mathbf{u} \cdot \nabla) u$ represents the inertial acceleration due to fluid transport. Here ν is the kinematic viscosity, given by the ratio between the fluid viscosity and its density. The case $\nu = 0$ yields

$$\begin{cases} \mathbf{u}_t + (\mathbf{u} \cdot \nabla) u + \frac{1}{\rho} \nabla p = 0, \\ \operatorname{div} \mathbf{u} = 0, \end{cases}$$

which is called the *Euler equations* for inviscid ("ideal") fluid flow. Euler equations have the form of a *system of conservation laws*, see also (9). These systems will be studied in more detail later in the lectures.

1.5. Well-posedness of PDEs

Baseball players or cricketers do not need to be able to solve explicitly the non-linear differential equations which govern the flight of the ball. They just catch it.

— Paul Ormerod

When studying PDEs, one of the most important questions that concerns us is, does the PDE have a unique solution? Sure, we need to have a solution, otherwise, there is nothing to work on. But why unique? Because if there exist, say, two solutions. Often it is the case that one solution represents correctly the underlying phenomenon while the others do not, or even do not make any physical sense – we do not want that. In such a case, to be able to find the correct solution, one often needs extra conditions. Besides, it is a desirable feature that this unique solution should not be too sensitive to subtle changes in the inputs (such as initial data, boundary data or the source term).

According to Hadamard[7], a PDE is called *well-posed* if

- existence: it has at least one solution;
- uniqueness: the solution is unique; and
- continuous dependence: the solution depends continuously on the data (e.g. initial or boundary data) given for the problem.

If at least one of these conditions is not satisfied, the problem is called *ill-posed*. Of course, we need to assign appropriate initial data or boundary data for the PDE before studying any of these conditions.

1.5.1. Solving PDEs. Weak solutions. In the case of the heat transfer problem, we have seen that we can find a solution to the PDE, moreover, an explicit formula for the solution is given. Furthermore, the solution is twice continuously differentiable and satisfies the equation at every point. This solution is called the *classical solution*. Note however that this situation is very rare in the world of PDEs. In fact, most of the time it is impossible to write down a solution formula to a PDE[8] or even to prove a classical solution exists globally (i.e. for all

[7]Jacques Hadamard (1865–1963), a French mathematician who made several contributions in number theory, complex analysis, differential geometry and partial differential equations.

[8]This is where some approximation techniques such as finite difference/finite element/finite volume methods come into play!

time). To make the task feasible, one often has to relax the assumptions on the unknown, such as assumptions about the differentiability or continuity of the unknown. In other words, to find the solutions, we extend the class of the function space where the solutions (are supposed to) live in to some bigger one[9]. We typically call the solutions found (if any) in the new function space *weak solutions*, whose exact definition depends on the form of each PDE, and we will revisit this notion again when dealing with them.

As an example, consider the Navier-Stokes equation (1.6) with *divergence-free* initial data, that is, $\mathbf{u}(x,0) = \mathbf{u}_0(x)$ with

$$\operatorname{div} \mathbf{u}_0 = 0.$$

Assume that u_0 is smooth and has appropriate growth properties at ∞. Then, Leray proved in 1934 ([**Ler34**]) that, one can find a global solution $\mathbf{u}$ with

$$\begin{cases} \mathbf{u}(\cdot,t) \in L^2(\mathbb{R}^3) \\ D\mathbf{u}(\cdot,t) \in L^2(\mathbb{R}^3), \end{cases}$$

where $D\mathbf{u}(\cdot,t) = (Du_1(\cdot,t), Du_2(\cdot,t), Du_3(\cdot,t))$ is understood in the sense that

$$\int_{\mathbb{R}^3} Du_i(\cdot,t)\phi \mathrm{d}x = -\int_{\mathbb{R}^3} u_i(\cdot,t) D\phi \mathrm{d}x \qquad (i=1,2,3)$$

for all $\phi \in C_c^\infty(\mathbb{R}^3)$. (Note there is no requirement on differentiability or even continuity on $\mathbf{u}$.) This remains the best result on attempting to find solutions to the Navier-Stokes question (1.6) in the last 90 years. Whether we can find a better solution is still an open problem, and is part of the Clay Millennium Prize Problems, see [**Fef00**].

1.5.2. Regularity. After proving the existence of some sort of weak solutions, one tries to deduce various properties of these weak solutions. One may ask questions such as, does *classical solution*, the one that satisfies all the necessary differentiability conditions, exist? If so, is it one of the weak solutions? Under which conditions do weak solutions become the classical ones? The process that answers these questions is called *regularity* (or *smoothness*). Again, unlike the heat equation when some formula for the solution can be found, this regularity procedure often relies on many intricate techniques of calculus or functional analysis. It is, therefore, wise to split the study of any PDE into separate parts (existence, uniqueness, regularity). The complexity of each part increases when the equation is nonlinear, or when it comes to a system of equations.

[9]The idea here is quite similar to the one used in solving algebraic equations, to which we use the notion of field extensions.

1.6. Exercises

Do not worry about your difficulties in mathematics. I can assure you mine are still greater.

— Albert Einstein

Exercise 1.1 (Fundamental solution of the heat equation)**.** Let

$$\Phi(x,t) = \begin{cases} \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|x|^2}{4t}}, & x \in \mathbb{R}^n, t > 0 \\ 0, & x \in \mathbb{R}^n, t = 0. \end{cases}$$

(i) Prove that $\Phi \in C^\infty(\mathbb{R}^n \times (0,\infty))$.

(ii) Prove that, for each $t > 0$,

$$\int_{\mathbb{R}^n} \Phi(x,t)\mathrm{d}x = 1.$$

Hint: You may use the Gaussian integral (see Appendix A.3).

(iii) Verify that Φ solves the heat equation

$$u_t - \Delta u = 0 \text{ in } \mathbb{R}^n \times (0,\infty).$$

(iv) Given $g \in C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$. Let u be defined as the *convolution* of $\Phi(\cdot,t)$ and g

$$u(x,t) = (\Phi(\cdot,t) * g)(x) := \int_{\mathbb{R}^n} \Phi(x-y,t)g(y)\mathrm{d}y \tag{1.7}$$

Prove that $u \in C^\infty(\mathbb{R}^n \times (0,\infty))$ and that u solves the heat equation in $\mathbb{R}^n \times (0,\infty)$. (This verifies the fact that the solution to the heat equation becomes smooth instantly.)
Hint: Use (iii).

(v) Prove that u given in (1.7) indeed "attains" its initial value at g, that is, for each $x_0 \in \mathbb{R}^n$,

$$\lim_{x \to x_0, t \to 0^+} u(x,t) = g(x_0).$$

Hint: Use (ii) and the definition of limit.

Exercise 1.2 (Infinite speed of propagation)**.** Let u be given by (1.7). Prove that if g is continuous, non-negative and not identically zero ($g \not\equiv 0$) then $u(x,t) > 0$ for all $x \in \mathbb{R}^n$ and $t > 0$. Carefully explain your answer!

Exercise 1.3 (Scale invariance)**.** Let u be a smooth solution of the heat equation

$$u_t - \Delta u = 0 \text{ in } \mathbb{R}^n \times (0,\infty)$$

For each $\lambda \in \mathbb{R}$, denote $u_\lambda(x,t) := u(\lambda x, \lambda^2 t)$.

(i) Prove that u_λ solves the heat equation.

(ii) Let $w(x,t) := x \cdot Du(x,t) + 2tu_t(x,t)$. Show that w also solves the heat equation.
Hint: Use (i).

Exercise 1.4. Verify (1.3) in detail, that is,

$$\mathcal{F}^{-1}\left(e^{-t|\xi|^2}\right) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{ix\cdot\xi} e^{-t|\xi|^2} \mathrm{d}\xi = \frac{1}{(2t)^{n/2}} e^{\frac{-|x|^2}{4t}},$$

for $x \in \mathbb{R}^n, t > 0$.

Hint: You may want to use Fubini's theorem.

Exercise 1.5. Prove that the solution of the heat equation obtained by the Fourier transform vanishes at infinity, that is, $|u(x,t)| \to 0$ as $|x| \to \infty$. (This implies that the solution obtained via Fourier transform is unique, see Remark 1.5).

1.7. Summary and notes

The partial differential equation entered theoretical physics as a handmaid, but has gradually become mistress.

— Albert Einstein

This chapter introduces the principles of mathematical modelling using PDEs and executes these principles in detail by considering the problem of heat transfer. While only touching the surface, we also introduce the Fourier transform and its application in solving the heat equation, together with some important concepts such as infinite speed of propagation, maximum principle, uniqueness, energy method and regularity for the heat equation. For a more thorough treatment of the subject, the reader is encouraged to consult [**Eva10, Joh82**]. Section 1.4 and Section 1.5 explained several notions (linear, nonlinear, well-posedness, weak solution, etc.) in studying PDEs. A list of well-known PDEs (mostly taken from [**Sal16b**]) is also provided. A more complete list of interesting PDEs can be found in [**ZD98**].

CHAPTER 2

Nonlinear conservation laws

"Begin at the beginning," the King said gravely, "and go on till you come to the end: then stop."

— Lewis Carroll, *Alice in Wonderland*

This chapter enters the realm of the theory of nonlinear scalar conservation laws in an attempt to understand the underlying phenomena that they model. One of the fundamental features of nonlinear scalar conservation laws (and hyperbolic conservation laws in general) that distinguish them from other types of partial differential equations is the formation of shocks. Starting with even smooth initial data, the wave-like solutions of conservation laws will generally get steeper and steeper, and eventually break, leading to the development of shock waves. These waves then continue to travel at their own speeds. This striking phenomenon makes the study of conservation laws challenging since many modern techniques (for instance those from functional analysis) for partial differential equations are not applicable. The solutions are only sought in the space of discontinuous functions and therefore, working with weak solutions is inevitable. The difficulty is exacerbated by the fact that weak solutions of conservation laws are not unique. To single out the right solution, some additional restrictions, called entropy conditions, need to be imposed. These are essentials in the mathematical theory of hyperbolic conservation laws.

In the first section, we give a derivation for scalar conservation laws and highlight a few examples of equations of this type. Section 2.2 introduces the method of characteristics to solve (locally) quasilinear PDEs, including scalar conservation laws. Section 2.3 reflects the most significant property of nonlinear conservation laws, the formation of shocks, as well as defines the weak formulation of solutions. The Rankine-Hugoniot condition on jumps is discussed here. Entropy conditions and entropy solutions, with motivations and related discussions, are contents of Section 2.4. In the next section, Section 2.5, we

present the existence and uniqueness of entropy solutions. Two of the most popular and fundamental methods for constructing solutions of conservation laws, the front tracking method and the vanishing viscosity method, are also discussed here. In the last section, we highlight a few results regarding fine structures of the entropy solution.

Many results in this chapter do not include proofs, either because they are too technically complicated or too simple. However, we will refer to appropriate references where relevant. Throughout this chapter, we assume that $v_0 : \mathbb{R}^n \to \mathbb{R}$ is a measurable bounded function. That is, there exists some constant M such that

$$\|v_0\|_{L^\infty(\mathbb{R}^n)} \leq M.$$

The function f is assumed to be smooth enough so that all the calculations make sense (for example, f is differentiable almost everywhere). Some additional conditions on f might be imposed in specific situations. For simplicity, some results will be stated in one (space) dimension, even though most of them can be extended to higher dimensions.

2.1. Derivation and examples

> *Motivation isn't enough. If you have an idiot and you motivate him, now you have a motivated idiot.*
>
> — Jim Rohn

Scalar conservation laws take the form as follows

$$\frac{\partial v}{\partial t}(x,t) + \nabla \cdot \mathbf{F}(v(x,t)) = 0, \tag{2.1}$$

where $v : \mathbb{R}^n \times [0,\infty) \to \mathbb{R}$ is the density of some *conserved quantity*. The function $\mathbf{F} = (f_1, \cdots, f_n) : \mathbb{R} \to \mathbb{R}^n$ is called *flux*, $\nabla \cdot \mathbf{F} = \sum_{i=1}^n \frac{\partial f_i}{\partial x_i}$. If we assign an initial condition for v at $t = 0$,

$$v(x,0) = v_0(x), \tag{2.2}$$

then we obtain an *initial value problem* (or *Cauchy problem*). When $n = 1$, we will use the normal letter f to denote the flux and write

$$v_t + f(v)_x = 0. \tag{2.3}$$

To have an idea where does the name "conservation law" come from, we consider in one dimension a (sufficiently smooth) quantity v

that satisfies (2.3). By integrating (2.3) in x over $[a, b]$ ones get

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\int_a^b v(x,t)\mathrm{d}x &= \int_a^b f'(v(x,t))\mathrm{d}x \\ &= f(v(a,t)) - f(v(b,t)). \end{aligned} \tag{2.4}$$

The term $\int_a^b v(x,t)\mathrm{d}x$ represents the total amount of conserved quantity v inside the interval $[a, b]$. The right-hand side of (2.4) describes the quantity flow going in and going out of the interval $[a, b]$ (which explains the name "flux" for f). The equation (2.4) therefore presents a law of conservation: the rate of change in the total amount of conserved quantity over time in a domain equals the amount of quantity flowing in and out through the boundary of that domain. (See also (1.1)).

2.1.1. Burgers equation. Consider the Burgers equation

$$\partial_t u(x,t) + \partial_x\left(\frac{1}{2}u^2(x,t)\right) = 0. \tag{2.5}$$

This equation was introduced by Bateman [**Bat15**] and Burgers [**Bur48**] (in the viscous form) and studied thoroughly by Hopf[1] [**Hop50**]. It is normally known as a simplification of the Navier-Stokes equations and is fundamental in the theory of fluid mechanics. Moreover, it is also a prototype example of scalar conservation laws due to the fact that it retains every typical feature of scalar conservation laws. One way to see the formation of Burgers' equation is to look at some particle $x(t)$ moving with velocity $u(x, t)$ for which we know that the following relationship holds

$$\dot{x}(t) = u(x(t), t).$$

Assume that u is smooth enough, we have

$$\ddot{x}(t) = \frac{\mathrm{d}}{\mathrm{d}t}u(x(t), t) = u_x\dot{x}(t) + u_t = uu_x + u_t.$$

If there are no external forces then the Newton law of motion ensures that the acceleration $a = \ddot{x}(t)$ equals zero, leading to

$$u_t + uu_x = 0, \tag{2.6}$$

which is equivalent to (2.5) since u is smooth. The equation (2.6) is sometimes termed the *quasilinear Burgers equation*.

[1]Eberhard Hopf (1902–1983, a German mathematician and astronomer.)

2.1.2. Traffic flow. Another typical example of scalar conservation laws is the so-called *LWR model* of traffic flow on the highway. This model is first described in the seminal paper [**LW55**] by Lighthill and Whitham where the authors develop it via the method of kinetic waves. One year later, Richards [**Ric56**] independently comes up with a similar model.

The key idea of the LWR model initiates from an observation that the traffic streams at a macroscopic level behave like fluid streams, where the distribution of cars can be replaced by a "continuous" function. Let ρ be the density and v be the average velocity of cars. The LWR assumes that the traffic velocity v depends only on its density, $v = v(\rho)$. Then $\rho v(\rho)$ represents the car flux. Since the change in the total amount of cars in a road segment, say $[a, b]$, is given by the flux of cars flowing in at a subtracting the flux of cars flowing out at b (assuming that there are no entries and exits on this road segment), then

$$\begin{aligned}\int_a^b \partial_t \rho(x,t)\mathrm{d}x = \frac{\mathrm{d}}{\mathrm{d}t}\int_a^b \rho(x,t)\mathrm{d}x &= \rho(a,t)v(\rho(a,t)) - \rho(b,t)v(\rho(b,t)) \\ &= \int_a^b \partial_x(\rho(x,t)v(\rho(x,t)))\mathrm{d}x.\end{aligned}$$

Therefore, since a and b are arbitrary,

$$\rho_t + (\rho v(\rho))_x = 0. \tag{2.7}$$

The flux function in this case is $f(\rho) = \rho v(\rho)$. In the simplest case, the velocity may be assumed to depend linearly on the density, that is

$$v(\rho) = v_{max}\left(1 - \frac{\rho}{\rho_{max}}\right),$$

where v_{max} and ρ_{max} denote the maximum velocity and density of cars, respectively. The flux is then given by

$$f(\rho) = v_{max}\rho\left(1 - \frac{\rho}{\rho_{max}}\right),$$

which is a strictly concave and decreasing function. Moreover, if we make a scaling $\rho_{max} = 1, v_{max} = 1$ then $f(\rho) = \rho(1-\rho)$ and we obtain the normalised traffic flow equation

$$\rho_t + (\rho(1-\rho))_x = 0. \tag{2.8}$$

This normalised model is often used as a toy model in studying scalar conservation laws. We note that, in this case, $v = 1 - \rho$ satisfies the

following extra conservation law

$$v_t + (v(v-1))_x = 0, \tag{2.9}$$

with a strictly convex flux function $\tilde{f}(v) = v(v-1)$. We note, moreover, that by setting $\rho = \frac{1-u}{2}$ we recover the Burgers equation (2.5).

2.2. The method of characteristics

For every problem, there is always a solution that is simple, neat, but wrong.

H. L. Mencken

In this section, we present a method to solve PDEs using the theory of ODEs. The rough idea is this. Assume that we have to solve an initial or a boundary value problem. We want to calculate a solution $u(x)$ by finding a curve that connects x with some point x_0 (on the boundary) along which we can compute u. Since we already know the value of u on the boundary, in particular, at the endpoint x_0, we are then able to calculate u at the other endpoint x, that is, $u(x)$. For the purpose of illustration, we restrict ourselves to $\mathbb{R}^2$ and consider a first-order quasilinear equation of the form

$$a(x,y,u)\frac{\partial u}{\partial x} + b(x,y,u)\frac{\partial u}{\partial y} = c(x,y,u). \tag{2.10}$$

We assume that the coefficients a, b and c are at least C^1 w.r.t x, y and u. A solution $u(x,y)$ is presented as a graph $(x,y,u(x,y))$ in $(x,y,z) \in \mathbb{R}^3$ (just like a solution to $y = x^2$ is presented as a graph (x,y) in $\mathbb{R}^2$). The normal[2] of this graph $z = u(x,y)$ is

$$N = (\partial_x u, \partial_y u, -1).$$

The coefficients of equation (2.10) are interpreted as a vector field $T = (a(x,y,z), b(x,y,z), c(x,y,z))$ in $(x,y,z) \in \mathbb{R}^3$ (the *characteristic vector field*), and equation (2.10) becomes

$$N \cdot T = 0.$$

The situation is illustrated in Figure 2.1. This implies that the *integral curves* (the *characteristic curves*) $(x(s), y(s), z(s))$ of the characteristic

[2]Remember that the normal vector of the surface $f(x,y,u) = 0$ is given by ∇f.

vector field

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}s} &= a(x,y,z)\\ \frac{\mathrm{d}y}{\mathrm{d}s} &= b(x,y,z)\\ \frac{\mathrm{d}z}{\mathrm{d}s} &= c(x,y,z)\end{aligned} \tag{2.11}$$

lie on the graph $z = u(x,y)$. A solution to (2.10) is given as the union of a parameter family of such characteristic curves. The set of ODEs (2.11) is also called *characteristic equations*. The projection of a characteristic curve on the (x,y) plane ($z=0$) is called a *characteristic base curve*, or *projected characteristic curve*, or simply *characteristic*. Note that we can write (2.11) as

$$\frac{\mathrm{d}x}{a(x,y,z)} = \frac{\mathrm{d}y}{b(x,y,z)} = \frac{\mathrm{d}z}{c(x,y,z)}, \tag{2.12}$$

which are often referred to as the *Lagrange–Charpit equations*. Finally solving the ODEs (2.11) or (2.12), with appropriate initial or boundary conditions, provides a solution to the PDE (2.10). To double check this, we can work it out from the chain rule

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s}u(x(s),y(s)) &= \frac{\mathrm{d}u}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\mathrm{d}u}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}s}\\ &= \frac{\partial u}{\partial x}a(x,y,u) + \frac{\partial u}{\partial y}b(x,y,u)\\ &= c(x,y,u),\end{aligned}$$

where we use (2.10) and (2.11).

2.2.1. Examples.

2.2.1.1. *Transport equation.* We start with a linear PDE, the *transport equation* in 1D,

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0, \tag{2.13}$$

equipped with an initial condition

$$u(x,0) = F(x).$$

We can transform this PDE into an ODE along some appropriate curve, that is, something of the form

$$\frac{\mathrm{d}}{\mathrm{d}s}u(x(s),t(s)) = G(x(s),t(s),u).$$

By the chain rule

$$\frac{\mathrm{d}}{\mathrm{d}s}u(x(s),t(s)) = \frac{\partial u}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\partial u}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}s}.$$

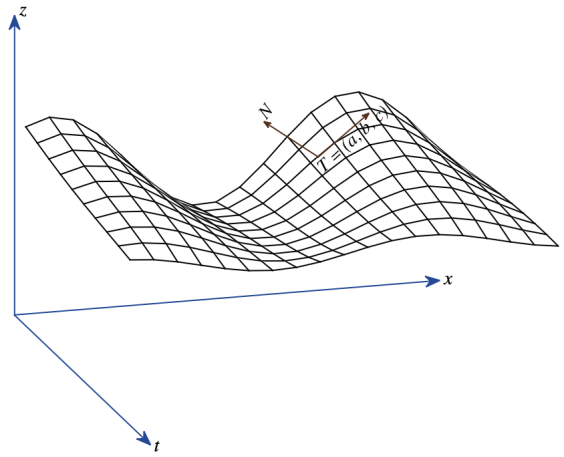


FIGURE 2.1. A solution to (2.10) as a graph ([**Cra18**]).

The characteristic equations for (2.13) are

$$\frac{\mathrm{d}t}{\mathrm{d}s} = 1, \quad \frac{\mathrm{d}x}{\mathrm{d}s} = a, \quad \frac{\mathrm{d}z}{\mathrm{d}s} = 0.$$

Hence, characteristics are straight lines. They can be solved explicitly

$$s = t, \quad x(s) = x(0) + as, \quad z(s) = z(0). \tag{2.14}$$

The projected characteristic is $(x(t), t) = (x(0) + at, t)$. (2.14) implies that along the characteristics the solution u is constant, $u(x(s), s) = z(s) = z(0) = u(x(0), 0)$. Now since $s = t, x(0) = x(t) - at$, $u(x(t), t) = u(x(t) - at, 0) = F(x(t) - at)$. Therefore,

$$u(x, t) = F(x - at).$$

The solution of PDE (2.13) corresponds to transporting the initial profile $F(x)$ unaltered (preserving the shape of initial waveform) along the characteristics with a speed $\mathrm{d}x/\mathrm{d}t = a$. See Figure 2.2.

2.2.1.2. *Burgers' equation.* Consider now the Burgers' equation

$$u_t + uu_x = 0,$$

with initial data

$$u(x, 0) = g(x).$$

The characteristic equations are

$$\frac{\mathrm{d}t}{\mathrm{d}s} = 1, \quad \frac{\mathrm{d}x}{\mathrm{d}s} = z, \quad \frac{\mathrm{d}z}{\mathrm{d}s} = 0.$$

These can be solved explicitly

$$s = t, \quad x(s) = x(0) + sz, \quad z(s) = z(0).$$

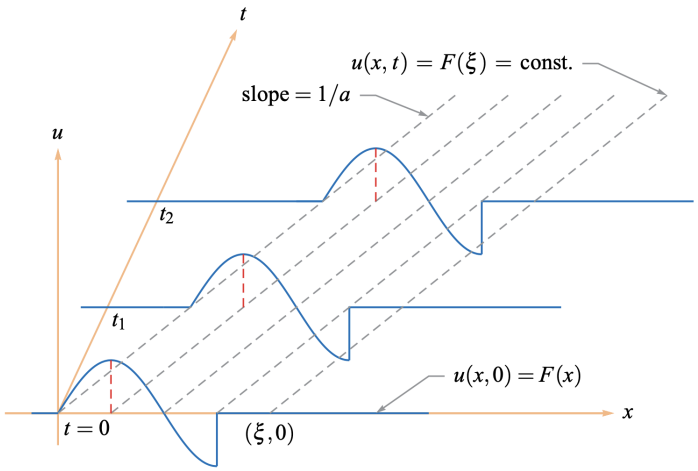


FIGURE 2.2. Solution to transport equation ([**Sal16a**]).

Hence, the solution is given by

$$u(x,t) = g(x - ut),$$

as an implicit formula. It represents a travelling wave (or a signal, a disturbance) propagating with speed u which depends on the solution. Because of this, some parts of the wave move faster than the others. This leads to the notion of wave breaking that we will see shortly (Figure 2.4). Moreover, since the characteristic speed $\mathrm{d}x/\mathrm{d}t = u$ depends on the solution, two characteristics based at different points $(x_1, 0)$ and $(x_2, 0)$ may intersect at some point (x, t) and the value $u(x, t)$ is not uniquely determined as soon as $g(x_1) \neq g(x_2)$, see Figure 2.3. We will come back to this in the next section on how to continue further this intersection point.

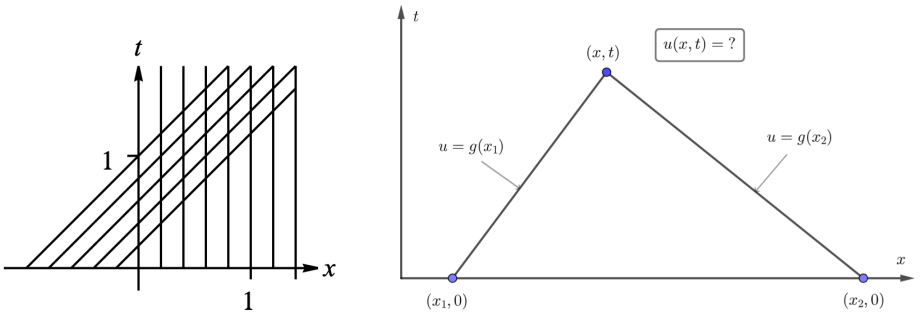


FIGURE 2.3. Crossing of two characteristics.

2.2.1.3. *General scalar conservation laws.* In the same manner, we can use the characteristic method to solve implicitly the general nonlinear scalar conservation law

$$u_t + \nabla \cdot \mathbf{F}(u(x,t)) = 0, \quad u(x,0) = g(x) \quad (x \in \mathbb{R}^n).$$

Indeed, similar to Burgers' equation, we can find that

$$x(s) = \mathbf{F}'(u)s + x(0) = \mathbf{F}'(g(x(0)))s + x(0),$$

and that the projected characteristic $y(s) = (x(s), s) = (\mathbf{F}'(g(x(0)))s + x(0), s)$ $(s \geq 0)$ is a straight line, along which u is a constant. Hence, the solution is given as

$$u = g(x - t\mathbf{F}'(u)),$$

in implicit form. It is standard to check that[3], provided that $1 + tDg(x - t\mathbf{F}'(u)) \cdot \mathbf{F}''(u) \neq 0$, u solves the scalar conservation law.

2.3. Shock wave phenomena and weak solutions

Whenever a new scientific concept comes into prominence, it sends shock waves of surprise to the scholars contributing to that field.

E. A. Carlson

2.3.1. The formation of shocks. To illustrate shock wave phenomena and how shocks are formed, we consider the quasilinear Burgers equation

$$u_t + uu_x = 0, \tag{2.15}$$

with some smooth initial data

$$u(x,0) = u_0(x).$$

From the previous section, we find that, at least for a short time,

$$u(x,t) = u_0(x - ut).$$

Differentiation with respect to x and t gives

$$u_x = (1 - u_x t)u_0', \quad u_t = (-u - u_t t)u_0'.$$

Thus,

$$u_x = \frac{u_0'}{1 + tu_0'}, \quad u_t = \frac{-u_0' u}{1 + tu_0'}.$$

[3]Exercise!

If $u_0'(x) < 0$ for some x then u_x and u_t will blow up when $t = t_0 = -\frac{1}{u_0'} > 0$. Thus, u must break down in finite time and wave breaking will essentially happen, forming *shock waves*, regardless of the smoothness of u_0. Therefore, in contrast to the heat equation that we encountered earlier whose solution is smooth for all time, our solution in this case is usually "rougher" than the initial data. This is a remarkable property of Burgers' equation (and nonlinear conservation laws in general). On one hand, it reflects the failure of the characteristic method. On the other hand, it implies that *global solutions* (solutions that exist for all time) for nonlinear conservation laws, can only be found in the space of discontinuous functions. See Figure 2.4 for an illustration of the wave breaking phenomenon and Figure 2.5 for a similar situation observed in the ocean.

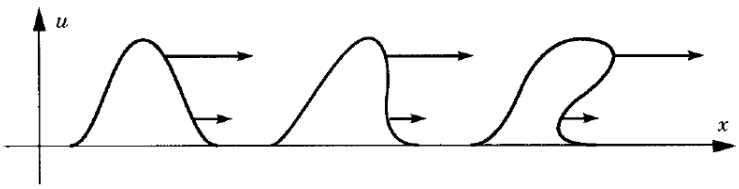


FIGURE 2.4. Wave breaking phenomenon ([**Str08**]).



FIGURE 2.5. Wave breaking phenomenon in the ocean (credit: Matt Clark, via Imgur).

2.3.2. Weak solutions. We have seen that solutions to nonlinear conservation laws will generally be discontinuous and the method of characteristics only provides a solution locally, which exists until the

first breaking point. To extend the solution further, we need to relax the differentiability (or even the continuity) assumptions on the solution, but the solution still satisfies the equation in some sense. The idea is like this. Assume v is smooth, we multiply the equation (2.3) by some smooth, compactly supported function ϕ (called *test function*) and use integration by parts to "transfer" the derivatives from v to ϕ,

$$\begin{aligned} 0 &= \int_0^\infty \int_{\mathbb{R}} [v_t + f(v)_x]\phi \mathrm{d}x\mathrm{d}t \\ &= -\int_{\mathbb{R}} v\phi|_{t=0}\mathrm{d}x - \int_0^\infty \int_{\mathbb{R}} v\phi_t \mathrm{d}x\mathrm{d}t - \int_0^\infty \int_{\mathbb{R}} f(v)\phi_x \mathrm{d}x\mathrm{d}t. \end{aligned}$$

Thus

$$\int_0^\infty \int_{\mathbb{R}} [v\phi_t + f(v)\phi_x]\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} v_0\phi(t=0)\mathrm{d}x = 0,$$

which makes perfectly good sense even if v is only a bounded function.

DEFINITION 2.1. A function $v \in L^\infty(\mathbb{R}^n \times [0,\infty))$ is called a weak solution for (2.1)-(2.2) if the identity

$$\int_{\mathbb{R}^n} \int_0^\infty [v(x,t)\phi_t + \mathbf{F}(v(x,t))\cdot\nabla\phi]\mathrm{d}t\mathrm{d}x + \int_{\mathbb{R}^n} v_0(x)\phi(x,0)\mathrm{d}x = 0 \quad (2.16)$$

holds for any test function $\phi \in C_0^\infty(\mathbb{R}^n \times [0,\infty))$.

Notice that if we choose the test function to be in $C_0^\infty(\mathbb{R}^n \times (0,\infty))$ (i.e. excluding time 0), then the second term in (2.16) disappears (no initial condition involved). However, it can be proved that these choices of test functions are equivalent. Note also that v is only required to be in $L^\infty(\mathbb{R}^n \times [0,\infty))$, which includes discontinuous functions, for (2.16) to make sense. However, not every discontinuous function is permitted since (2.16) does impose severe restrictions on the curves of discontinuity as we shall see from the following result, known as the *Rankine–Hugoniot jump condition*. Consider for simplicity $n = 1$ and assume that we have an isolated discontinuity (that is, $v(x,t)$ is a classical solution in a sufficiently small neighbourhood of $x(t)$) that moves along a smooth curve $\Gamma : x = x(t)$.

THEOREM 2.2 (Rankine–Hugoniot). *The discontinuity curve $x = x(t)$ of the weak solution v satisfies the following constraint*

$$x'(t)[v_l - v_r] = [f(v_l) - f(v_r)], \quad (2.17)$$

where v_l and v_r are the left and the right states of v on the shock curve x. If we denote $s := x'(t)$ (s is called the shock speed*) and use the notation $[\![v]\!] = v_r - v_l$ (called the* shock strength*) then* (2.17) *becomes*

$$s[\![v]\!] = [\![f]\!].$$

PROOF. Let $\phi \in C_0^\infty$ be supported in a small ball centering on Γ. The ball is so small that it does not intersect with the x-axis and u is smooth everywhere in the ball except along Γ. Denote this ball by D, which is divided into two parts D_1, D_2 by Γ (see Figure 2.6). As $\phi = 0$

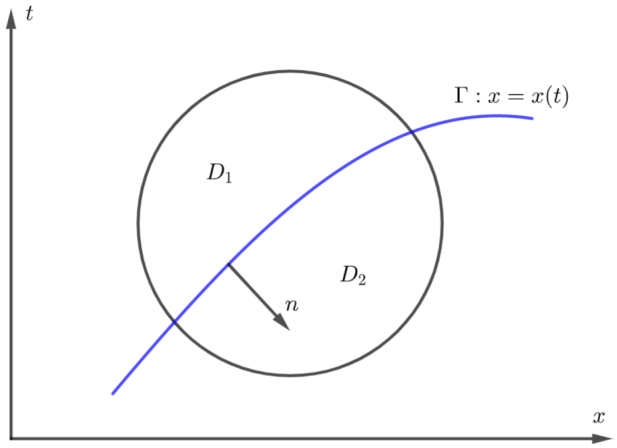


FIGURE 2.6. Rankine-Hugoniot condition along a shock curve.

along the x-axis, the definition of weak solutions becomes

$$\int_0^\infty \int_D [v\phi_t + f(v)\phi_x]\mathrm{d}x\mathrm{d}t = 0.$$

We write the left hand side as $\int\int_{D_1} + \int\int_{D_2}$ and use integration by parts. Since v is smooth in D_1, D_2, we have

$$\int_0^\infty \int_{D_1} [v\phi_t + f(v)\phi_x]\mathrm{d}x\mathrm{d}t = -\int_0^\infty \int_{D_1} [v_t + f(v)_x]\phi\mathrm{d}x\mathrm{d}t + \oint_{\partial D_1} [vn_t + f(v)n_x]\phi\mathrm{d}S,$$

where $n = (n_x, n_t)$ is the unit normal to the curve Γ, pointing from D_1 to D_2. Since v solves the equation classically in D_1, we have

$$\int_0^\infty \int_{D_1} [v\phi_t + f(v)\phi_x]\mathrm{d}x\mathrm{d}t = \oint_{\partial D_1} [vn_t + f(v)n_x]\phi\mathrm{d}S.$$

Similarly

$$\int_0^\infty \int_{D_2} [v\phi_t + f(v)\phi_x]\mathrm{d}x\mathrm{d}t = \oint_{\partial D_2} [vn_t + f(v)n_x]\phi\mathrm{d}S.$$

Since ϕ vanishes on $\partial D_1, \partial D_2$ except along Γ, we obtain

$$\int_\Gamma [(v_l - v_r)n_t + (f(v_l) - f(v_r))n_x]\phi\mathrm{d}S = 0.$$

This holds for all test function ϕ as above, so along Γ,

$$[v_l - v_r]n_t + [f(v_l) - f(v_r)]n_x = 0.$$

Now since Γ is given by $x = x(t)$, we have $\frac{n_t}{n_x} = -\dot{x} = s$, which gives

$$s[\![v]\!] = [\![f]\!],$$

as required. □

The result of Theorem 2.2 has a great impact on gas dynamics and the theory of hyperbolic conservation laws. The formula (2.17) can be used to find explicitly weak solutions to some initial value problems that are not classically solvable.

EXAMPLE 2.3. Consider the Burgers equation (2.5) with initial data

$$u_0(x) = \begin{cases} 1 & \text{if } x < 0 \\ 1 - x & \text{if } 0 \le x < 1 \\ 0 & \text{if } x \ge 1. \end{cases} \tag{2.18}$$

A weak solution u for (2.5) can be calculated up to $t < 1$ by using the method of characteristics (see Section 2.2). The implicit formula is

$$u = u_0(x - ut).$$

From the initial condition,

- $u = 1$ if $x - ut < 0$, which gives $u = 1$ if $x < t$,
- $u = 1 - (x - ut)$ if $0 \le x - ut < 1$, which gives $u = \frac{1-x}{1-t}$ if $t \le x < 1$,
- $u = 0$ if $x - ut \ge 1$, which gives $u = 0$ if $x \ge 1$.

Therefore

$$u(x,t) = \begin{cases} 1 & \text{if } x < t, 0 \le t < 1 \\ \frac{1-x}{1-t} & \text{if } t \le x < 1, 0 \le t < 1 \\ 0 & \text{if } x \ge 1, 0 \le t < 1. \end{cases} \tag{2.19}$$

For $t \ge 1$, the method of characteristics breaks down, we cannot proceed further using this. By applying the Rankine-Hugoniot jump condition

$$s = \frac{[\![f]\!]}{[\![u]\!]} = \frac{u_l^2 - u_r^2}{2(u_l - u_r)} = \frac{u_l + u_r}{2}.$$

For $u_l = 1, u_r = 0$, we have that $s = \frac{1}{2}$. Thus we can define

$$u(x,t) = \begin{cases} 1 & \text{if } x < 1 + \frac{1}{2}(t-1) \\ 0 & \text{if } x \ge 1 + \frac{1}{2}(t-1), \end{cases} \tag{2.20}$$

to be a weak solution of (2.5) for $t \ge 1$. See Figure 2.7.

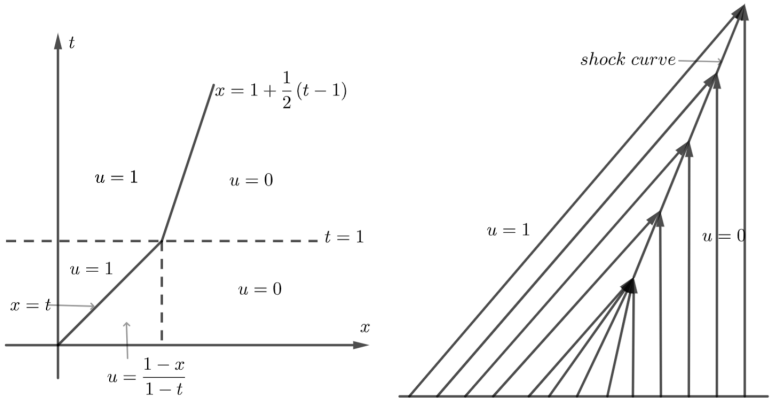


FIGURE 2.7. Shock wave solution to Burgers' equation.

A serious problem with weak solutions is that they are not unique as the following examples show.

EXAMPLE 2.4. Consider the Burgers' equation (2.5) with initial data

$$u_0(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0. \end{cases}$$

Nonlinear conservation laws with initial data given in this form are called *Riemann problems*. The method of characteristics determines the solution everywhere in $t > 0$ except in the sector $0 < x < t$. The following two functions differ precisely in this sector

$$u_1(x,t) = \begin{cases} 0 & \text{if } x < t/2 \\ 1 & \text{if } x > t/2, \end{cases} \qquad u_2(x) = \begin{cases} 0 & \text{if } x < 0 \\ x/t & \text{if } 0 < x < t \\ 1 & \text{if } x > t. \end{cases}$$

The characteristics are depicted in Figure 2.8. It is easy to check that

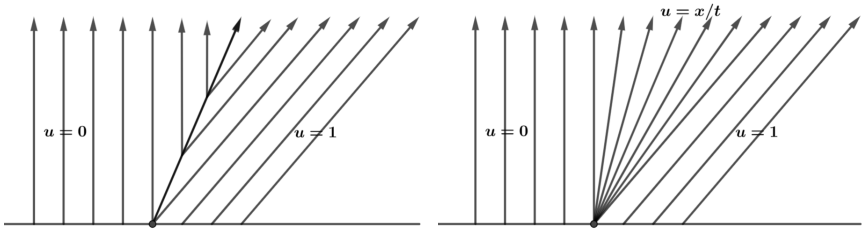


FIGURE 2.8. Different solutions to Burgers' equation.

these are both solutions to our problem. It is interesting to note that u_2 is a continuous solution. The fact that a continuous solution can

have discontinuous initial values is again a distinct feature of nonlinear conservation laws!

EXAMPLE 2.5. We consider again the Burgers equation (2.5) with initial data

$$u_0(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0. \end{cases} \tag{2.21}$$

Then for any $\alpha \in [0, 1]$, each u_α provides a solution[4] for (2.5) where

$$u_\alpha(x) = \begin{cases} -1 & \text{if } x < -t \\ x/t & \text{if } -t < x \leq -\alpha t \\ -\alpha & \text{if } -\alpha t < x \leq 0 \\ \alpha & \text{if } 0 < x \leq \alpha t \\ x/t & \text{if } \alpha t < x \leq t \\ 1 & \text{if } t < x. \end{cases} \tag{2.22}$$

Hence, there exists a continuum set of weak solutions to (2.5)-(2.21)! It is therefore required to impose more restrictions on weak solutions to single out the correct one.

2.4. Entropy conditions

This algorithm is admittedly complicated, but no simpler mechanism seems to do nearly as much.

— D. E. Knuth, *The TEXbook*

2.4.1. Derivation. One of the criteria for selecting solutions is coming from the fact that the desired solution should be physically meaningful. In other words, it should be the limit state (in a sense to be made precisely) of a physical model where one adds an extra diffusion term to the right-hand side of (2.1),

$$\frac{\partial v}{\partial t}(x, t) + \nabla \cdot f(v(x, t)) = \epsilon \Delta v(x, t). \tag{2.23}$$

where Δ is the Laplacian operator with respect to spacial variables, that is, $\Delta v(x, t) = \sum_{i=1}^{n} \frac{\partial^2 v}{\partial x_i^2}$. The equation (2.23) may reflect a more physically meaningful model. Let us consider the traffic flow equation

$$\rho_t + (\rho v(\rho))_x = 0.$$

[4]See Exercise 2.5.

One may argue that in reality, the car drivers would naturally reduce their speed as they are approaching a traffic jam (or as they are seeing a relatively larger density of cars ahead). In that case, $v(\rho)$ may be replaced by

$$\tilde{v}(\rho) = v(\rho) - \epsilon \frac{\rho_x}{\rho},$$

and hence the equation becomes

$$\rho_t + (\rho v(\rho))_x = \epsilon \rho_{xx}.$$

This example shows the natural appearance of a diffusion term on the right-hand side of (2.23). It does not imply however that this should be the case for other situations. In general, one may consider (2.23) even without any physical explanations to the diffusive term on the right-hand side. This may be because its presence could be beneficial for analytical or computational purposes. Therefore, as mentioned by Dafermos in his famous book [**Daf16**], the term $\epsilon \Delta v$ should be viewed as "artificial viscosity".

We denote by η some convex function, $\eta : \mathbb{R} \to \mathbb{R}$, and define the associated function $q : \mathbb{R} \to \mathbb{R}^n$ by

$$q(u) = \int^u \eta'(s) f'(s) ds, \tag{2.24}$$

that is, $q'(u) = \eta'(u) f'(u)$.

REMARK 2.6. The function η is called an *entropy* function and the corresponding function q is called an *entropy flux*. The pair (η, q) is called an *entropy-entropy flux pair*.

Let us come back to (2.23) and consider v^ϵ to be a solution to the initial value problem (2.23)-(2.2).[5] Multiplying both sides of (2.23) by $\eta'(v^\epsilon(x,t))$ and using (2.24), we get

$$\partial_t \eta(v^\epsilon) + \partial_x q(v^\epsilon) = \epsilon \Delta \eta(v^\epsilon) - \epsilon \eta''(v^\epsilon) |\nabla v^\epsilon|^2. \tag{2.25}$$

Since η is convex, $\eta'' \geq 0$, we can deduce from (2.25) that

$$\partial_t \eta(v^\epsilon) + \partial_x q(v^\epsilon) \leq \epsilon \Delta \eta(v^\epsilon).$$

Thus if we let $\epsilon \to 0$ and assume that,

$$v^\epsilon \to v \quad \text{almost everywhere,}$$

then v satisfies

$$\partial_t \eta(v) + \partial_x q(v) \leq 0. \tag{2.26}$$

[5]This is possible thanks to the theory of parabolic equations, of which (2.23) is a special case. In fact, there exists a unique smooth solution to (2.23) (with f regular enough, see [**Tay11**, e.g. Proposition 1.5, Chapter 15]).

We just arrive at the following definition (where we relax the regularity of η and q a bit and allow them to be merely Lipschitz continuous).

DEFINITION 2.7 (Entropy solution). A function $v \in L^\infty(\mathbb{R}^n \times [0,\infty))$ is called an entropy solution for (2.1)-(2.2) if (2.26) holds in the sense of distributions, that is

$$\int_{\mathbb{R}^n}\int_0^\infty \eta(v(x,t))\phi_t + q(v(x,t))\phi_x \mathrm{d}t\mathrm{d}x + \int_{\mathbb{R}^n} \phi(x,0)\eta(v_0(x))\mathrm{d}x \geq 0 \tag{2.27}$$

for any convex entropy-entropy flux pair (η, q) and any non-negative test function $\phi \in C_0^\infty(\mathbb{R}^n \times [0,\infty))$.

Again, the second integral in (2.27) disappears if we choose the test function in $C_0^\infty(\mathbb{R}^n \times (0,\infty))$. It is also not hard to see that a classical solution is an entropy solution, where (2.27) is satisfied as an identity. The inequality (2.27) is sometimes called *Kruzkov's entropy condition*, to recognise the celebrated work by Kruzkov in [**Kru70**], where the entropy condition is stated in an abstract form as above. In that work, Kruzkov shows that we can actually choose the entropy-entropy flux pair to be very specific, namely

$$\eta(v) = |v-k|, \quad q(v) = \mathrm{sign}(v-k)(f(v)-f(k)), \tag{2.28}$$

for any $k \in \mathbb{R}$. This will be referred to as the *Kruzkov entropy-entropy flux pair*.

2.4.2. Shock admissibility conditions. When dealing with discontinuous solutions, it may be convenient (for example in numerical approximations) to test the entropy condition at every discontinuity curve. A weak solution will be an entropy solution if each of its shock curves satisfies certain *admissibility conditions*. Let us consider the equation (2.3). Recall that the corresponding characteristics are given by

$$x(t) = f'(v)t + x(0), \tag{2.29}$$

for some $x(0)$ on the x-axis. As we already know, characteristic lines generally intersect and shock waves form. In other words, they collide on the shock curve. If we interpret characteristics as carrying "information", we lose information when a shock forms. This is in contrast with the solution u_1 in Example 2.4 where characteristics go out from the shock curve (Figure 2.8), in other words, "information" is created at the shock, which does not make sense physically. This will be avoided if we impose the condition that, if we go back on any characteristic, we do not run into a shock. By virtue of (2.29), this happens when the

characteristic speed on the left is always bigger than on the right, that is,

$$f'(v_l) > s > f'(v_r), \tag{2.30}$$

where s is the shock speed, $s = \dot{x}(t)$. The condition (2.30) is called the *Lax's shock admissibility condition*, a criterion introduced in the seminal paper [**Lax57**]. Equivalently, if f is convex, then (2.30) reduces to

$$v_l > v_r. \tag{2.31}$$

A shock curve that satisfies either of the above conditions is called an *admissible shock*. In other words, we only accept a shock if the wave from the left moves faster than the wave from the right of the curve. This admissibility criterion may be illustrated in the traffic flow model, where shocks are present as traffic jams, the car flow approaching a traffic jam moves faster than the one after joining in the traffic jam.

With more general f (not necessarily convex), for v to be an entropy solution, one requires the following condition, known as *Oleinik's shock admissibility condition* [**Ole59**], along every shock curve,

$$\frac{f(v_r) - f(v)}{v_r - v} \leq \frac{f(v_r) - f(v_l)}{v_r - v_l} \leq \frac{f(v_l) - f(v)}{v_l - v}, \tag{2.32}$$

for v in between v_l and v_r.

It is easy to see that Oleinik's shock admissibility condition implies Lax's shock admissibility condition. However, if the flux function is genuinely nonlinear ($f'' \neq 0$), they are equivalent. Moreover, we have

THEOREM 2.8. *Assume that f is convex. Then the Kruzkov entropy condition, the Lax entropy condition and the Oleinik entropy condition are equivalent.*

Discussions leading to a proof of this result can be found, for instance, in Chapter 2 in [**HR15**] and in Chapter VIII in [**Daf16**]. See also Exercise 2.8.

REMARK 2.9.

- Even though Lax or Oleinik entropy condition may be easier to work with, they do not indicate whether or not the considered solution is a weak solution. The Kruzkov notion of entropy condition is convenient in the sense that it includes the notion of weak solutions. Indeed, by applying (2.27) with $\eta(v) = \pm v$ and $q(v) = \pm f(v)$ we derive Definition 2.1 of weak solutions.
- There are other entropy conditions, such as Liu entropy condition, that are particularly useful in the case of systems of hyperbolic conservation laws, hence not discussed here. For an extensive presentation of this topic, see [**Daf16**].

REMARK 2.10 (Entropy, irreversibility and entropy production). The entropy condition (2.26) has some connection to the Second Law of Thermodynamics (which says that the entropy[6] of a system is always increasing in time and the process is irreversible). In fact, (2.26) induces a time irreversibility condition on solutions. Indeed, assume that $v(x,t)$ is an entropy solution to (2.1) and denote

$$\tilde{v}(x,t) := v(-x,-t).$$

Then it can be easily verified that $\tilde{v}$ still satisfies (2.1) in the weak sense. However, $\tilde{v}$ is not an entropy solution except when it is a classical solution, since the inequality (2.26) reverses its sign. When v develops shocks, the left hand side of (2.26) (viewed in the sense of distributions)

$$\mu := \partial_t \eta(v) + \partial_x q(v)$$

is a strictly negative measure. The measure defined above is called the *entropy production measure*. An interesting property of this measure is that it concentrates on the set of points of jump discontinuity[7](see, for instance, [**DLR03, DLOW03, COW08, BM17**]). We refer to [**Daf16**, Section 4.5, 3.3] for further discussions on this topic.

2.5. Existence and uniqueness of entropy solutions

Faith is an island in the setting sun. But proof, yes. Proof is the bottom line for everyone.

— Paul Simon

2.5.1. Functions of bounded variation. We first need some definitions regarding functions of *bounded variation* which play a fundamental role in the theory of hyperbolic conservation laws.

DEFINITION 2.11. Let $\Omega \subset \mathbb{R}^n$ be open. The *total variation* of $u \in L^1(\Omega)$ is given by

$$TV(u,\Omega) := \sup\left\{ \int_\Omega u(x) \mathrm{div}\psi(x) \mathrm{d}x : \psi \in C_c^\infty(\Omega, \mathbb{R}^n), \|\psi\|_{L^\infty(\Omega)} \leq 1 \right\}. \tag{2.33}$$

[6]We note that the "entropy" notion η used in the context of hyperbolic conservation laws takes an opposite sign with the counterpart "entropy" used in physics.

[7]This is still an open problem for scalar conservation laws in $\mathbb{R}^n$ with $n > 1$. See recent work [**Sil19**] for more information.

(We use the notation $TV(u)$ instead of $TV(u, \Omega)$ when Ω is clear from the context.) We say that u is a *function of bounded variation* if $TV(u) < \infty$, and we denote

$$BV(\Omega) := \{u : \Omega \to \mathbb{R}, TV(u) < \infty\}, \tag{2.34}$$

the space of functions of bounded variation on Ω. If $u \in BV(\Omega')$ for all $\Omega' \subset\subset \Omega$ (i.e., Ω' is compactly contained in Ω), we say that u is of locally bounded variation and use $BV_{\text{loc}}(\Omega)$ to denote the space of all these functions.

REMARK 2.12.

- If $u \in L^1(\Omega)$ is a function of bounded variation then its distributional derivative, denoted by Du or grad u, is a finite (vector-valued) signed measure, and the total variation of this measure, denoted by $|Du|$, equals the total variation of u. We refer to [**AFP00**] for detailed discussion and treatment of BV functions.
- In one dimension ($n = 1$), the classical definition of functions of bounded variation is as follows. $u : I \to R$ is of bounded variation in an interval $I \subset \mathbb{R}$ if $TV(u) < \infty$, where

$$TV(u) := \sup \left\{ \sum_{i=1}^{k-1} |u(x_{i+1}) - u(x_i)| : k \geq 2, x_1 < \cdots < x_k, x_i \in I \right\}, \tag{2.35}$$

and if $u \in C^1(I)$,

$$TV(u) := \int_I u'(x) \mathrm{d}x.$$

This agrees with (2.33) in Definition 2.11. We note however that in one dimension one does not normally require u to be integrable. Nevertheless, if I is bounded then u is automatically integrable since it must be bounded thanks to (2.35). Again, we refer to [**AFP00**, Section 3.2] (see also [**Leo17**, Chapter 2 and 7]) for more information.

Functions of bounded variation possess certain properties that nicely suit the study of nonlinear conservation laws. One example of such properties is if $f : D \to \mathbb{R}$ is of bounded variation, f has one-sided limits (that is, limits from the left and the right) at every point[8] in D. Furthermore, the derivative f' exists almost everywhere.

In loose terms, a function of bounded variation is allowed to be discontinuous but not so badly discontinuous, in the sense that the set

[8]So that for instance the one-sided limits in the Rankine-Hugoniot condition, Theorem 2.2, are well-defined.

of discontinuity should be at most countable. On the other hand, not every continuous function is of bounded variation, for instance, if it oscillates too much around certain points. The following example will give some ideas.

EXAMPLE 2.13. Consider the following functions on $[0, 2/\pi]$:

$$f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

$$g(x) = \begin{cases} x\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

$$h(x) = \begin{cases} x^2\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Their graphs are given in Figure 2.9. The function f is not continuous nor of bounded variation, while g is continuous but not of bounded variations, and h is both continuous and of bounded variation[9]. We can see that all these functions oscillate around 0 but with different scales. If we take away 0 from the domain and only consider f, g, h on $[a, b]$ with $b > a > 0$, then they will be of bounded variation!

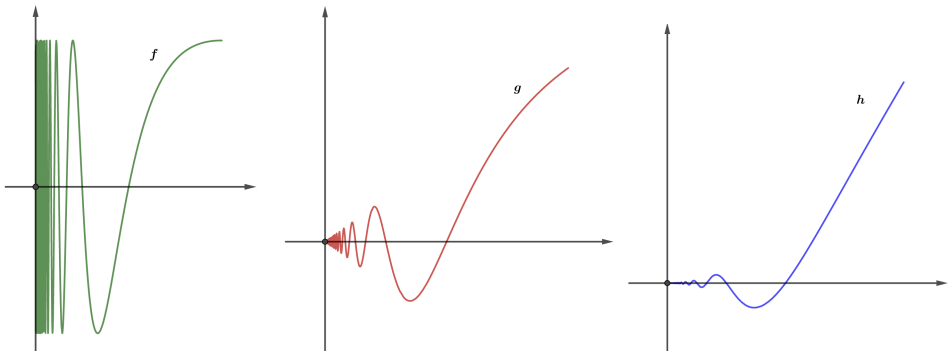


FIGURE 2.9. Graphs of f, g and h.

2.5.2. Kruzkov's theorem. Given the entropy conditions introduced in Section 2.4, one recovers the uniqueness. Moreover, the entropy solution enjoys other nice properties recorded in the following theorem.

[9]Exercise: verify these facts!

THEOREM 2.14 (Kruzkov). *For every $v_0 \in L^\infty(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$, there exists a unique entropy solution v to (2.1)-(2.2) in $C([0,\infty); L^1_{\text{loc}}(\mathbb{R}^n))$ that satisfies the following properties, for every $t > 0$:*

(i) *Maximum principle*:

$$\|v(\cdot,t)\|_{L^\infty(\mathbb{R}^n)} \leq \|v_0(\cdot)\|_{L^\infty(\mathbb{R}^n)}.$$

(ii) *Total variation diminishing*: *if $v_0 \in BV(\mathbb{R}^n)$,*

$$TV(v(\cdot,t)) \leq TV(v_0). \tag{2.36}$$

Moreover, if $\bar{v}$ is the entropy solution to (2.1) corresponding to the initial data $\bar{v}_0$ with $\bar{v}_0 \in L^1(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, then the following hold for all $t > 0$:

(iii) *L^1-contractility*:

$$\|v(\cdot,t) - \bar{v}(\cdot,t)\|_{L^1(\mathbb{R}^n)} \leq \|v_0(\cdot) - \bar{v}_0(\cdot)\|_{L^1(\mathbb{R}^n)}. \tag{2.37}$$

(iv) *Monotonicity*:

$$v_0 \leq \bar{v}_0 \quad \textit{implies} \quad v(\cdot,t) \leq \bar{v}(\cdot,t).$$

The L^1-contraction property (2.37) is probably the most remarkable about the entropy solution recorded in Kruzkov's theorem. It implies immediately that the entropy solution is unique. We also want to emphasise (2.36), another important property of the entropy solution: if v_0 is a function of bounded variation, then $v(\cdot,t)$ is also a function of bounded variation, for all $t > 0$. In fact, this property can be proved quite easily using (2.37). Indeed, we consider the total variation of v, regarded as the total variation of the ($\mathbb{R}^n-$valued) measure grad$v$ and notice that

$$\begin{aligned}|\text{grad}_\alpha(v(\cdot,t))|(\mathbb{R}^n) &= \lim_{h\to 0} \frac{1}{h} \int_{\mathbb{R}^n} |v(x+he_\alpha,t) - v(x,t)| \mathrm{d}x \\ &\leq \lim_{h\to 0} \frac{1}{h} \int_{\mathbb{R}^n} |v_0(x+he_\alpha) - v_0(x)| \mathrm{d}x = |\text{grad}_\alpha v_0|(\mathbb{R}^n),\end{aligned}$$

for $\alpha = 1, \ldots, n$, thanks to (2.37).

For proofs of other properties, we direct the reader to [**Daf16**].

The existence theory for scalar conservation laws is well-developed. In the next sections, we outline two of the most used methods to construct an entropy solution for (2.1). But before that, we consider the entropy solution to the Riemann problem, which is very important in the development of the theory of nonlinear conservation laws.

2.5.3. Entropy solution to the Riemann problem. We have encountered briefly in Example 2.4 and Example 2.3.2 the Riemann problem for Burgers' equation when we deal with initial data given by a two-state shock wave. Let us now generalise this and consider the Riemann problem for the scalar conservation law (2.3)

$$v(x,0) = g(x) = \begin{cases} v_l & \text{if } x < 0 \\ v_r & \text{if } x > 0. \end{cases} \tag{2.38}$$

This seemingly simple problem is very important in the theory of nonlinear conservation laws, in particular, it is the building block of the front tracking method that we will explain in the next section. We will assume that the flux function f is C^2 and uniformly convex (i.e., $f'' > 0$). Denote[10] $G = (f')^{-1}$.

THEOREM 2.15. (i) *If $v_l > v_r$, the unique entropy solution of the Riemann problem* (2.3)-(2.38) *is given by a* shock wave

$$v(x,t) = \begin{cases} v_l & \text{if } x < \sigma t \\ v_r & \text{if } x > \sigma t \end{cases} \quad (x \in \mathbb{R}, t > 0),$$

with constant speed σ, where

$$\sigma := \frac{f(v_l) - f(v_r)}{v_l - v_r}.$$

(ii) *If $v_l < v_r$, the unique entropy solution of the Riemann problem* (2.3)-(2.38) *is given by a* rarefaction wave

$$v(x,t) = \begin{cases} v_l & \text{if } x < f'(v_l)t \\ G(\frac{x}{t}) & \text{if } f'(v_l)t < x < f'(v_r)t \\ v_r & \text{if } x > f'(v_r)t \end{cases} \quad (x \in \mathbb{R}, t > 0).$$

Illustrations of these wave solutions can be seen in Figure 2.10 and Figure 2.11. The proof of this theorem relies on the fact that the function v given in both cases solves the equation and satisfies the entropy conditions, hence it is the only one. We refer to [**Eva10**] for the details.

2.5.4. Front tracking method. The idea is to approximate the initial function by a step function and the flux by a piecewise linear function. An approximate solution of (2.1)-(2.2) is then given by solving a finite set of *Riemann problems*, with each initial data consisting of a single discontinuity wave. For the purpose of illustration, let us consider the 1D problem (2.3)-(2.2) where $f'' \geq 0$. Let v_0 be

[10] G is well-defined due to the uniform convexity of f. Why? Exercise!

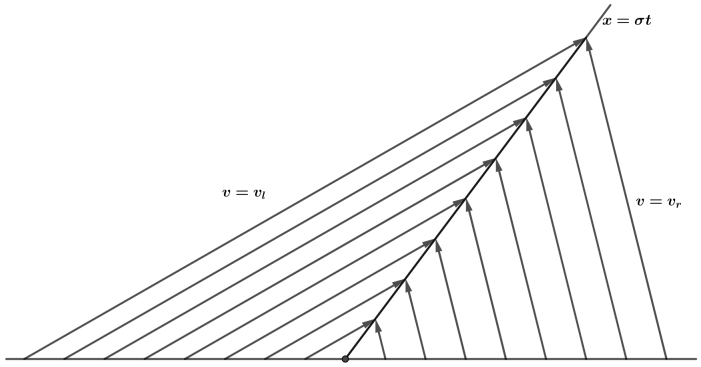


FIGURE 2.10. Shock wave solution to the Riemann problem.

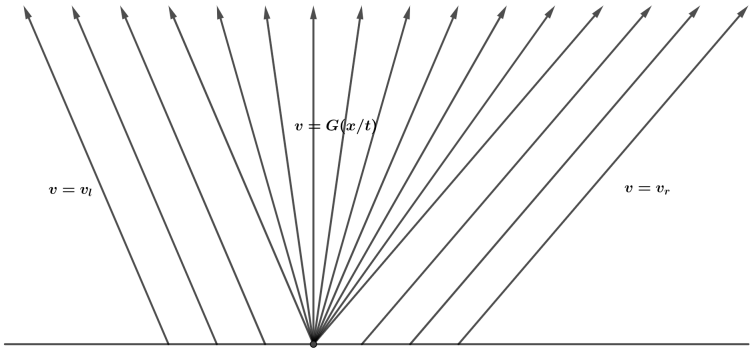


FIGURE 2.11. Rarefaction wave solution to the Riemann problem.

a function of bounded variation and M be a positive number such that $\|v_0\|_{L^\infty(\mathbb{R})} \leq M$. For some natural number N, define the set $A_N \subset [-M, M]$ by

$$A_N := \{v_{j,N} = j\frac{M}{2^N},\ j = -2^N, \ldots, 2^N\}.$$

Then define the piecewise linear function f^N by (see Figure 2.12)

$$\begin{gathered} f^N(s) = f(v_{j,N}) + \frac{2^N}{M}(s - v_{j,N})(f(v_{j+1,N}) - f(v_{j,N})), \\ \text{for } s \in (v_{j,N}, v_{j+1,N}], \quad j = -2^N, \ldots, 2^N. \end{gathered} \tag{2.39}$$

Let $v_{0,N}$ be a step function taking values in the set A_N,

$$v_{0,N} = v_{0,N}^j \chi_{(x_{j-1}, x_j]}, \quad v_{0,N}^j \in A_N$$

such that $v_{0,N}$ approximates v_0 in L^1 (see Figure 2.13),

$$\|v_{0,N} - v_0\|_{L^1(\mathbb{R})} \to 0, \quad \text{as } N \to \infty.$$

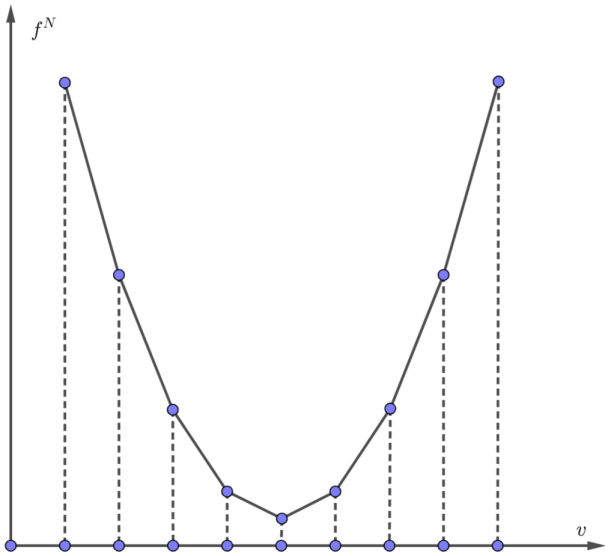


FIGURE 2.12. Piecewise linear approximation f^N of f.

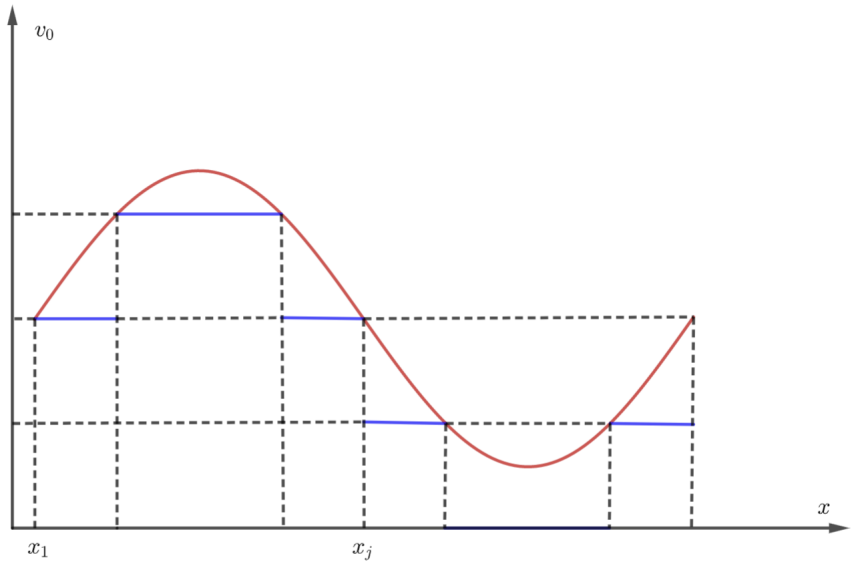


FIGURE 2.13. A piecewise constant approximation (the blue pieces) of v_0 (the red curve).

Consider the Cauchy problem

$$v_t + f^N(v)_x = 0, \quad v(x,0) = v_{0,N}(x). \tag{2.40}$$

Since $v_{0,N}$ is a step function with break points at, say $\{x_j\}_{j=1}^J$, a local solution to (2.40), defined for sufficiently small $t > 0$, is given by solving

a finite number of *Riemann problems*, each of them with initial data of the form

$$v(x,0) = \begin{cases} v_l & \text{if } x < x_j \\ v_r & \text{if } x > x_j, \end{cases} \tag{2.41}$$

where v_l and v_r denote the values of v_0 at the left and the right limits $v_{0,N}(x_j-), v_{0,N}(x_j+) \in A_N$. One claims that Riemann problems obey a maximum principle, meaning the solution to (2.40) with initial data (2.41) remains in between v_l and v_r, and that the solution of (2.40) will take values in the set $\{v_{j,N}\} \cup \{\text{break points of } f^N\}$.[11] This solution is defined up to some time $t = t_1$ where two or more jump discontinuities (coming from nearby Riemann problems) collide, forming a new Riemann problem. Since we already know how to solve Riemann problems, we can define the solution until the new collision time $t = t_2$. This procedure continues for the new Riemann problem and the solution is prolonged up to the next collision time $t = t_3$, and so on. Hence, we can find the solution for all time. Note that, this process does not go on forever thanks to a fact that the number of interactions is finite, see for example [**Bre00**] for a proof. This is understandable because each time a new collision forms, two or more discontinuities collapse to produce a single discontinuity. The wave pattern is hence simplified as the number of jump discontinuities is decreasing over time. In fact, if the initial condition v_0 is a non-negative differentiable function with compact support and the flux function is smooth and uniformly convex, then after a certain time, all shocks will finally be merged and continue as a single shock (see [**Whi75**]).

The method of front tracking approximations provides an effective tool in establishing the existence of entropy solutions for the scalar conservation laws. Moreover, one can use this method to derive the stability property of the entropy solution with respect to the flux function, as the following result shows. Hereafter we denote

$$\|f\|_{Lip} := \sup_{u \neq v} \left| \frac{f(u) - f(v)}{u - v} \right|, \tag{2.42}$$

the Lipschitz constant for f.

THEOREM 2.16. *Let $v_0 \in L^1(\mathbb{R}) \cap BV(\mathbb{R})$ and $v, \bar{v}$ be the entropy solutions to* (2.3)-(2.2) *with respect to Lipschitz continuous flux functions*

[11] This still works with nonconvex f but with some modification, more precisely, the solution will take values in $\{v_{j,N}\} \cup \{\text{break points of } f^N_{\smile} \text{ or } f^N_{\frown}\}$, where $f_{\smile}$ (or $f_{\frown}$) denotes the convex envelope (or concave envelope) of f.

f, g (respectively), then there exists a constant $C > 0$ such that

$$\|v(\cdot, t) - \bar{v}(\cdot, t)\|_{L^1(\mathbb{R})} \leq Ct\|f - g\|_{Lip}, \tag{2.43}$$

for every $t > 0$.

The constant C in (2.43) depends only on v_0 and can be chosen to be $TV(v_0)$. A proof of this result using the front tracking method can be found in [**HR15**, Chapter 2]. We note that the assumption $v_0 \in BV(\mathbb{R})$ is needed in establishing *a priori* a bound on $TV(v(\cdot, t))$ which is an essential part in the proof of existence (typically via Helly's theorem).

The front tracking method is considerably more intricate for higher dimensions, we refer to the monograph [**Bre00**] for the detailed treatment.

2.5.5. Vanishing viscosity method. Let ϵ be a (small) positive number. Consider the "viscous" equation

$$\frac{\partial v}{\partial t}(x, t) + \nabla \cdot f(v(x, t)) = \epsilon \Delta v(x, t), \tag{2.44}$$

with some initial data

$$v(x, 0) = v_0(x), \tag{2.45}$$

where $v_0 \in L^\infty(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$. The discussions in Section 2.4.1 lead to the following result.

THEOREM 2.17. *Let v^ϵ be the solution of* (2.44)-(2.45). *Assume that for some sequence $\{\epsilon_k\}$, with $\epsilon_k \to 0$ as $k \to \infty$,*

$$v^{\epsilon_k} \to v \quad \textit{boundedly almost everywhere on } \mathbb{R}^n \times [0, \infty).$$

for some function v. Then v is an entropy solution of (2.1)-(2.2).

Note that since (2.44) is a parabolic equation, the Cauchy problem (2.44)-(2.45) always has a unique smooth solution (see, for example, [**Tay11**]). Moreover, this solution also enjoys the L^1-*contraction property*, thanks to the following result.

THEOREM 2.18. *Let v^ϵ and $\bar{v}^\epsilon$ be solutions of* (2.44) *with respect to initial data v_0 and $\bar{v}_0$ that are in $L^1(\mathbb{R}^n)$ and bounded. Then*

$$\|v^\epsilon(\cdot, t) - \bar{v}^\epsilon(\cdot, t)\|_{L^1(\mathbb{R}^n)} \leq \|v_0(\cdot) - \bar{v}_0(\cdot)\|_{L^1(\mathbb{R}^n)},$$

for all $t > 0$.

The result of Theorem 2.17 implies that the definition of entropy solution agrees with the notion of "viscosity" solution obtained via vanishing viscosity method. Finally, the following theorem shows that this viscosity solution does exist.

THEOREM 2.19. *The solution v^ϵ of* (2.44)-(2.45) *converges as $\epsilon \to 0$ boundedly almost everywhere on $\mathbb{R}^n \times [0,\infty)$ to the unique entropy solution v of* (2.1)-(2.2).

The following lemma establishes an *equicontinuity in average* property (some sort of "integral" equicontinuity) for $\{v^\epsilon\}$ that comes in useful in proving Theorem 2.19 where a compactness argument in L^p is needed.

LEMMA 2.20. *Let v^ϵ be the solution of* (2.44)-(2.45). *Then there exist a constant c, independent of ϵ, and a nondecreasing function ω on $[0,\infty)$, with $\omega(r) \to 0$ as $r \to 0$, such that the followings hold for any $t > 0$,*

$$\int_{\mathbb{R}^n} |v^\epsilon(x+y,t) - v^\epsilon(x,t)|dx \le \omega(|y|), \quad y \in \mathbb{R}^n,$$
$$\int_{\mathbb{R}^n} |v^\epsilon(x,t+h) - v^\epsilon(x,t)|dx \le c\omega(h^{1/3}), \quad h > 0.$$

PROOF OF THEOREM 2.19. The sequence $\{v^\epsilon(\cdot,t)\}$ is equicontinuous in average thanks to Lemma 2.20. Hence by the Kolmogorov-Riesz theorem (a version of Arzelà-Ascoli's theorem in L^p, see [**Bre10**, Theorem 4.26]), $\{v^\epsilon(\cdot,t)\}$ lies in a compact set of $L^1_{\rm loc}(\mathbb{R})$. Therefore, for every sequence $\{\epsilon_k\}$, $\epsilon^k \to 0$ as $k \to \infty$, there exists a subsequence, still denoted by $\{\epsilon_k\}$, and a function $v \in C^0([0,\infty); L^1_{\rm loc}(\mathbb{R}^n))$, such that

$$v^{\epsilon_k}(\cdot,t) \to v(\cdot,t), \quad \text{in } L^1_{\rm loc}(\mathbb{R}^n),$$

uniformly in $t \in [0,T]$ for any $T > 0$. This implies that, by passing to a further subsequence if necessary, v^{ϵ_k} converges to v boundedly almost everywhere on $\mathbb{R}^n \times [0,\infty)$. By Theorem 2.17, v is an entropy solution of (2.1)-(2.2). Due to the uniqueness of entropy solutions, the whole family $\{v^\epsilon\}$ must converge to v as $\epsilon \to 0$. This completes the proof. □

2.6* Regularity of entropy solutions

The Devil is in the details.

— L. Mies Van Der Rohe

We highlight in this section a few fine properties of the entropy solution for scalar conservation laws. Consider the one-space dimensional scalar conservation law

$$\partial_t v(x,t) + \partial_x f(v(x,t)) = 0, \quad (x,t) \in \mathbb{R} \times [0,\infty). \tag{2.46}$$

Assume that f is genuinely nonlinear, $f''(v) > 0$, $v \in \mathbb{R}$. First of all, v, being the entropy solution to (2.46), instantaneously becomes a

function of locally bounded variation and enjoys the one-sided Lipschitz property, even if v_0 is merely in $L^\infty(\mathbb{R})$.

THEOREM 2.21 (Oleinik). *Let $v_0 \in L^\infty(\mathbb{R})$. Then the entropy solution v of (2.46), with initial data v_0, belongs to $BV_{\text{loc}}(\mathbb{R} \times (0,\infty))$. Moreover, for every $t > 0$ and $x > y$, we have*

$$\frac{v(x,t) - v(y,t)}{x - y} \leq \frac{1}{\kappa t}, \tag{2.47}$$

where $\kappa = \min\{f''(v) : |v| \leq \|v\|_{L^\infty(\mathbb{R})}\}$.

See Exercise 2.7 for a proof of a weaker version of Theorem 2.21. A reverse result of this theorem is also true, if a weak solution of (2.46) satisfies the one-sided Lipschitz property (2.47) then it is an entropy solution. Because of this reverse property, the inequality (2.47) is sometimes used as an admissibility condition (called *Oleinik admissibility condition*) to single out or characterise the entropy solution.

Theorem 2.21 records some sort of *regularising effect* ([**COW08**]) that is due to the genuine nonlinearity: entropy solutions generally become more regular than their initial data[12]. This regularising phenomenon shows up again in the following result, whose proof can be found in [**Daf16**], saying that the continuity is upgraded automatically to Lipschitz continuity once v becomes an entropy solution.

THEOREM 2.22. *Assume that the set $\mathcal{C}$ of continuity points of an entropy solution v has nonempty interior $\mathcal{C}^0$. Then v is locally Lipschitz on $\mathcal{C}^0$.*

Finally, we mention here a *generic property* of scalar conservation laws in one space dimension that features the finiteness of shocks.

THEOREM 2.23. *Generically, entropy solutions of (2.46) with smooth initial data are piecewise smooth. More precisely, there exists a set $\mathcal{S} \in C^k(\mathbb{R})$, $k \geq 3$, of first category such that for each initial data in $v_0 \in C^k(\mathbb{R}) \setminus \mathcal{S}$, the solution v of (2.46) has only a finite number of shock curves and v is C^k in the complementary of the shocks.*

The set $\mathcal{S}$ in Theorem 2.23 is also called a *meagre set*. This set is, roughly speaking, quite small or negligible, hence the complement set $C^k(\mathbb{R}) \setminus \mathcal{S}$ covers almost all $C^k(\mathbb{R})$. In particular, the interior set of $C^k(\mathbb{R}) \setminus \mathcal{S}$ is an open dense set in $C^k(\mathbb{R})$. In other words, for most initial data, the total number of shocks is finite. This result was first

[12]This is only applicable if the initial data is relatively rough (i.e., in L^∞), but not anymore when the initial data is smoother as we have seen previously, smooth initial data may result in discontinuous, hence less smooth, solutions!

established, in a weaker version, by Schaeffer in [**Sch73**] and then improved by Dafermos in [**Daf85**] and in [**Daf16**], to which we refer for the proof. See also Tadmor and Tassa [**TT93**, Section 4] for a characterization of the set $\mathcal{S}$. We note however that this generic property fails to hold for the case of system, see Caravenna and Spinolo [**CS17**].

2.7. Exercises

Our problems are manmade, therefore they may be solved by man.

— John F. Kennedy

Exercise 2.1. Solve the following PDEs using the method of characteristics:

(i) For $x \in \mathbb{R}, t > 0$,
$$2\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} = 0, \quad u(x,0) = \frac{1}{1+x^2}.$$

(ii) For $x, y \in \mathbb{R}$,
$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = xe^{-u}, \quad u(x,x^2) = 0.$$

(iii) For $x, y \in \mathbb{R}$,
$$u\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 1, \quad u(x,x) = \frac{x}{2}.$$

Exercise 2.2.

(i) Prove that if $E(t)$ satisfies $E'(t) \leq G(t) + E(t)$ for $t > 0$ and some non-negative G, and $E(0) = 0$ then
$$E(t) \leq e^t \int_0^t G(s)\mathrm{d}s \quad \text{for all } t \geq 0.$$

(ii) Let $a > 0, R > 0$. Consider the initial-boundary value problem of the transport equation
$$\begin{cases} u_t + au_x = f(x,t), & 0 < x < R, t > 0, \\ u(0,t) = 0, & t > 0, \\ u(x,0) = 0, & 0 < x < R, \end{cases}$$
with a source term f. Use (1) to prove the stability estimate (for $t > 0$)
$$\int_0^R u^2(x,t)\mathrm{d}x \leq e^t \int_0^t \int_0^R f^2(x,s)\mathrm{d}x\mathrm{d}s.$$

This inequality implies that the solution depends stably on the source term, in L^2-norm, at least for finite times.
Hint: Using energy method. Multiply by u the equation, integrate over $(0, R)$ and use integration by parts.

Exercise 2.3. Verify by direct calculations that the formula given in the lecture notes $u = g(x - t\mathbf{F}'(u))$ provides an implicit solution for the Cauchy problem

$$u_t + \operatorname{div} \mathbf{F}(u) = 0, \quad u(x, 0) = g(x) \quad (x \in \mathbb{R}^n).$$

(Note: $\operatorname{div} \mathbf{F}(u) = \mathbf{F}'(u) \cdot Du$).

Exercise 2.4.

(i) Using the method of characteristics and Rankine-Hugoniot condition, solve explicitly for all $t > 0$ the initial value problem for Burgers' equation

$$u_t + \left(\frac{1}{2}u^2\right)_x = 0, \quad u_0(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0. \end{cases}$$

Hint: The characteristic method does not provide any solution in the strip $-t < x < t$. To fill this strip, find a solution to the problem with $u(x, 0) = u_0^\epsilon(x)$, where u_0^ϵ is smooth and $u_0^\epsilon \to u_0$ when $\epsilon \to 0$ pointwise. (You may think about some u_0^ϵ that "connects" two discontinuous parts of u_0 by a continuous line).

(ii) Now find a solution for all $t > 0$ when the initial data is

$$u_0(x) = \begin{cases} 1 & \text{if } x < 0 \\ -1 & \text{if } x > 0. \end{cases}$$

Hint: The method of characteristics fails as characteristics crossed immediately. How to find one? Does u_0 satisfy the jump condition? If yes, it is one.

(iii) If we multiply Burgers' equation by u, we formally find that u satisfies

$$\left(\frac{1}{2}u^2\right)_t + \left(\frac{1}{3}u^3\right)_x = 0. \tag{2.48}$$

Are the solutions to the Burgers equation you found in parts (i) and (ii) weak solutions to (2.48)? If not, find a weak solution to (2.48).
Hint: What is the Rankine-Hugoniot condition for (2.48)? You may mimic the proof in the lecture notes to find it out. See if the condition is satisfied.

Exercise 2.5. Consider the Burgers' equation with initial data u_0,

$$u_t + \left(\frac{1}{2}u^2\right)_x = 0, \quad u_0(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0. \end{cases} \tag{2.49}$$

Prove that for any $\alpha \in [0,1]$, each u_α provides a weak solution for (2.49) where

$$u_\alpha(x) = \begin{cases} -1 & \text{if } x < -t \\ x/t & \text{if } -t < x \leq -\alpha t \\ -\alpha & \text{if } -\alpha t < x \leq 0 \\ \alpha & \text{if } 0 < x \leq \alpha t \\ x/t & \text{if } \alpha t < x \leq t \\ 1 & \text{if } t < x. \end{cases} \tag{2.50}$$

Hint: Verify that u_α satisfies the Rankine-Hugoniot condition at each discontinuous curve.

Exercise 2.6.

(i) Assume that the entropy solution to the Burgers equation

$$u_t + \left(\frac{1}{2}u^2\right)_x = 0, \quad u(x,0) = u_0(x),$$

is smooth and compactly supported. Prove that

$$\|u(\cdot,t)\|_{L^2(\mathbb{R})} \leq \|u_0\|_{L^2(\mathbb{R})}.$$

Hint: Multiply the equation with u.

(ii) If v is the entropy solution to the Burgers equation corresponding to the initial data v_0. Does the following estimate hold

$$\|u(\cdot,t) - v(\cdot,t)\|_{L^2(\mathbb{R})} \leq \|u_0 - v_0\|_{L^2(\mathbb{R})}?$$

Hint: No. Try to find some counterexamples (thinking along the lines of the Riemann problem may help).

Exercise 2.7. Show that if $u \in C^1$ is an entropy solution to $u_t + f(u)_x = 0$, $u(x,0) = u_0(x)$ where u_0 is an increasing function and f is uniformly convex, then u satisfies

$$\frac{u(x+a,t) - u(x,t)}{a} \leq \frac{E}{t}, \quad a > 0, t > 0,$$

for some $E > 0$ independent of x, t and a.

Hint: Mean Value Theorem may help.

Exercise 2.8. Let u be a weak solution of the form

$$u(x,t) = \begin{cases} u_l & \text{if } x < st \\ u_r & \text{if } x > st, \end{cases} \quad \left(s = \frac{f(u_l) - f(u_r)}{u_l - u_r} \right),$$

of the conservation law $u_t + f(u)_x = 0$ ($x \in \mathbb{R}, t > 0$), where $u_l \neq u_r$ and f is smooth. Prove that if u is an entropy solution, then it satisfies the Oleinik condition

$$\frac{f(u_r) - f(u)}{u_r - u} \leq s \leq \frac{f(u_l) - f(u)}{u_l - u},$$

for all u in between u_l and u_r.

Hint: Start with the definition of entropy solutions and use integration by parts.

2.8. Summary and notes

If you're an underdog, mentally disabled, physically disabled, if you don't fit in, if you're not as pretty as the others, you can still be a hero, read TEXBook!

— Bug Buster

This chapter is loosely adapted and extended from Chapter 2 of the author's PhD thesis [**Duo21**].

The theory of nonlinear scalar conservation laws, especially in one-space dimension, is well-developed. This chapter only highlights a few aspects of this rich theory. In particular, we present the method of characteristics for finding implicit solutions and its limitations, the formation of shock waves, weak solutions and nonuniqueness of weak solutions, the entropy conditions and the well-posedness theory of entropy solutions. Two strategies for the existence of the entropy solution, the front tracking approximation and the vanishing viscosity method, are discussed.

The front tracking approximation method (also known as *Dafermos's method*) was first studied by Dafermos in [**Daf72**] to show the existence of solutions for 1D scalar conservation laws. It was then developed further by Holden, Holden and Hoegh-Krohn in [**HHHK88**] as a numerical tool.

The vanishing viscosity method in the context of scalar conservation laws is due to Kruzkov [**Kru70**]. Proofs of Theorem 2.18 and Lemma

2.20 can be found in there, or in [**Daf16**] (where the proof of Theorem 2.19 is adapted from) with somewhat simplified versions. Also in [**Kru70**], the ingenious *doubling of variables* technique was introduced in order to establish the contraction property (2.37). This technique can also be used, for instance, in establishing the uniqueness of entropy solutions of Hamilton-Jacobi equations in [**CEL84**].

Other methods for constructing entropy solutions include the finite difference scheme [**Smo94**], the nonlinear contraction semigroup [**CL71, Cra72**], and the kinetic formulation theory [**LPT94, TT07**], to name a few. We refer to the monographs [**Daf16, Bre00, LeF02, Ser99**] for more information.

The one-sided Lipschitz property, Theorem 2.21 was first established by Oleinik [**Ole57**] and the inequality (2.47) is known in the literature as the *Oleinik decay estimate*. Some extensions of this nice result to the case of non-convex flux, multi-dimensional space and convex balance laws can be found in [**Gla08, AGN12, Hof83**].

In the case of systems, much less is known. Some results in this chapter can be extended to the case of systems, but mostly not trivial. The first global existence result was established in the landmark paper by Glimm [**Gli65**] for initial data with small total variation, in which the so-called *random choice method* was introduced. Later on, DiPerna [**DiP76**] extended Dafermos' front tracking method in the scalar case to 2×2 systems. Based on these works, Bressan [**Bre92**] and Risebro [**Ris93**] extended the front tracking method for systems of any size, providing an alternative existence proof. The construction of entropy solutions using the vanishing viscosity method (the extension of results in Section 2.5.5, in particular, the system version of Theorem 2.19) had been an outstanding open problem in hyperbolic conservation laws and was finally solved recently by Bressan and Bianchini in [**BB05**].

The general theory of multidimensional systems of conservation laws is *terra incognita*, with several obstacles and open problems. We refer to [**BGS07**] for some aspects of the theory, with emphasis on the boundary-valued problems.

CHAPTER 3

Sobolev spaces

What is important is to deeply understand things and their relations to each other. This is where intelligence lies. The fact of being quick or slow isn't really relevant.

— Laurent Schwartz

We have seen from the previous chapters that L^2, and in general L^p, is a natural space for studying PDEs (in which the 'kinetic' energy is defined). However, since we are dealing with differential equations, we need to assign an appropriate meaning to the "derivative" of an L^p function, called a *weak derivative*. In a nutshell, *Sobolev spaces* consist of functions whose weak derivatives belong to L^p. These spaces provide one of the most useful settings for the analysis of PDEs: they allow us to formulate equations and boundary conditions when pointwise derivatives do not exist, and they provide the compactness needed to construct weak solutions.

Throughout this chapter, Ω is an open subset of $\mathbb{R}^n$. When an embedding, extension or trace theorem is used, we will assume that Ω is bounded with Lipschitz boundary. A smooth boundary is more than sufficient for all the results stated here.

3.1. Weak derivatives

Classical derivatives require pointwise smoothness, which is too restrictive for many problems in PDEs. For example, solutions of elliptic or parabolic equations may fail to be differentiable everywhere, yet still satisfy the equation in an integral sense. Weak derivatives capture exactly the amount of differentiability that is visible through integration by parts.

DEFINITION 3.1 (Weak derivative). Let $u \in L^1_{\text{loc}}(\Omega)$. A function $v \in L^1_{\text{loc}}(\Omega)$ is called the *weak derivative* of u with respect to x_i, written

$v = D_i u$, if

$$\int_\Omega v\phi \,\mathrm{d}x = -\int_\Omega u\partial_{x_i}\phi \,\mathrm{d}x$$

for all test functions $\phi \in C_c^\infty(\Omega)$.

More generally, for a multi-index $\alpha \in \mathbb{N}^n$ (see Notations in page 1), we say that v is the *α-th weak derivative* of u, written $v = D^\alpha u$, if

$$\int_\Omega v\phi \,\mathrm{d}x = (-1)^{|\alpha|}\int_\Omega u D^\alpha\phi \,\mathrm{d}x$$

for all $\phi \in C_c^\infty(\Omega)$.

REMARK 3.2.
- Weak derivatives, if they exist, are unique up to sets of measure zero.
- If u is classically differentiable and $\partial_{x_i} u \in L^1_{\mathrm{loc}}(\Omega)$, then the weak and classical derivatives coincide.
- The notion of weak derivative is a special case of the *distribution derivative*. For more details, we refer to [**Rob01**].

EXAMPLE 3.3. Consider the following function on $(0, 2)$

$$u(x) = \begin{cases} x & \text{if } 0 < x \le 1, \\ 1 & \text{if } 1 < x < 2. \end{cases}$$

Obviously $u \in L^1(0, 2)$. For any $\phi \in C_c^\infty(0, 2)$ we have

$$\begin{aligned} -\int_0^2 u\phi' \mathrm{d}x &= -\int_0^1 x\phi' \mathrm{d}x - \int_1^2 \phi' \mathrm{d}x \\ &= -[x\phi]|_{x=0}^{x=1} + \int_0^1 \phi \mathrm{d}x - \phi(2) + \phi(1) \\ &= \int_0^1 \phi \mathrm{d}x. \end{aligned}$$

Hence, by Definition 3.1, u has a weak derivative

$$v(x) = \begin{cases} 1 & \text{if } 0 < x \le 1, \\ 0 & \text{if } 1 < x < 2. \end{cases}$$

3.2. Sobolev spaces

DEFINITION 3.4. Let $k \in \mathbb{N}$ and $1 \le p \le \infty$. The Sobolev space $W^{k,p}(\Omega)$ consists of all functions $u \in L^p(\Omega)$ such that $D^\alpha u \in L^p(\Omega)$ for all $|\alpha| \le k$. For $1 \le p < \infty$, the norm is defined by

$$\|u\|_{W^{k,p}(\Omega)} = \left(\sum_{|\alpha|\le k} \|D^\alpha u\|^p_{L^p(\Omega)}\right)^{1/p}. \tag{3.1}$$

For $p = \infty$, we use

$$\|u\|_{W^{k,\infty}(\Omega)} = \max_{|\alpha|\le k} \|D^\alpha u\|_{L^\infty(\Omega)}.$$

THEOREM 3.5. *$W^{k,p}(\Omega)$ is a Banach space.*

SKETCH. Suppose (u_j) is Cauchy in $W^{k,p}(\Omega)$. For every $|\alpha| \le k$, the sequence $(D^\alpha u_j)$ is Cauchy in $L^p(\Omega)$. Since $L^p(\Omega)$ is complete, there are functions $v_\alpha \in L^p(\Omega)$ such that

$$D^\alpha u_j \longrightarrow v_\alpha \qquad \text{in } L^p(\Omega).$$

In particular, $u_j \to v_0$ in $L^p(\Omega)$. For any $\phi \in C_c^\infty(\Omega)$, Hölder's inequality and the definition of weak derivative give

$$\begin{aligned}\int_\Omega v_\alpha \phi \,\mathrm{d}x &= \lim_{j\to\infty} \int_\Omega D^\alpha u_j \phi \,\mathrm{d}x \\ &= (-1)^{|\alpha|} \lim_{j\to\infty} \int_\Omega u_j D^\alpha \phi \,\mathrm{d}x = (-1)^{|\alpha|} \int_\Omega v_0 D^\alpha \phi \,\mathrm{d}x.\end{aligned}$$

Thus $v_\alpha = D^\alpha v_0$ weakly, so $v_0 \in W^{k,p}(\Omega)$ and $u_j \to v_0$ in $W^{k,p}(\Omega)$. The same argument works for $p = \infty$. See also [**Eva10**, Theorem 5.1]. □

The case when $p = 2$ receives special treatment and we will denote $W^{k,2} = H^k$. The reason for this notation is that H^k is a Hilbert space with the inner product

$$\langle f, g\rangle_{H^k} = \sum_{|\alpha|\le k} \int_\Omega (D^\alpha f)(D^\alpha g) \,\mathrm{d}x.$$

It is worth noting that, on a domain with boundary, $C_c^\infty(\Omega)$ is generally not dense[1] in $H^k(\Omega)$ for $k \ge 1$ (although $C_c^\infty(\Omega)$ is dense in $L^2(\Omega)$). Therefore the completion of $C_c^\infty(\Omega)$ with respect to the H^k norm is different from $H^k(\Omega)$. This motivates the following definition.

DEFINITION 3.6. The space $H_0^k(\Omega)$ is the completion of the space $C_c^\infty(\Omega)$ in $H^k(\Omega)$.

The previous definition can be used to prove the following inequality. It provides a bound on the norm of a function in terms of the norm of its gradient, under suitable boundary conditions.

[1] $A \subset B$ is called dense in B if for any $\varepsilon > 0$ and $b \in B$, there exists $a \in A$ such that $\|a - b\|_B < \varepsilon$. Equivalently, any element in B can be approximated by some sequence in A.

PROPOSITION 3.7 (Poincaré's inequality). *Let $\Omega \subset \mathbb{R}^n$ be bounded in the x_1 direction, so that $|x_1| \leq K < \infty$ for $x \in \Omega$, and let $u \in H_0^1(\Omega)$. Then there exists a constant $C > 0$, depending only on Ω, such that*

$$\|u\|_{L^2(\Omega)} \leq C\|\nabla u\|_{L^2(\Omega)}. \tag{3.2}$$

PROOF. Since $C_c^\infty(\Omega)$ is dense in $H_0^1(\Omega)$, we only need to show (3.2) for elements of $C_c^\infty(\Omega)$. By integration by parts in the x_1 variable and the Cauchy–Schwarz inequality,

$$\|u\|_{L^2(\Omega)}^2 = -\int_\Omega x_1 \frac{\partial}{\partial x_1}|u(x)|^2 \mathrm{d}x \leq 2K\|u\|_{L^2(\Omega)}\|\partial_{x_1} u\|_{L^2(\Omega)}.$$

Therefore

$$\|u\|_{L^2(\Omega)} \leq 2K\|\partial_{x_1} u\|_{L^2(\Omega)} \leq 2K\|\nabla u\|_{L^2(\Omega)},$$

as required. The general case follows by approximation. □

3.3. General Sobolev embedding theorems

The Sobolev embedding theorems describe how functions in Sobolev spaces $W^{k,p}(\Omega)$ embed into other function spaces. By the very definition, $W^{k,p}(\Omega) \subset L^p(\Omega)$. But this is not the end of the story. Roughly speaking, the more weak derivatives that are controlled, the more regular the function. When k and p reach certain thresholds, these functions become more integrable, bounded or even continuous. These results are essential because they guarantee that weak solutions possess enough regularity to make sense of nonlinear terms and to obtain qualitative information about the solutions.

We begin with some general embedding results. In the statement below, $C^{m,\alpha}(\overline{\Omega})$ denotes the Hölder space of functions whose derivatives up to order m are continuous and whose derivatives of order m are Hölder continuous with exponent α.

THEOREM 3.8 (Sobolev Embedding). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with Lipschitz boundary, let $k \in \mathbb{N}$ and let $1 \leq p < \infty$.*

(1) ***Subcritical case:*** *Suppose $kp < n$ and define*

$$p_k^* = \frac{np}{n-kp}, \qquad \text{or equivalently} \qquad \frac{1}{p_k^*} = \frac{1}{p} - \frac{k}{n}.$$

The exponent p_k^* is called the Sobolev conjugate *associated with k and p. Then*

$$W^{k,p}(\Omega) \hookrightarrow L^{p_k^*}(\Omega), \qquad \|u\|_{L^{p_k^*}(\Omega)} \leq C\|u\|_{W^{k,p}(\Omega)}.$$

Since Ω is bounded, the same conclusion holds with p_k^* replaced by any q satisfying $1 \leq q \leq p_k^*$.

(2) ***Critical case:*** *Suppose $kp = n$ and $p > 1$. Then*

$$W^{k,p}(\Omega) \hookrightarrow L^q(\Omega) \qquad \text{for every } 1 \le q < \infty.$$

In general there is no embedding into $L^\infty(\Omega)$.

(3) ***Supercritical case:*** *Suppose $kp > n$ and $p > 1$, and set*

$$s = k - \frac{n}{p} > 0.$$

If $s \notin \mathbb{N}$, set $m = \lfloor s \rfloor$ and $\alpha = s - m$. Then

$$W^{k,p}(\Omega) \hookrightarrow C^{m,\alpha}(\overline{\Omega}).$$

If $s \in \mathbb{N}$, then

$$W^{k,p}(\Omega) \hookrightarrow C^{s-1,\alpha}(\overline{\Omega}) \qquad \text{for every } 0 < \alpha < 1.$$

In particular, when $k = 1$ and $p > n$, Morrey's inequality gives

$$W^{1,p}(\Omega) \hookrightarrow C^{0,1-n/p}(\overline{\Omega}).$$

The constants in these estimates depend on k, p, n and Ω, but not on u.

Sketch of proof. We explain the main ideas of the proof. They are useful well beyond this particular theorem.

First suppose that $u \in C_c^\infty(\mathbb{R}^n)$. A representation formula obtained from the fundamental solution of the Laplacian gives

$$|u(x)| \le C \int_{\mathbb{R}^n} \frac{|\nabla u(y)|}{|x-y|^{n-1}} \, \mathrm{d}y. \tag{3.3}$$

If $1 < p < n$, the Hardy–Littlewood–Sobolev inequality applied to the right-hand side yields

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \le C \|\nabla u\|_{L^p(\mathbb{R}^n)}, \qquad p^* = \frac{np}{n-p}. \tag{3.4}$$

The endpoint $p = 1$ follows from the classical $W^{1,1}$ Sobolev inequality, proved for instance by applying the fundamental theorem of calculus in each coordinate and then using Hölder's inequality. Thus the first-order subcritical estimate is valid for $1 \le p < n$.

For a Lipschitz domain there is a bounded extension operator

$$E : W^{k,p}(\Omega) \longrightarrow W^{k,p}(\mathbb{R}^n), \qquad Eu|_\Omega = u.$$

Applying (3.4) to the extension, and successively to its derivatives, gives

$$\|u\|_{L^{p_k^*}(\Omega)} \le C \|u\|_{W^{k,p}(\Omega)}, \qquad \frac{1}{p_k^*} = \frac{1}{p} - \frac{k}{n},$$

whenever $kp < n$. This proves the first part. Approximation by smooth functions allows us to pass from smooth u to every $u \in W^{k,p}(\Omega)$.

When $kp = n$, choose $r < p$ sufficiently close to p. Since Ω is bounded, $W^{k,p}(\Omega) \subset W^{k,r}(\Omega)$, and $kr < n$. The first part therefore gives

$$W^{k,p}(\Omega) \hookrightarrow L^{nr/(n-kr)}(\Omega).$$

By taking r arbitrarily close to p, the exponent on the right can be made arbitrarily large. This proves the critical statement, together with the elementary inclusions between L^q spaces on a bounded domain. The failure of the L^∞ endpoint is demonstrated by logarithmically singular examples.

Finally, for $p > n$, Morrey's estimate states that a smooth function satisfies

$$|u(x) - u(y)| \le C|x - y|^{1-n/p}\|u\|_{W^{1,p}(\Omega)}. \tag{3.5}$$

It follows by approximation that each element of $W^{1,p}(\Omega)$ has a Hölder continuous representative. For higher k, apply the subcritical embeddings successively to the weak derivatives until their integrability exponent is larger than n, and then apply (3.5). If $s = k - n/p$ is not an integer, this gives $C^{m,\alpha}$ with $m = \lfloor s \rfloor$ and $\alpha = s - m$. If s is an integer, the iteration passes through the critical case; one obtains every exponent $\alpha < 1$, but not necessarily the Lipschitz endpoint. This proves the final part. □

The Sobolev embedding theorem tells us that higher values of k and p yield functions that are more regular in terms of integrability, continuity or differentiability. Notice, however, that the critical case is delicate: controlling exactly n/p derivatives does not usually give boundedness.

Example 3.9. Consider $W^{1,p}(\Omega)$ on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^n$.

(1) If $p < n$, then $W^{1,p}(\Omega) \subset L^{p^*}(\Omega)$, where $p^* = \frac{np}{n-p}$ is called the *Sobolev conjugate* of p.
(2) If $p = n > 1$, then $W^{1,n}(\Omega) \subset L^q(\Omega)$ for every finite q, but in general $W^{1,n}(\Omega) \not\subset L^\infty(\Omega)$. A stronger endpoint result is the Trudinger–Moser inequality: after an appropriate normalization, functions in $W^{1,n}_0(\Omega)$ have exponential integrability.
(3) If $p > n$, then $W^{1,p}(\Omega) \subset C^{0,\alpha}(\overline{\Omega})$ for $\alpha = 1 - \frac{n}{p}$.

3.4. Compactness and further tools for Sobolev spaces

3.4.1. Compact embeddings. An important refinement of Sobolev embedding is the concept of a *compact embedding*. A continuous embedding $X \hookrightarrow Y$ maps bounded sets in X to bounded sets in Y. It is

compact if every bounded sequence in X has a subsequence that converges strongly in Y. This distinction is invaluable for demonstrating the existence of weak solutions to nonlinear PDEs.

THEOREM 3.10 (Rellich–Kondrachov Compactness Theorem). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with Lipschitz boundary, and let $1 \le p < \infty$.*

(1) *If $p < n$, then the embedding*
$$W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$$
is compact for every $1 \le q < p^*$, where $p^* = np/(n-p)$.
(2) *If $p = n$, then $W^{1,n}(\Omega) \hookrightarrow L^q(\Omega)$ compactly for every $1 \le q < \infty$.*
(3) *If $p > n$, then $W^{1,p}(\Omega) \hookrightarrow C^{0,\beta}(\overline{\Omega})$ compactly for every $0 \le \beta < 1 - n/p$. Here $C^{0,0}(\overline{\Omega})$ means $C(\overline{\Omega})$.*

SKETCH OF PROOF. Let (u_j) be bounded in $W^{1,p}(\Omega)$. Using the extension theorem, followed by multiplication by a fixed smooth cut-off function, extend u_j to functions $Eu_j \in W^{1,p}(\mathbb{R}^n)$, supported in a fixed bounded set, in such a way that
$$\|Eu_j\|_{W^{1,p}(\mathbb{R}^n)} \le C\|u_j\|_{W^{1,p}(\Omega)}.$$
For a smooth function v, the fundamental theorem of calculus gives
$$v(x+h) - v(x) = \int_0^1 h \cdot \nabla v(x+th)\,\mathrm{d}t.$$
Minkowski's inequality and translation invariance of the L^p norm therefore imply
$$\|v(\,\cdot + h) - v\|_{L^p(\mathbb{R}^n)} \le |h|\|\nabla v\|_{L^p(\mathbb{R}^n)}. \tag{3.6}$$
By density, the same estimate holds for every $v \in W^{1,p}(\mathbb{R}^n)$. Hence translations of the family (Eu_j) tend to zero uniformly in j as $h \to 0$. The functions are also uniformly bounded in L^p and supported in a common bounded set. The Fréchet–Kolmogorov compactness criterion now shows that (Eu_j) has a subsequence converging strongly in $L^p(\mathbb{R}^n)$. Restricting to Ω, we obtain
$$u_j \longrightarrow u \qquad \text{strongly in } L^p(\Omega)$$
along a subsequence.

If $p < n$, then for $1 \le q < p$, strong convergence in $L^q(\Omega)$ follows from Hölder's inequality and the boundedness of Ω. If $p < q < p^*$, choose $0 < \theta < 1$ such that
$$\frac{1}{q} = \frac{\theta}{p} + \frac{1-\theta}{p^*}.$$

The Sobolev embedding gives a uniform L^{p^*} bound, and interpolation yields

$$\|u_j - u\|_{L^q} \leq \|u_j - u\|_{L^p}^{\theta} \|u_j - u\|_{L^{p^*}}^{1-\theta} \longrightarrow 0.$$

The case $q = p$ has already been proved. Thus the embedding is compact for every $q < p^*$. The strict inequality is essential: concentration of mass at a point gives bounded sequences in $W^{1,p}$ with no strongly convergent subsequence in L^{p^*}.

If $p = n > 1$, strong convergence in L^q for $q \leq p$ follows from strong convergence in L^p and the boundedness of Ω. If $q > p$, choose $r > q$ and interpolate between strong convergence in L^p and the uniform L^r bound supplied by the Sobolev embedding theorem. When $p = n = 1$, the one-dimensional estimate $\|u\|_{L^\infty} \leq C\|u\|_{W^{1,1}}$ supplies the required upper bound instead. This gives strong convergence in L^q for every finite q.

If $p > n$, Morrey's inequality shows that (u_j) is uniformly bounded and equicontinuous on $\overline{\Omega}$. The Arzelà–Ascoli theorem gives a uniformly convergent subsequence. Combining uniform convergence with the uniform $C^{0,1-n/p}$ bound gives convergence in $C^{0,\beta}(\overline{\Omega})$ for every $\beta < 1-n/p$. This proves the remaining cases. □

The two embedding theorems above are the central results of this chapter. We now record several consequences that will be used in the chapters on wave and Navier–Stokes equations. They explain more precisely how boundary values, integration by parts, compactness and differentiation in time are handled for weak solutions.

3.4.2. Approximation, extensions and traces.

THEOREM 3.11 (Extension and smooth approximation). *Let Ω be a bounded Lipschitz domain, $k \in \mathbb{N}$ and $1 \leq p < \infty$. There exists a bounded linear operator*

$$E : W^{k,p}(\Omega) \longrightarrow W^{k,p}(\mathbb{R}^n)$$

such that $Eu = u$ almost everywhere on Ω. Consequently, restrictions to Ω of functions in $C_c^\infty(\mathbb{R}^n)$ are dense in $W^{k,p}(\Omega)$.

SKETCH. For a half-space, one first reflects a function across the boundary, using a suitable linear combination of reflections when derivatives of order greater than one must be preserved. A Lipschitz boundary can be locally represented as the graph of a Lipschitz function. Flattening the boundary, applying the half-space construction and then combining the local extensions with a partition of unity produces Eu. The norm of Eu is controlled by the norm of u. Finally, convolve Eu

with a standard mollifier and restrict the resulting smooth functions to Ω. $\square$

The weak derivative is tested only against functions compactly supported inside Ω, so it does not by itself specify the value of a Sobolev function on $\partial\Omega$. Boundary values are instead provided by the trace operator.

THEOREM 3.12 (Trace theorem). *Let Ω be a bounded Lipschitz domain. There is a unique bounded linear map*

$$\operatorname{Tr} : H^1(\Omega) \longrightarrow L^2(\partial\Omega)$$

which agrees with the usual restriction $u \mapsto u|_{\partial\Omega}$ for $u \in C^\infty(\overline{\Omega})$. Moreover,

$$H^1_0(\Omega) = \{u \in H^1(\Omega) : \operatorname{Tr} u = 0\}.$$

Thus the notation $u = 0$ on $\partial\Omega$ has a precise meaning for $u \in H^1(\Omega)$.

SKETCH. For smooth functions on a half-space, the fundamental theorem of calculus in the normal direction gives an estimate of the form

$$\|u|_{\partial\Omega}\|_{L^2(\partial\Omega)} \le C\|u\|_{H^1(\Omega)}.$$

Flattening a Lipschitz boundary and using a partition of unity gives the estimate on Ω. The density result in Theorem 3.11 then defines $\operatorname{Tr} u$ for every $u \in H^1(\Omega)$. Every element of $H^1_0(\Omega)$ clearly has zero trace. Conversely, a function with zero trace can be approximated in H^1 by cutting it off near the boundary and mollifying; hence it belongs to $H^1_0(\Omega)$. $\square$

PROPOSITION 3.13 (Sobolev integration by parts). *Let $u, v \in H^1(\Omega)$. For $i = 1, \dots, n$,*

$$\int_\Omega (D_i u)v \,\mathrm{d}x = -\int_\Omega u D_i v \,\mathrm{d}x + \int_{\partial\Omega} (\operatorname{Tr} u)(\operatorname{Tr} v)\nu_i \,\mathrm{d}S, \tag{3.7}$$

where ν is the outward unit normal. In particular, if $v \in H^1_0(\Omega)$, then the boundary term vanishes.

PROOF. The formula is the classical integration-by-parts identity for smooth functions. Approximate u and v in $H^1(\Omega)$ by smooth functions, use the continuity of the trace operator, and pass to the limit in every term of the identity. $\square$

3.4.3. Sobolev functions on an interval and calculus rules. In one space dimension, membership in $W^{1,p}$ has a particularly concrete interpretation. This also explains why a function with a jump does not have an L^p weak derivative.

THEOREM 3.14 (One-dimensional characterization). *Let $I = (a, b)$ be a bounded interval and let $1 \le p \le \infty$. A function $u \in L^p(I)$ belongs to $W^{1,p}(I)$ if and only if it has an absolutely continuous representative $\widetilde{u}$ whose classical derivative belongs to $L^p(I)$. For this representative,*

$$\widetilde{u}(y) - \widetilde{u}(x) = \int_x^y Du(s)\,\mathrm{d}s, \qquad a \le x \le y \le b. \tag{3.8}$$

In particular, every element of $W^{1,1}(I)$ has a continuous representative, and every element of $W^{1,\infty}(I)$ has a Lipschitz representative.

PROOF. If u has such a representative, ordinary integration by parts shows that its classical derivative is also its weak derivative. Conversely, let $v = Du \in L^p(I)$ and define

$$U(x) = \int_a^x v(s)\,\mathrm{d}s.$$

Then U is absolutely continuous and has weak derivative v. Hence $u - U$ has weak derivative zero. A locally integrable function on an interval whose weak derivative is zero is constant almost everywhere; this follows, for example, by mollifying it on compact subintervals. Therefore u agrees almost everywhere with $U + c$ for some constant c, and (3.8) follows. □

Sobolev functions also obey most familiar calculus rules, although the derivatives are interpreted almost everywhere.

PROPOSITION 3.15 (Product and chain rules). *Let $1 \le p \le \infty$.*

(1) *If $u, v \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$, then $uv \in W^{1,p}(\Omega)$ and*
$$D_i(uv) = uD_iv + vD_iu \qquad \text{almost everywhere.}$$

(2) *If $u \in W^{1,p}(\Omega)$ and $F \in C^1(\mathbb{R})$ has bounded derivative, then $F(u) \in W^{1,p}(\Omega)$, provided $F(u) \in L^p(\Omega)$, and*
$$D_i(F(u)) = F'(u)D_iu \qquad \text{almost everywhere.}$$

(3) *In particular, $|u|$, $u_+ = \max\{u, 0\}$ and $u_- = \max\{-u, 0\}$ belong to $W^{1,p}(\Omega)$ whenever $u \in W^{1,p}(\Omega)$, and*
$$D_iu_+ = \mathbf{1}_{\{u>0\}}D_iu, \qquad D_iu_- = -\mathbf{1}_{\{u<0\}}D_iu$$
almost everywhere.

SKETCH. For smooth functions these are the usual calculus rules. Approximate Sobolev functions locally by mollification and pass to almost-everywhere limits of suitable subsequences. For the last part, approximate the Lipschitz functions $s \mapsto |s|$ and $s \mapsto \max\{s, 0\}$ by smooth functions with uniformly bounded derivatives. The formulas on the level set $\{u = 0\}$ follow from the fact that $D_i u = 0$ almost everywhere on any level set of a Sobolev function. See [**Eva10, Leo17**] for the full argument and more general versions. □

3.4.4. Poincaré's inequality with zero mean. The boundary condition in Proposition 3.7 can be replaced by a normalization that removes the constant functions.

PROPOSITION 3.16 (Poincaré–Wirtinger inequality). *Let Ω be a bounded, connected Lipschitz domain and set*

$$u_\Omega = \frac{1}{|\Omega|} \int_\Omega u \, \mathrm{d}x.$$

Then there exists $C = C(\Omega) > 0$ such that

$$\|u - u_\Omega\|_{L^2(\Omega)} \leq C \|\nabla u\|_{L^2(\Omega)} \qquad \text{for every } u \in H^1(\Omega). \tag{3.9}$$

Consequently, $\|\nabla u\|_{L^2}$ is an equivalent norm on $H^1_0(\Omega)$.

PROOF. Suppose (3.9) were false. Then there would be a sequence $(u_j) \subset H^1(\Omega)$ with $(u_j)_\Omega = 0$, $\|u_j\|_{L^2} = 1$ and $\|\nabla u_j\|_{L^2} \to 0$. This sequence is bounded in H^1, so after passing to a subsequence it converges strongly in L^2 to some u and weakly in H^1. We have $\nabla u = 0$, so connectedness of Ω implies that u is constant almost everywhere. The zero-mean condition gives $u = 0$, contradicting $\|u\|_{L^2} = 1$. The norm equivalence on H^1_0 follows from Proposition 3.7. □

3.4.5. Weak and strong compactness. Let X be a Banach space. A sequence (u_j) converges *weakly* to u in X, written $u_j \rightharpoonup u$, if

$$\ell(u_j) \longrightarrow \ell(u) \qquad \text{for every continuous linear functional } \ell \in X^*.$$

Weak convergence is easier to obtain than norm (or *strong*) convergence, but compact embeddings often turn a weakly convergent sequence into a strongly convergent one in a less regular space.

COROLLARY 3.17. *Let Ω be a bounded Lipschitz domain. If (u_j) is bounded in $H^1(\Omega)$, then there exist a subsequence, still denoted by (u_j), and $u \in H^1(\Omega)$ such that*

$$u_j \rightharpoonup u \quad \text{in } H^1(\Omega), \qquad u_j \longrightarrow u \quad \text{in } L^2(\Omega).$$

After possibly taking a further subsequence, $u_j(x) \to u(x)$ for almost every $x \in \Omega$.

PROOF. The Hilbert space $H^1(\Omega)$ is reflexive, so every bounded sequence has a weakly convergent subsequence. The compact embedding $H^1(\Omega) \hookrightarrow L^2(\Omega)$ then gives strong convergence in L^2. Finally, every strongly convergent sequence in L^2 has an almost-everywhere convergent subsequence. □

PROPOSITION 3.18 (Weak lower semicontinuity). *Let X be a Banach space and suppose $u_j \rightharpoonup u$ weakly in X. Then*

$$\|u\|_X \le \liminf_{j\to\infty} \|u_j\|_X.$$

In particular, if $u_j \rightharpoonup u$ in $H^1(\Omega)$, then

$$\|u\|_{L^2} \le \liminf_{j\to\infty} \|u_j\|_{L^2}, \qquad \|\nabla u\|_{L^2} \le \liminf_{j\to\infty} \|\nabla u_j\|_{L^2}.$$

PROOF. By the Hahn–Banach theorem, for every $u \neq 0$ there is $\ell \in X^*$ such that $\|\ell\|_{X^*} = 1$ and $\ell(u) = \|u\|_X$. Weak convergence gives

$$\|u\|_X = \lim_{j\to\infty} \ell(u_j) \le \liminf_{j\to\infty} \|u_j\|_X.$$

The statements for u and ∇u follow by applying this result in $L^2(\Omega)$ and $[L^2(\Omega)]^n$, respectively. This simple inequality is a central tool in the direct method of the calculus of variations. □

3.4.6. A two-dimensional interpolation inequality. The following inequality is a special case of the Gagliardo–Nirenberg inequalities. Its particular exponents are one of the main reasons why the uniqueness theory of the Navier–Stokes equations is better in two dimensions than in three (see).

PROPOSITION 3.19 (Ladyzhenskaya's inequality). *Let $\Omega \subset \mathbb{R}^2$ be open and let $u \in H^1_0(\Omega)$. Then*

$$\|u\|_{L^4(\Omega)} \le C \|u\|^{1/2}_{L^2(\Omega)} \|\nabla u\|^{1/2}_{L^2(\Omega)}. \tag{3.10}$$

The same estimate holds for vector-valued functions, with a possibly different constant.

PROOF. It is enough to prove the estimate for $u \in C^\infty_c(\Omega)$ and then use density. Extend u by zero to $\mathbb{R}^2$. The fundamental theorem of calculus gives

$$|u(x_1,x_2)|^2 \le 2 \int_{\mathbb{R}} |u(s,x_2)||\partial_1 u(s,x_2)| \, \mathrm{d}s =: A(x_2),$$
$$|u(x_1,x_2)|^2 \le 2 \int_{\mathbb{R}} |u(x_1,s)||\partial_2 u(x_1,s)| \, \mathrm{d}s =: B(x_1).$$

Consequently,

$$\begin{aligned}\|u\|_{L^4(\mathbb{R}^2)}^4 &\le \int_{\mathbb{R}^2} A(x_2)B(x_1)\,\mathrm{d}x_1\mathrm{d}x_2 \\ &\le 4\|u\|_{L^2}^2\|\partial_1 u\|_{L^2}\|\partial_2 u\|_{L^2} \le 2\|u\|_{L^2}^2\|\nabla u\|_{L^2}^2,\end{aligned}$$

where the Cauchy–Schwarz inequality was used in the second line. Taking fourth roots proves (3.10). Applying the scalar estimate componentwise proves the vector-valued case. □

3.4.7. A useful result for time-dependent equations. The dual of $H_0^1(\Omega)$ is denoted by

$$H^{-1}(\Omega) := (H_0^1(\Omega))^*.$$

By identifying $f \in L^2(\Omega)$ with the functional

$$v \longmapsto \int_\Omega f v\,\mathrm{d}x,$$

we obtain the continuous and dense embeddings

$$H_0^1(\Omega) \hookrightarrow L^2(\Omega) \hookrightarrow H^{-1}(\Omega). \tag{3.11}$$

More generally, let V be a Hilbert space densely and continuously embedded in a Hilbert space H. Identifying H with its dual gives the evolution triple

$$V \hookrightarrow H \hookrightarrow V^*.$$

THEOREM 3.20 (Energy identity in an evolution triple). *Let $V \hookrightarrow H \hookrightarrow V^*$ be as above. Suppose*

$$u \in L^2(0,T;V), \qquad \partial_t u \in L^2(0,T;V^*),$$

where the time derivative is understood in the distributional sense. Then u has a representative in $C([0,T];H)$, the function $t \mapsto \|u(t)\|_H^2$ is absolutely continuous, and

$$\frac{1}{2}\|u(t)\|_H^2 - \frac{1}{2}\|u(s)\|_H^2 = \int_s^t \langle \partial_\tau u(\tau), u(\tau)\rangle_{V^*,V}\,\mathrm{d}\tau \tag{3.12}$$

for all $0 \le s \le t \le T$.

SKETCH. Extend u slightly beyond $(0,T)$ and mollify it in time. For the time-smoothed function u_ε, the ordinary Hilbert-space product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|u_\varepsilon(t)\|_H^2 = \langle \partial_t u_\varepsilon(t), u_\varepsilon(t)\rangle_{V^*,V}.$$

Integrate between s and t. Time mollification gives convergence in $L^2(0,T;V)$ for u_ε and in $L^2(0,T;V^*)$ for $\partial_t u_\varepsilon$. Hence the duality products converge in $L^1(0,T)$, and a standard completion argument yields an H-continuous representative of u and the identity (3.12). This result is what rigorously permits a weak solution to be used as a test function in many energy estimates. □

These results ensure that weak solutions have enough regularity and integrability to make sense of boundary conditions and the integral terms that arise in weak formulations. For example, when $n > 2$, the embedding $H^1(\Omega) \hookrightarrow L^{2^*}(\Omega)$ with $2^* = 2n/(n-2)$ helps us control nonlinear terms. In two dimensions, $H^1(\Omega)$ embeds into $L^q(\Omega)$ for every finite q, while the sharper inequality (3.10) supplies the interpolation structure required in the Navier–Stokes uniqueness estimate (see Chapter 5).

For further details and additional embeddings, we refer to [**Bre10, Eva10, Leo17**] and to the supplemental notes available on the course platform[2].

3.5. Exercises

Exercise 3.1. (i) Using the definition, find a weak derivative of the function $u(x) = |x|$ for $x \in \mathbb{R}$.

(ii) Prove that the following function does not have a weak derivative in $L^1_{\mathrm{loc}}(0,2)$:

$$u(x) = \begin{cases} x & \text{if } 0 < x \le 1, \\ 2 & \text{if } 1 < x < 2. \end{cases}$$

Hint: Show that there is no function $v \in L^1_{\mathrm{loc}}(0,2)$ such that $\int_0^2 u\phi' \mathrm{d}x = -\int_0^2 v\phi \mathrm{d}x$. The jump at $x = 1$ produces a point mass, which cannot be represented by an L^1_{loc} function.

Exercise 3.2. For $a \in \mathbb{R}$, define the function $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \frac{1}{|x|^a}.$$

(i) Suppose that $\epsilon > 0$ and let $\phi^\epsilon \in C_c^\infty(\mathbb{R}^n)$ be a cut-off function that is equal to one in $B_\epsilon(0)$ and zero outside $B_{2\epsilon}(0)$. Let

$$f^\epsilon(x) = \frac{1 - \phi^\epsilon(x)}{|x|^a}.$$

[2]In particular, these notes `https://moodle.lut.fi/pluginfile.php/2349968/course/section/411884/sobolev-spaces_notes.pdf`

Show that, for every $i = 1, \dots, n$, there exists a constant $C > 0$ independent of ϵ such that

$$|\partial_{x_i} f^\epsilon(x)| \leq \frac{C}{|x|^{a+1}}.$$

(ii) Use (i) to show that f is weakly differentiable if $a + 1 < n$, with weak derivative

$$\partial_{x_i} f(x) = -\frac{a}{|x|^{a+1}} \frac{x_i}{|x|}.$$

Hint: The Dominated Convergence Theorem may help.

Exercise 3.3. Let $U = B_1(0)$ be the unit ball in $\mathbb{R}^2$. Let $u : U \to \mathbb{R}$ be defined by

$$u(x) = \log \log \left(1 + \frac{1}{|x|}\right).$$

(i) Is u a bounded function in U? Prove that $u \in L^2(U)$.
(ii) Prove that $u \in H^1(U)$.

What does this example say about the critical embedding $H^1(U) \hookrightarrow L^\infty(U)$ in two dimensions?

Exercise 3.4. Let $\Omega \subset \mathbb{R}^n$ be open with smooth boundary. Prove that

$$\|\nabla u\|_{L^2(\Omega)} \leq C \|u\|_{L^2(\Omega)}^{1/2} \|\nabla^2 u\|_{L^2(\Omega)}^{1/2},$$

for all $u \in H^2(\Omega) \cap H_0^1(\Omega)$. Hint: Integrate by parts in $\|\nabla u\|_{L^2}^2$ and use the Cauchy–Schwarz inequality.

Exercise 3.5. Let Ω be a bounded Lipschitz domain. Use Corollary 3.17 to prove Poincaré's inequality by contradiction. More precisely, assume that the inequality fails and construct a sequence $(u_j) \subset H_0^1(\Omega)$ such that

$$\|u_j\|_{L^2} = 1, \qquad \|\nabla u_j\|_{L^2} \longrightarrow 0.$$

Identify the strong L^2 limit and obtain a contradiction.

Exercise 3.6. Let $\Omega \subset \mathbb{R}^2$ and let $u, v, w \in H_0^1(\Omega)$ be vector fields. Use Ladyzhenskaya's inequality and Hölder's inequality to prove

$$\left| \int_\Omega (u \cdot \nabla) v \cdot w \, \mathrm{d}x \right| \leq C \|u\|_{L^2}^{1/2} \|\nabla u\|_{L^2}^{1/2} \|\nabla v\|_{L^2} \|w\|_{L^2}^{1/2} \|\nabla w\|_{L^2}^{1/2}.$$

This is the estimate used for the nonlinear term in the two-dimensional Navier–Stokes equations.

Exercise 3.7. Let $1 \le p < n$, choose $x_0 \in \Omega$, and let $\phi \in C_c^\infty(B_1(0))$ be nonzero. For all sufficiently large j, define

$$u_j(x) = j^{(n-p)/p}\phi(j(x - x_0)).$$

Show that (u_j) is bounded in $W^{1,p}(\Omega)$ and that

$$\|u_j\|_{L^{p^*}(\Omega)}$$

is independent of j, while $u_j(x) \to 0$ for almost every $x \in \Omega$. Conclude that the critical embedding

$$W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega)$$

is continuous but not compact.

3.6. Summary and notes

This chapter introduces weak derivatives and Sobolev spaces — the modern language of PDEs. The one-dimensional characterization shows that $W^{1,p}$ functions are not arbitrary equivalence classes: on an interval they possess absolutely continuous representatives and satisfy the fundamental theorem of calculus. Smooth approximation, product and chain rules then allow many classical calculations to survive in the weak setting.

Poincaré's inequality shows how a boundary condition, or alternatively a zero-mean normalization, controls the full H^1 norm. The Sobolev embedding theorem converts weak differentiability into higher integrability or continuity. The three regimes $kp < n$, $kp = n$ and $kp > n$ should be kept distinct, since the critical endpoints often fail. The Rellich–Kondrachov theorem adds the decisive fact that the embeddings below the critical exponent are compact. Weak compactness, strong convergence and weak lower semicontinuity together form the standard compactness mechanism used to construct solutions: derive a uniform Sobolev bound, extract a weakly convergent subsequence, use compactness to obtain strong convergence where needed, and then pass to the limit.

The trace theorem gives a rigorous meaning to boundary conditions, and Sobolev integration by parts is the basis of weak formulations. Ladyzhenskaya's inequality supplies the particular two-dimensional estimate needed for the nonlinear term in the Navier–Stokes equations. Finally, the evolution triple $V \hookrightarrow H \hookrightarrow V^*$ and Theorem 3.20 explain when a time-dependent weak solution has meaningful time traces and when the familiar energy calculation is justified. These tools will be used repeatedly for the energy estimates in the wave equation and

for the weak formulation and uniqueness theory of the Navier–Stokes equations.

Our presentation concerns integer-order Sobolev spaces on sufficiently regular domains. Fractional Sobolev spaces, sharp endpoint embeddings, capacities, embeddings on unbounded or irregular domains, and the full theory of negative-order spaces are beyond the scope of these notes. For a concise PDE-oriented account we refer to [**Eva10**]; [**Bre10**] emphasizes the functional-analytic viewpoint; and [**Leo17**] provides a detailed treatment of approximation, calculus rules, extension, traces and embeddings. The evolution-space result is also discussed in the PDE framework of [**Rob01**].

CHAPTER 4

Wave equations

Newton's first law of graduation: A grad student in procrastination tends to stay in procrastination unless an external force is applied to it.

— Jorge Cham, phdcomics

Wave equations describe the propagation of waves, such as sound, light, or water waves, in various media. These equations are fundamental in physics and engineering and serve as prototypes for hyperbolic partial differential equations. This chapter introduces the wave equation and explores methods for finding solutions such as d'Alembert's formula, Fourier transform and separation of variables. We also discuss other properties such as uniqueness, energy conservation and the finite speed of propagation.

4.1. The wave equation

The classical wave equation in n-dimensional space is given by

$$\frac{\partial^2 u}{\partial t^2} - c^2 \Delta u = 0, \quad (x,t) \in \Omega \times (0,T), \tag{4.1}$$

where

- $u(x,t)$ is the unknown function of space $x \in \Omega \subseteq \mathbb{R}^n$ and time $t \in (0,T) \subseteq \mathbb{R}_+$,
- $c > 0$ is the wave speed,
- Δ is the Laplace operator in the spatial variable x.

The wave equation is often accompanied by initial and boundary conditions:

$$u(x,0) = f(x), \qquad \text{(initial displacement)} \tag{4.2}$$

$$\frac{\partial u}{\partial t}(x,0) = g(x), \qquad \text{(initial velocity)} \tag{4.3}$$

$$u(x,t) = 0, \qquad \text{(boundary condition on } \partial\Omega). \tag{4.4}$$

The conditions (4.2)-(4.3) are commonly called *Cauchy initial conditions*, while (4.4) is called *Dirichlet boundary condition*.

4.2. D'Alembert's formula in one dimension

For the one-dimensional case $n = 1$, where $\Omega = \mathbb{R}$, the wave equation reads

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0,$$

which can be factorized into

$$\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right) u = 0.$$

Denote

$$v(x,t) = \left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right) u \tag{4.5}$$

then v satisfies the transport equation

$$v_t + cv_x = 0.$$

Recall the method of characteristics and solution to the transport equation (2.13), we arrive at

$$v(x,t) = G(x - ct),$$

where we denote $G(x) := v(x,0)$. From (4.5), u solves

$$u_t - cu_x = G(x - ct), \tag{4.6}$$

which is another (inhomogeneous) transport equation but with a nonzero right-hand side. Similarly, the method of characteristics (exercise!) gives

$$u(x,t) = F(x + ct) + \frac{1}{2c}\int_{x-ct}^{x+ct} G(y)\mathrm{d}y, \tag{4.7}$$

where we denote $F(x) := u(x,0)$. We now compute F, G in terms of f, g. From (4.2),

$$F(x) = f(x), \quad x \in \mathbb{R}.$$

From (4.3), (4.5) and (4.7)

$$G(x) = v(x,0) = u_t(x,0) - cu_x(x,0) = g(x) - cf'(x).$$

By substituting F, G into (4.7), we get

$$u(x,t) = f(x + ct) + \frac{1}{2c}\int_{x-ct}^{x+ct} g(y) - cf'(y)\mathrm{d}y,$$

and thus

$$u(x,t) = \frac{1}{2}\left[f(x-ct) + f(x+ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(y)\mathrm{d}y, \quad x \in \mathbb{R}, t > 0. \tag{4.8}$$

This is known as *d'Alembert's formula*[1].

Another way to see the derivation of this formula is by changing of variables $\mu = x + ct$ and $\eta = x - ct$ to transform the PDE into $u_{\mu\eta} = 0$. The general solution of this PDE is $u(\mu, \eta) = F_1(\mu) + G_1(\eta)$ where F_1 and G_1 are functions of C^1. Transform back in x, t coordinates,

$$u(x,t) = F_1(x+ct) + G_1(x-ct), \tag{4.9}$$

where F_1 and G_1 can be recovered by using the initial conditions.

REMARK 4.1.
- In light of (4.9), the solution is decomposed into two parts: a left moving component F_1 and a right moving component G_1. Any solution of the wave equation in 1D can be written in this way.
- Although the equation is a second-order PDE, the formula (4.8) is well-defined for continuous functions $f, g \in C(\mathbb{R})$. On one hand, this gives some sense of a weak solution to the wave equation. On the other hand, it implies that the wave equation does not pose a regularity effect (the solution does not become instantaneously smooth) as does the heat equation, see Section 1.3.3.

4.3. Method of separation of variables

Consider the wave equation in 1D

$$\partial_t^2 u - \partial_x^2 u = 0, \quad x \in (0, 2\pi) \tag{4.10}$$

as given in (4.1) for $c = 1$, together with initial data (4.2)-(4.3) and also with Dirichlet boundary conditions

$$u(t,0) = 0 = u(t, 2\pi). \tag{4.11}$$

The method of *separation of variables* seeks special solutions in the form of a product

$$u(t,x) = v(t)w(x). \tag{4.12}$$

Substituting this into the wave equation yields the following

$$\partial_t^2 v \cdot w = v \cdot \partial_x^2 w, \tag{4.13}$$

[1]Jean le Rond d'Alembert (1717–1783), French mathematician. He derived this formula in 1747 as a way to represent solutions to the vibrating string problem.

which implies the existence of a separation constant λ (ignoring possible zeros of v and w), such that

$$\frac{\partial_t^2 v}{v} = \lambda = \frac{\partial_x^2 w}{w}. \tag{4.14}$$

Since the left-hand side depends only on t and the right-hand side only on x, λ must be a constant. This decomposition reduces the problem to two ordinary differential equations (ODEs)

(1) an evolution problem for $v(t)$, determined by the initial conditions; and
(2) an eigenvalue problem for $w(x)$, constrained by the boundary conditions.

The eigenvalue problem for the pair $(w(x), \lambda)$ is given by

$$\partial_x^2 w = \lambda w, \quad w(0) = 0 = w(2\pi). \tag{4.15}$$

Solving this boundary value problem yields a countable set of eigenfunctions and eigenvalues

$$w_k(x) = \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right), \quad \lambda_k = -\frac{k^2}{4}, \quad k = 1, 2, \dots \tag{4.16}$$

For each eigenvalue λ_k, the evolution equation for $v(t)$ is

$$\frac{d^2}{dt^2} v = \lambda v, \tag{4.17}$$

with initial conditions $v(0) = a$, $\dot{v}(0) = b$ to be specified. The general solution takes the form

$$v_k(t) = a_k \cos(\omega_k t) + \frac{b_k}{\omega_k} \sin(\omega_k t), \tag{4.18}$$

where the temporal frequencies are

$$\omega_k = \sqrt{-\lambda_k} = \frac{k}{2}. \tag{4.19}$$

For each $k \in \mathbb{N}$, this forms a two-parameter family of such special solutions, that is

$$u_k(x,t) = \left(a_k \cos(\omega_k t) + \frac{b_k}{\omega_k} \sin(\omega_k t)\right) \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right)$$

solves the wave equation (4.10) for each $k \in \mathbb{N}$. Since the wave equation is linear, any linear combination of these solutions solves the initial value problem (4.10) with general initial data (4.2)-(4.3) and satisfies the Dirichlet boundary conditions (4.11), as established by the following theorem.

THEOREM 4.2. *The general solution of the initial-boundary value problem* (4.10) *is given by a linear superposition of the special solutions*

$$u(t,x) = \sum_{k=1}^{\infty} \left(a_k \cos(\omega_k t) + \frac{b_k}{\omega_k} \sin(\omega_k t) \right) \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right), \qquad (4.20)$$

where the initial data is specified by

$$u(0,x) = f(x) = \sum_{k=1}^{\infty} a_k \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right) = \sum_{k=1}^{\infty} a_k w_k(x), \qquad (4.21)$$

$$\partial_t u(0,x) = g(x) = \sum_{k=1}^{\infty} b_k \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right) = \sum_{k=1}^{\infty} b_k w_k(x). \qquad (4.22)$$

The sine series used to describe $f(x)$ and $g(x)$ forms a complete orthonormal basis of the Hilbert space $L^2(0,2\pi)$. This means that any initial data $f(x), g(x) \in L^2(0,2\pi)$ can be represented in terms of their Fourier sine series coefficients $(a_k, b_k)_{k\in\mathbb{N}}$, leading to a well-defined solution $u(t,x)$.

Given the initial data $f(x)$ and $g(x)$, the next step is to compute the coefficients a_k and b_k from their eigenfunction expansions. This is where the usefulness of an orthonormal basis becomes evident. Indeed, for the sine series

$$w_k(x) = \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right),$$

we can easily see (exercise!) that

$$\int_0^{2\pi} w_k(x) w_m(x)\,\mathrm{d}x = \frac{1}{\pi}\int_0^{2\pi} \sin\left(\frac{k}{2}x\right)\sin\left(\frac{m}{2}x\right)\mathrm{d}x = 0, \text{ for } k \neq m,$$

$$\int_0^{2\pi} w_k^2(x)\,\mathrm{d}x = \frac{1}{\pi}\int_0^{2\pi} \sin^2\left(\frac{k}{2}x\right)\mathrm{d}x = 1.$$

This confirms that the eigenfunctions $w_k(x)$ form an orthonormal set in $L^2([0,2\pi])$, meaning

$$\langle w_k, w_m\rangle = 0, \quad \text{for } k \neq m, \quad \text{and} \quad \langle w_k, w_k\rangle = \|w_k\|_{L^2}^2 = 1.$$

To determine the coefficients a_k and b_k, we take the inner product of $f(x)$ and $g(x)$ with $w_k(x)$ and use the orthonormality to get

$$\langle f, w_k\rangle = a_k, \quad \langle g, w_k\rangle = b_k.$$

That is

$$a_k = \int_0^{2\pi} \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right) f(x)\,\mathrm{d}x,$$

$$b_k = \int_0^{2\pi} \frac{1}{\sqrt{\pi}} \sin\left(\frac{k}{2}x\right) g(x)\,\mathrm{d}x.$$

Thus, the coefficients $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ are fully characterized by the initial conditions and hence the solution $u(t,x)$ is well-defined.

4.4. Wave solution by Fourier transform

Assuming first that $f(x), g(x) \in \mathcal{S}(\mathbb{R}^n)$, that is, they belong to the Schwartz class (see Appendix B), we take the Fourier transform of the wave equation (4.1) (in the x variable) to obtain

$$\partial_t^2 \hat{u}(t,\xi) + |\xi|^2 \hat{u}(t,\xi) = 0. \tag{4.23}$$

Solutions of this second-order ODE are linear combinations of $e^{\pm i|\xi|t}$. Taking into account the initial conditions $\hat{u}(0,\xi) = \hat{f}(\xi)$ and $\partial_t \hat{u}(0,\xi) = \hat{g}(\xi)$, we derive

$$\hat{u}(t,\xi) = \cos(|\xi|t)\hat{f}(\xi) + \frac{\sin(|\xi|t)}{|\xi|}\hat{g}(\xi). \tag{4.24}$$

Applying the inverse Fourier transform, the solution is given by

$$u(t,x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^n} e^{i\xi\cdot x} \left[\cos(|\xi|t)\hat{f}(\xi) + \frac{\sin(|\xi|t)}{|\xi|}\hat{g}(\xi)\right] \mathrm{d}\xi. \tag{4.25}$$

From (4.24), we observe that $\hat{u}(t,\xi) \in \mathcal{S}(\mathbb{R}^n)$ for each time t, and therefore, the solution $u(t,x)$ given in (4.25) also belongs to the Schwartz class. However, this high level of smoothness is not necessary, and expression (4.25) remains well-defined whenever $\hat{f}(\xi), \hat{g}(\xi)/|\xi| \in L^2(\mathbb{R}^n)$, though it may not possess classical derivatives.

THEOREM 4.3. *For $f, g \in \mathcal{S}$, the expression* (4.25) *provides a solution of the wave equation $u(t,x) \in \mathcal{S}(\mathbb{R}^n)$ for each $t \in \mathbb{R}$. Furthermore, if $\hat{f}(\xi), \hat{g}(\xi)/|\xi| \in L^2(\mathbb{R}^n)$, then* (4.25) *defines a weak solution to the wave equation in the sense that* (4.23) *is satisfied.*

4.5. Energy methods

4.5.1. Uniqueness theorem. Let $U \subset \mathbb{R}^n$ be a bounded, open set with a smooth boundary ∂U, and define the parabolic cylinder and boundary

$$U_T = U \times (0,T], \quad \Gamma_T := \overline{U_T} \setminus U_T$$

where $T > 0$.

We consider the initial/boundary-value problem for the wave equation

$$\begin{aligned} u_{tt} - \Delta u &= f \quad \text{in } U_T, \\ u &= g \quad \text{on } \Gamma_T, \end{aligned}$$

$$u_t = h \quad \text{on } U \times \{t = 0\}.$$

THEOREM 4.4 (Uniqueness for the wave equation). *There exists at most one function $u \in C^2(U_T)$ solving the above problem.*

PROOF. Suppose u and $\tilde{u}$ are two solutions. Define $w = u - \tilde{u}$, which satisfies the homogeneous problem

$$w_{tt} - \Delta w = 0 \quad \text{in } U_T,$$
$$w = 0 \quad \text{on } \Gamma_T,$$
$$w_t = 0 \quad \text{on } U \times \{t = 0\}.$$

Define the energy function

$$E(t) = \frac{1}{2} \int_U \left(w_t^2 + |\nabla w|^2 \right) dx.$$

Differentiating $E(t)$ with respect to t and integrating by parts, we obtain

$$\frac{d}{dt} E(t) = \int_U w_t (w_{tt} - \Delta w) \, dx.$$

Since w satisfies the wave equation, we have

$$\frac{d}{dt} E(t) = 0.$$

Thus, $E(t)$ is constant in time, and since the initial data satisfies $E(0) = 0$, it follows that $E(t) = 0$ for all $0 \le t \le T$.

Since $E(t) = 0$ implies $w_t = 0$ and $\nabla w = 0$ in U_T, we conclude that $w \equiv 0$. Hence, $u = \tilde{u}$, proving uniqueness. □

REMARK 4.5. If we consider the wave equation in $\mathbb{R}^n$ with initial conditions (4.2)-(4.3), the above proof can be mimicked to show that the total energy of u in $\mathbb{R}^n$ is also conserved, that is

$$E(t) = \frac{1}{2} \int_{\mathbb{R}^n} \left(u_t^2 + |\nabla u|^2 \right) dx = E(0) = \frac{1}{2} \int_{\mathbb{R}^n} \left(g^2 + |\nabla f|^2 \right) dx.$$

4.5.2. Domain of dependence. As another illustration of energy methods, let us examine the domain of dependence of solutions to the wave equation in all of space. Suppose $u \in C^2$ solves

$$u_{tt} - \Delta u = 0 \quad \text{in } \mathbb{R}^n \times (0, \infty).$$

Fix $x_0 \in \mathbb{R}^n$, $t_0 > 0$, and consider the *backwards wave cone* with apex (x_0, t_0)

$$K(x_0, t_0) := \{(x, t) \mid 0 \le t \le t_0, |x - x_0| \le t_0 - t\}.$$

See Figure 4.1 for an illustration of $K(x_0, t_0)$.

Recall that $B(x, r)$ is a closed ball centred at x with radius r (see Notations on page 1). We have the following result.

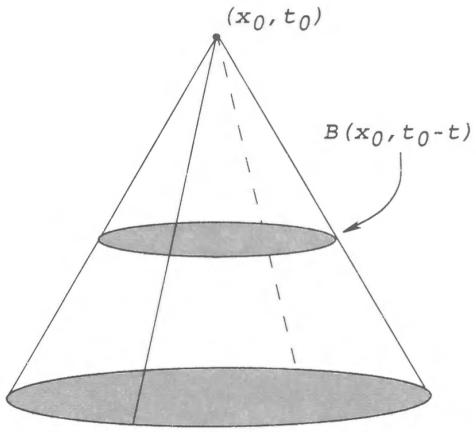


FIGURE 4.1. Cone of dependence $K(x_0, t_0)$.

THEOREM 4.6 (Finite speed of propagation). *If*

$$u = u_t = 0 \quad \text{on } B(x_0, t_0) \times \{t = 0\},$$

then $u = 0$ *within the cone* $K(x_0, t_0)$.

PROOF. Define the local energy:

$$E(t) = \frac{1}{2} \int_{B(x_0, t_0 - t)} \left(u_t^2 + |\nabla u|^2\right) dx.$$

Differentiating $E(t)$ with respect to t, we apply the Leibniz integral rule (or differentiation formula for moving regions, see Appendix A)

$$\frac{d}{dt}E(t) = \frac{1}{2} \int_{B(x_0, t_0 - t)} \frac{d}{dt} \left(u_t^2 + |\nabla u|^2\right) \mathrm{d}x - \frac{1}{2} \int_{\partial B(x_0, t_0 - t)} \left(u_t^2 + |\nabla u|^2\right) \mathrm{d}S.$$

Using the divergence theorem, we get

$$\begin{aligned} \frac{d}{dt}E(t) = &\int_{B(x_0, t_0 - t)} u_t (u_{tt} - \Delta u)\, dx + \int_{\partial B(x_0, t_0 - t)} u_t \frac{\partial u}{\partial \nu}\, dS \\ &- \frac{1}{2} \int_{\partial B(x_0, t_0 - t)} \left(u_t^2 + |\nabla u|^2\right) dS. \end{aligned}$$

Since u satisfies the wave equation, the volume integral vanishes, leaving

$$\frac{d}{dt}E(t) = \int_{\partial B(x_0, t_0 - t)} u_t \frac{\partial u}{\partial \nu}\, dS - \frac{1}{2} \int_{\partial B(x_0, t_0 - t)} \left(u_t^2 + |\nabla u|^2\right) dS.$$

Applying the Cauchy-Schwarz and Cauchy inequalities (see Appendix B),

$$\left\| u_t \frac{\partial u}{\partial \nu} \right\|_{L^1} \leq \|u_t\|_{L^2} \left\| \frac{\partial u}{\partial \nu} \right\|_{L^2} \leq \frac{1}{2} \left(\|u_t\|_{L^2}^2 + \left\| \frac{\partial u}{\partial \nu} \right\|_{L^2}^2 \right),$$

we obtain

$$\frac{d}{dt} E(t) \leq 0.$$

Since $E(0) = 0$, it follows that $E(t) \leq 0$ for all $0 \leq t \leq t_0$, implying that

$$u_t = 0, \quad \nabla u = 0.$$

Thus, u must be identically zero within the cone $K(x_0, t_0)$, completing the proof. □

REMARK 4.7. The previous theorem has a physical interpretation: any disturbance originating outside $B(x_0, t_0)$ has no effect on the solution u within the cone $K(x_0, t_0)$. This exhibits the *finite speed of propagation* of the wave equation, in contrast to the infinite speed of propagation for the heat equation (see Section 1.3).

4.6. Exercises

Exercise 4.1. Solve the following initial-boundary value problems using the method of separation of variables (write down the solution explicitly in the sine series)

$$u_{tt} - u_{xx} = 0, \quad x \in [0, 1],\ t > 0,$$

with $u(x, 0) = f(x),\ u_t(x, 0) = 0,\ u(0, t) = u(1, t) = 0$, where

$$f(x) = \begin{cases} x, & \text{if } 0 \leq x \leq \frac{1}{2}, \\ 1 - x, & \text{if } \frac{1}{2} \leq x \leq 1. \end{cases}$$

Exercise 4.2. Using the formula given by the method of separation of variables (Theorem 4.2), prove that the solution of the wave equation $u_{tt} - u_{xx} = 0,\ x \in (0, 2\pi)$, is stable with respect to changes in initial data. That is, there exists a constant $C > 0$ such that

$$\sup_{t>0} \|u_1 - u_2\|_{L^2}^2 \leq C \left(\|f_1 - f_2\|_{L^2}^2 + \|g_1 - g_2\|_{L^2}^2 \right),$$

where u_1, u_2 are solutions of the wave equation w.r.t initial conditions f_1, g_1 and f_2, g_2, respectively.

Hint: Use Theorem 4.2 and the fact that $(w_k(x))_{k\in\mathbb{N}}$ form a complete orthonormal basis of $L^2(0,2\pi)$, you can write

$$u = \sum_{k=1}^{\infty} \langle u, w_k \rangle w_k,$$

for any $u \in L^2(0,2\pi)$.

Exercise 4.3. Let u solve the wave equation $u_{tt} - \Delta u = 0$ in $\mathbb{R}^n \times (0,\infty)$ with initial conditions $u(x,0) = f(x), u_t(x,0) = g(x)$. Assume that $f, g, \nabla f \in L^2(\mathbb{R}^n)$. Denote $E_K(t), E_P(t)$ to be the kinetic and potential energies of u, that is,

$$E_K(t) = \frac{1}{2}\int_{\mathbb{R}^n} u_t^2 \mathrm{d}x, \quad E_P(t) = \frac{1}{2}\int_{\mathbb{R}^n} |\nabla u|^2 \mathrm{d}x.$$

Prove that

$$\lim_{t\to\infty} E_P(t) = \lim_{t\to\infty} E_K(t).$$

Hint: First notice that the total energy $E(t) = E_K(t) + E_P(t)$ is conserved. Use the Fourier transform (equation (4.24) in Section 4.4 and Plancherel's identity (Theorem B.6 in Appendix) to prove that

$$\lim_{t\to\infty} E_P(t) = \frac{1}{4}\int_{\mathbb{R}^n} |\nabla f|^2 + |g|^2 \mathrm{d}x = \frac{1}{2}E(0),$$

which is one-half of the total energy.

4.7. Summary and notes

This chapter introduces the wave equation and several methods for solving it, including d'Alembert's formula, the method of separation of variables and the Fourier transform method. The techniques involving the energy of the solution are employed again, first to prove the uniqueness and then to deduce the domain of dependence. Other methods such as using spherical means and images method are not treated in here and can be found in relevant textbooks [**Cra18, Str08**].

CHAPTER 5

Fluid mechanics and Navier–Stokes equations

Love is like a fluid mechanics.
Every inch of your heart is
feeling pressure.

— Pie Pie

This chapter is devoted to the study of some of the most fundamental equations of physics, the Navier–Stokes equations. The Navier–Stokes equations of fluid dynamics are a formulation of Newton's laws of motion for a continuous distribution of matter in the fluid state, characterized by an inability to support shear stresses. We first present a way to derive the incompressible Navier–Stokes equations for a single-component Newtonian fluid. We then proceed to the analysis of Navier–Stokes equations in 2D, including the existence and uniqueness of weak and strong solutions. Some aspects of the existence and smoothness problem in 3D will be discussed.

5.1. Derivation of Navier–Stokes equations

There are several ways to derive Navier–Stokes equations, for instance from the microscopic description in terms of a Boltzmann equation. The following, taken from [**DG95**], deriving first Euler's equations and then adding the viscosity term to form Navier–Stokes equations, is quite natural and illustrates the elements of the physics contained in the equations.

5.1.1. Euler equations. We will begin our examination by considering an ideal fluid that has no viscosity. In the Eulerian perspective of fluid mechanics, the dependent variables are the fluid's density, which is denoted as $\rho(x,t)$, the velocity vector field, represented as[1] $u(x,t) = (u_1, u_2, u_3)$, and the pressure field, denoted as $p(x,t)$. In this context, the spatial coordinate, x, belongs to a d-dimensional space,

[1]In contrast to the previous chapters, for the sake of convenience, we use normal italic symbols to denote vector fields and do not distinguish them with the scalar cases. However, in the context of fluid mechanics, it is clear that u is a vector field, and ρ, p are scalars, so no unnecessary confusion should arise.

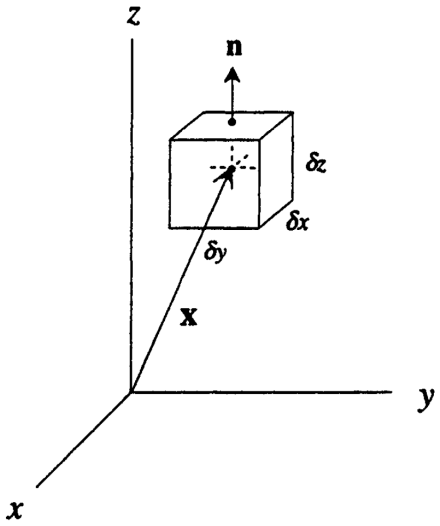


FIGURE 5.1. A fluid element of volume δV at x.

where d is typically either 2 or 3, and the default value for this chapter is 3. An infinitesimal volume of the fluid, δV, which is located at position x at time t, has a mass of $\delta m = p(x,t)\delta V$ and moves with a velocity of $u(x,t)$, possessing a momentum of $\delta m u(x,t)$. The normal force across an infinitesimal volume's face with an area of $n\delta a$, located at x and directed outward, is $-np(x,t)\delta a$. The pressure denotes the normal stress or force per unit area that neighbouring elements of the fluid impose on each other. These definitions are illustrated in Figure 5.1.

We first explain the concept of the *convective derivative*, which is essential to the kinematic principle. When considering the rate of change of a function $f(x,t)$ at a fixed point x in space, we can simply use the partial derivative with respect to time

$$\left(\frac{\mathrm{d}f(x,t)}{\mathrm{d}t}\right)_{\text{fixed position}} = \lim_{\delta t \to 0} \frac{f(x,t+\delta t) - f(x,t)}{\delta t} = \frac{\partial f(x,t)}{\partial t}.$$

However, when measuring the rate of change of the same quantity at x from the perspective of an observer moving with velocity u, we obtain a different expression

$$\begin{aligned}\left(\frac{\mathrm{d}f(x,t)}{\mathrm{d}t}\right)_{\text{moving}} &= \lim_{\delta t \to 0} \frac{f(x+u\delta t, t+\delta t) - f(x,t)}{\delta t} \\ &= \frac{\partial f(x,t)}{\partial t} + u \cdot \nabla f(x,t).\end{aligned}$$

This rate of change, which takes into account the movement of the observer with the fluid, is known as the *convective derivative* and is denoted by $\mathrm{d}/\mathrm{d}t$. Thus, for a function $f(x,t)$ that depends on both x and t, we can write

$$\frac{\mathrm{d}f(x,t)}{\mathrm{d}t} = \frac{\partial f(x,t)}{\partial t} + u \cdot \nabla f(x,t). \tag{5.1}$$

The fundamental equations of motion for a fluid system characterized by $\rho(x,t), u(x,t)$ and $p(x,t)$ come from three different distinct considerations: conservation of mass, Newton's second law, and material properties.

As the system evolves, let us consider the volume δV of an element of mass δm. According to the law of conservation of mass, the mass δm for this element remains unchanged. While the element compresses or expands, the volume and density will change, but the mass remains fixed, and the equation

$$\frac{\mathrm{d}\delta m}{\mathrm{d}t} = 0, \tag{5.2}$$

holds. The rate of change of the volume occupied by δm is obtained as follows. For a rectangular volume $\delta V = \delta x \delta y \delta z$ we write

$$\frac{\mathrm{d}\delta V}{\mathrm{d}t} = \frac{\mathrm{d}\delta x}{\mathrm{d}t}\delta y \delta z + \frac{\mathrm{d}\delta y}{\mathrm{d}t}\delta x \delta z + \frac{\mathrm{d}\delta z}{\mathrm{d}t}\delta x \delta y. \tag{5.3}$$

The length elements increase or decrease in accordance with the relative velocity of their endpoints. The rate of change of the length δx is

$$\frac{\mathrm{d}\delta x}{\mathrm{d}t} = u_1(x + \delta x/2, y, z, t) - u_1(x - \delta x/2, y, z, t)\frac{\partial u_1}{\partial x}\delta x,$$

and similar equations hold for the other components. Combined with equation (5.3), this gives

$$\frac{\mathrm{d}\delta V}{\mathrm{d}t} = (\nabla \cdot u)\delta V.$$

Thus, the divergence of the velocity vector field is the local rate of change of the volume of elements of mass. In terms of the density ρ,

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\delta m}{\delta V} = -\frac{\delta m}{(\delta V)^2}\frac{\mathrm{d}\delta V}{\mathrm{d}t} = -\rho \nabla \cdot u.$$

Using the definition of the convective derivative, conservation of mass can be expressed as the continuity equation

$$0 = \frac{\partial \rho}{\partial t} + u \cdot \nabla \rho + \rho \nabla \cdot u = \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u). \tag{5.4}$$

Newton's second law of motion, which states that the rate of change of momentum equals the net applied force, can be applied to each

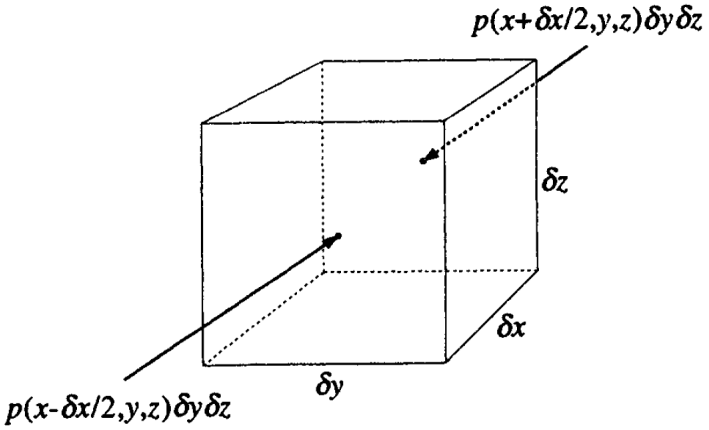


FIGURE 5.2. Pressure force of a fluid element.

element of mass in the fluid. When there are no externally applied forces, the net force δF acting on each element of mass is due to the pressure field. For instance, the x-component of the force, as shown in Figure 5.2, is

$$\delta F_1 = p(x - \hat{i}\delta x/2, t)\delta y\delta z - p(x + \hat{i}\delta x/2, t)\delta y\delta z = \frac{\partial p}{\partial x}\delta V,$$

and analogous expressions hold for the y and z components. Thus, Newton's second law for the fluid mass element δm at position δx is expressed as

$$\frac{\mathrm{d}}{\mathrm{d}t}(\delta m u(x,t)) = \delta F = -\delta V \nabla p.$$

Recalling the equation of conservation of mass (5.2) and the definition of the convective derivative (5.1) and dividing through by δm we find *Euler's equations*

$$\frac{\partial u}{\partial t} + u \cdot \nabla u = -\frac{1}{\rho}\nabla p. \tag{5.5}$$

Combined, the continuity equation (5.4) and Euler's equations (5.5) provide $d+1$ evolution equations for the $d+2$ dependent variables (ρ, p and the d components of u). What remains is to provide a connection between density and pressure. Typically this is in the form of a thermodynamic equation of state. For example, in an ideal gas at constant temperature, $p \sim \rho$. If temperature variations are to be accounted for, then the pressure may become a function of both the local density and the local temperature and a further evolution equation for the temperature must be supplied. A significant simplification is achieved by considering fluids which are effectively *incompressible*.

Mathematically the condition of *incompressibility* is simply

$$\nabla \cdot u = 0.$$

Physically, this constraint restricts applicability to problems where all the relevant velocities are much less than the speed of sound in the fluid. The continuity equation (5.2) then implies that the convective derivative of the density vanishes, so the density of each fluid element never changes from its initial value. This, in turn, implies that an initially homogeneous (constant density) fluid remains so

$$\rho(x,0) = \text{constant} \quad \Rightarrow \quad \rho(x,t) = \text{constant}.$$

Euler's equations for an incompressible homogeneous fluid are

$$\frac{\partial u}{\partial t} + u \cdot \nabla u + \frac{1}{\rho}\nabla p = 0, \tag{5.6}$$

$$\nabla \cdot u = 0, \tag{5.7}$$

where the density is now a parameter. These are $d+1$ equations for the $d+1$ unknowns (p and the d components of u). The pressure is determined by the velocity vector field, as is seen by taking the divergence of (5.6), commuting the space and time derivatives, and using the divergence-free condition on u from (5.7),

$$\Delta p = -\rho \nabla \cdot (u \cdot \nabla u) = -\rho u_{x_i x_j} u_{x_j x_i}.$$

Hence the pressure field is a solution of Poisson's equation with a source which is quadratic in the derivatives of the velocities.

5.1.2. Viscosity, the stress tensor and the Navier–Stokes equations. The third fundamental equation of motion arises from the fact that a fluid is a continuum with material properties. These properties describe the relationship between stress and strain in the fluid. In particular, the stress tensor σ relates the stress on a small element of fluid to its deformation. The stress tensor has nine components σ_{ij}, where $i, j = 1, 2, 3$, corresponding to the six independent components of stress and the three directions in which the stress can act.

The rate of deformation tensor D describes how the fluid is deforming at a point. It has six components D_{ij}, which correspond to the rate of change of the displacement vector in the i direction due to motion in the j direction. For an incompressible fluid, the rate of deformation tensor is symmetric and has zero trace, and can be expressed as the sum of a deviatoric part D' and a spherical part D_s:

$$D_{ij} = D'_{ij} + \frac{1}{3}\delta_{ij} D_s.$$

The deviatoric part of the rate of deformation tensor describes the shear deformation of the fluid and is related to the stress tensor by the constitutive relation

$$\sigma_{ij} = -p\delta_{ij} + 2\mu D'_{ij},$$

where μ is the dynamic viscosity of the fluid.

The equation of motion for an incompressible fluid can be obtained by considering the balance of forces on a small volume element of the fluid. The net force on the element is equal to the rate of change of its momentum. The momentum density of the element is ρu, where ρ is the density and u is the velocity vector. The rate of change of momentum density is

$$\frac{\partial}{\partial t}(\rho u) + \nabla \cdot (\rho u u) = -\nabla p + \nabla \cdot \sigma.$$

The first term on the left-hand side is the rate of change of momentum density due to changes in the velocity field, while the second term is the advective term due to the transport of momentum by the velocity field. The right-hand side represents the net force on the element, which is due to the pressure gradient and the stress tensor.

Using the continuity equation (5.4) and the fact that the rate of deformation tensor is symmetric and has zero trace, we can simplify the momentum equation to obtain the Navier–Stokes equation for an incompressible fluid:

$$\frac{\partial u}{\partial t} + u \cdot \nabla u = -\frac{1}{\rho}\nabla p + \nu \nabla^2 u, \tag{5.8}$$

where $\nu = \mu/\rho$ is the kinematic viscosity of the fluid. The term $\nu\nabla^2 u$ represents the diffusion of momentum due to viscosity.

In summary, the fundamental equations of motion for a fluid system are the continuity equation (5.4) and the Navier–Stokes equation (5.8). These equations describe how the density and velocity fields evolve in time due to the motion and deformation of the fluid.

Viscosity is a measure of the diffusion of momentum due to the microscopic molecular nature of fluids, and its effect is to produce a resistance to shearing motions. As such, it is a frictional force with its origins in the microscopic interactions between the atoms or molecules making up the fluid. Its net effect is to dissipate organized, macroscopic forms of energy – the kinetic energy in the flow field – and convert it to the disorganized, microscopic form of energy – heat. Shearing forces in continuum mechanical systems are described by the stress tensor. The tensorial nature of these forces results from the fact that there are two

directions associated with each such force, the direction of the force itself and the orientation of the area across which the force acts.

If shear stress tensor is denoted by $T = (T_{ij})$, the equation of motion for the velocity vector field u, referring to (5.5), is then

$$\frac{\partial u}{\partial t} + u \cdot \nabla u + \frac{1}{\rho}\nabla p = \frac{1}{\rho}\nabla \cdot T, \tag{5.9}$$

where we will refer to [**DG95**] for the detailed formation. The components of the viscous force per unit volume are then

$$(\nabla \cdot T)_i = \alpha \Delta u_i + \beta \frac{\partial}{\partial x_i} \nabla \cdot u,$$

where α, β are material parameters. Together with the incompressibility condition, $\nabla \cdot u = 0$, (5.9) then becomes the *incompressible Navier–Stokes equations*

$$\frac{\partial u}{\partial t} + u \cdot \nabla u + \frac{1}{\rho}\nabla p = \nu \Delta u, \tag{5.10}$$

$$\nabla \cdot u = 0. \tag{5.11}$$

The material parameter ν is called the *kinematic viscosity*. As with the incompressible Euler equations the Navier–Stokes equations are four coupled nonlinear partial differential equations for four unknown functions: the three components of u and the pressure p. Compared to the incompressible Euler equations, the net effect of the linear coupling between stress and rate of strain is to introduce the "diffusion" term in (5.10). The diffusion of momentum between neighbouring elements of the fluid is indeed a chief new ingredient in the incompressible Navier–Stokes equations. Whereas the Euler equations are of a hyperbolic type, the addition of viscosity changes the velocity evolution equation into a parabolic form. Being in this form, Navier–Stokes equations are *dissipative* in the sense that the *kinetic energy*, defined by

$$E(t) := \int_\Omega \left(\frac{1}{2}\rho |u|^2\right) \mathrm{d}x = \frac{1}{2}\rho \|u\|^2_{L^2(\Omega)}, \tag{5.12}$$

in some bounded domain Ω with appropriate boundary conditions is disappearing over time. More precisely, if we assume no-slip boundary condition $n \cdot u|_{\partial\Omega} = 0$, then[2]

$$0 \geq \frac{\mathrm{d}}{\mathrm{d}t} E(t) = -\rho\nu \int_\Omega |\nabla u|^2 \mathrm{d}x.$$

[2]see Exercise 5.1.

The integral on the right-hand side represents the dissipation of kinetic energy due to the viscosity in the Navier–Stokes equations. To understand this term, consider the velocity field $u(x,t)$ at a particular time t and a particular point x in the fluid. The gradient of the velocity field $\nabla u(x,t)$ represents the rate of change of the velocity field in space at that point. If the gradient of the velocity field is large, it means that the velocity field is changing rapidly in space at that point.

The integral of $|\nabla u|^2$ over the fluid domain represents the total amount of kinetic energy being dissipated due to the rapid changes of the velocity field in space. This is because when there are large gradients in the velocity field, the viscous forces in the Navier–Stokes equations cause the velocity field to smooth out, converting kinetic energy into heat. Physically, this dissipation term refers to the fluid's internal friction caused by its viscosity. The viscosity causes the fluid to resist deformation and tends to smooth out any velocity gradients. This dissipation of kinetic energy ultimately causes the fluid to come to rest in the absence of any external forces. Mathematically, this dissipation term plays a key role in analyzing the Navier–Stokes equations since it is responsible for their well-posedness and regularity properties. In particular, the dissipation term allows for energy estimates and control of the solution, which is essential for understanding the fluid's long-term behaviour.

5.2. Weak formulation of the Navier–Stokes equations.

For the rest of this chapter, we consider the Navier–Stokes equations (5.10) with a nomalized fluid density $\rho = 1$ and viscosity $\nu = 1$. The domain Ω is bounded with a smooth boundary[3], and we often omit the dependency of the function spaces on the domain when it is clear from the context. We will use the following notations for the inner product and norm in L^2

$$\langle \cdot, \cdot \rangle \quad \text{and} \quad \| \cdot \|.$$

We denote $C^\infty_{c,\sigma}$ to be the space of divergence-free smooth functions with compact support in Ω. We also denote

$$H := \text{ the completion of } C^\infty_{c,\sigma} \text{ in } L^2 \text{ norm}, \tag{5.13}$$

and equip H with the L^2-norm. Define

$$V := H \cap H^1_0, \tag{5.14}$$

and equip V with the norm $\| \cdot \|_V = \| \cdot \|_{H^1}$.

[3]Note that the theory of Navier–Stokes equations on a torus or in the whole $\mathbb{R}^n$ can be treated more or less in the same way, we refer to [**RRS16**] for details.

In this section, we will establish an appropriate notion of a weak solution of the Navier–Stokes equations. In the weak formulation of the Navier–Stokes equations, we not only transfer some derivatives onto the test function, as is common in the weak form of many PDEs, but we also eliminate one term in the equation, namely the pressure term. To achieve this, we introduce a special class of test functions. The following definition applies to all the domains considered in this chapter.

DEFINITION 5.1. The space of test functions D_σ on the space-time domain $\Omega \times [0,\infty)$ is defined as

$$D_\sigma = \{\varphi \in C_c^\infty(\Omega \times [0,\infty)) : \nabla \cdot \varphi(t) = 0 \text{ for all } t \in [0,\infty)\}.$$

This means that any test function $\varphi \in D_\sigma$ must be divergence-free but does not necessarily vanish at $t = 0$.

5.2.1. Derivation of the weak formulation. Suppose u is a smooth, divergence-free solution (a *classical solution*) of the Navier–Stokes equations

$$\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0. \tag{5.15}$$

Multiplying this equation by a test function $\varphi \in D_\sigma$, integrating over space, and integrating by parts in the second term, we obtain

$$\langle \partial_t u, \varphi \rangle + \langle \nabla u, \nabla \varphi \rangle + \langle (u \cdot \nabla)u, \varphi \rangle + \langle \nabla p, \varphi \rangle = 0. \tag{5.16}$$

Since φ is divergence-free, the last term vanishes

$$\langle \nabla p, \varphi \rangle = -\langle p, \nabla \cdot \varphi \rangle = 0. \tag{5.17}$$

Next we integrate in time from 0 to s and apply integration by parts to the first term

$$\int_0^s \langle \partial_t u, \varphi \rangle \, dt = \langle u(s), \varphi(s) \rangle - \langle u(0), \varphi(0) \rangle - \int_0^s \langle u, \partial_t \varphi \rangle \, dt. \tag{5.18}$$

Rearranging, we obtain the weak formulation

$$\int_0^s -\langle u, \partial_t \varphi \rangle + \int_0^s \langle \nabla u, \nabla \varphi \rangle + \int_0^s \langle (u \cdot \nabla)u, \varphi \rangle = \langle u(0), \varphi(0) \rangle - \langle u(s), \varphi(s) \rangle. \tag{5.19}$$

We first have the following property of the nonlinear term.

LEMMA 5.2. *Let $u \in V$ and $v, w \in H^1$. Then:*

$$\langle (u \cdot \nabla)v, w \rangle = -\langle (u \cdot \nabla)w, v \rangle. \tag{5.20}$$

In particular,

$$\langle (u \cdot \nabla)v, v \rangle = 0. \tag{5.21}$$

PROOF. We prove the identity using integration by parts. Assume $u \in C_c^\infty(\Omega)$ and $v, w \in C^1(\Omega)$. Then, using Einstein's summation convention,

$$\langle (u \cdot \nabla)v, w \rangle = \int_\Omega u_j (\partial_j v_i) w_i \, dx.$$

Integrating by parts and using that $\nabla \cdot u = 0$,

$$\int_\Omega u_j (\partial_j v_i) w_i \, dx = -\int_\Omega (\partial_j u_j) v_i w_i \, dx - \int_\Omega u_j v_i (\partial_j w_i) \, dx.$$

Since $\nabla \cdot u = 0$, the first term vanishes, leaving

$$\langle (u \cdot \nabla)v, w \rangle = -\langle (u \cdot \nabla)w, v \rangle.$$

For the second part,

$$\langle (u \cdot \nabla)v, v \rangle = -\langle (u \cdot \nabla)v, v \rangle,$$

which implies

$$2\langle (u \cdot \nabla)v, v \rangle = 0.$$

Thus, $\langle (u \cdot \nabla)v, v \rangle = 0$, completing the proof. □

In a similar spirit as in Chapter 2, to ensure the class of weak solutions is not too broad we need to impose some further conditions on the solutions, more precisely some energy constraints will be introduced. Let us assume for a moment that u is a classical solution, taking the L^2 inner product with u gives

$$\frac{1}{2}\frac{d}{dt}\|u\|^2 + \|\nabla u\|^2 + \langle (u \cdot \nabla)u, u \rangle = 0. \tag{5.22}$$

By Lemma 5.2, the nonlinear term vanishes, leading to

$$\frac{1}{2}\frac{d}{dt}\|u\|^2 + \|\nabla u\|^2 = 0. \tag{5.23}$$

Integrating over time,

$$\frac{1}{2}\|u(t)\|^2 + \int_0^t \|\nabla u(s)\|^2 \, ds = \frac{1}{2}\|u(0)\|^2, \tag{5.24}$$

for any $t > 0$. Thus, we require that for any weak solution, for all $T > 0$,

$$\sup_{0 \le t \le T} \|u(t)\| < \infty, \quad \int_0^T \|\nabla u(s)\|^2 ds < \infty. \tag{5.25}$$

This implies weak solutions belong to the space

$$L^\infty(0, T; H) \cap L^2(0, T; V). \tag{5.26}$$

DEFINITION 5.3. A function u is a weak solution of the Navier–Stokes equations with initial condition $u_0 \in H$ if

(i) $u \in L^\infty(0,T;H) \cap L^2(0,T;V)$ for all $T > 0$, and
(ii) u satisfies the equation

$$\int_0^s -\langle u, \partial_t \varphi\rangle + \int_0^s \langle \nabla u, \nabla \varphi\rangle + \int_0^s \langle (u\cdot\nabla)u, \varphi\rangle = \langle u_0, \varphi(0)\rangle - \langle u(s), \varphi(s)\rangle, \tag{5.27}$$

for all test functions $\varphi \in D_\sigma$ and almost every $s > 0$.

REMARK 5.4. The condition (ii) can be replaced by the following condition

$$\int_0^\infty -\langle u, \partial_t \varphi\rangle + \int_0^\infty \langle \nabla u, \nabla \varphi\rangle + \int_0^\infty \langle (u\cdot\nabla)u, \varphi\rangle = \langle u_0, \varphi(0)\rangle, \tag{5.28}$$

for all test functions $\varphi \in D_\sigma$. Moreover, the space of test functions D_σ can be relaxed to consist of functions with less regularity in time, see [**RRS16**].

Since a function in $L^\infty(0,T;H) \cap L^2(0,T;V)$ is an equivalence class of functions equal almost everywhere, and any such function can be modified on a set of measure zero without changing it in any essential way. Therefore, the regularity condition (i) alone is not sufficient to explain how a weak solution u fulfills the initial condition $u(0) = u_0$. In fact, condition (ii) of Definition 5.3 ensures that u is L^2-weakly continuous in time and, consequently, imposes that the initial condition makes sense, as the following theorem shows.

THEOREM 5.5. *Weak solutions of the Navier–Stokes equations are L^2-weakly continuous in time, i.e.,*

$$\lim_{t\to t_0} \langle u(t), v\rangle = \langle u(t_0), v\rangle \quad \text{for all } v \in L^2(\Omega). \tag{5.29}$$

In particular, $u(t) \rightharpoonup u_0$ weakly in L^2 as $t \to 0^+$.

We referred to [**RRS16**] for a proof of this result.

5.3. Uniqueness of weak solutions in 2D

We will now prove that weak solutions of the Navier–Stokes equations are unique in space dimension two. The situation is very different from the three-dimensional case, as we have better exponents in the Sobolev embedding theorem. Consequently, we can prove that $\partial_t u \in L^2(0,T;V^*)$. This regularity allows us to use weak solutions of the 2D Navier–Stokes equations as test functions and establish their uniqueness in a relatively straightforward manner, as shown below.

THEOREM 5.6. *Let $\Omega \subset \mathbb{R}^2$ be a smooth bounded domain, and let u and v be two weak solutions corresponding to the same initial condition $u_0 \in H(\Omega)$. Then $u(t) = v(t)$ for all $t > 0$.*

PROOF. Let $w = u - v$. Substituting $\varphi = w$ as a test function in the weak form of the equation (see (3.3)), we obtain

$$\int_0^s -\langle u, \partial_t w\rangle + \int_0^s \langle \nabla u, \nabla w\rangle + \int_0^s \langle (u\cdot\nabla)u, w\rangle = \langle u_0, w(0)\rangle - \langle u(s), w(s)\rangle.$$

Since in dimension two, all functions under the integral signs are integrable (exercise!), the functions $\langle u(s), w(s)\rangle$ and $\langle v(s), w(s)\rangle$ must be absolutely continuous, and hence differentiable for almost all $s > 0$. Differentiating the above equality with respect to time gives

$$-\langle u, \partial_t w\rangle + \langle \nabla u, \nabla w\rangle + \langle (u\cdot\nabla)u, w\rangle = -\frac{d}{dt}\langle u, w\rangle.$$

Similarly,

$$-\langle v, \partial_t w\rangle + \langle \nabla v, \nabla w\rangle + \langle (v\cdot\nabla)v, w\rangle = -\frac{d}{dt}\langle v, w\rangle.$$

Taking the difference of the two equations, we obtain

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\|w\|^2 + \|\nabla w\|^2 &\le |\langle (u\cdot\nabla)u - (v\cdot\nabla)v, w\rangle| \\ &= |\langle (u\cdot\nabla)u, w\rangle - \langle ((u-w)\cdot\nabla)(u-w), w\rangle| \\ &= |\langle (u\cdot\nabla)w, w\rangle + \langle (w\cdot\nabla)u, w\rangle - \langle (w\cdot\nabla)w, w\rangle|.\end{aligned}$$

By Lemma 5.2, the first and last terms vanish, so we obtain

$$\frac{1}{2}\frac{d}{dt}\|w\|^2 + \|\nabla w\|^2 \le |\langle (w\cdot\nabla)u, w\rangle|.$$

Since $w \in H_0^1(\Omega)$, we apply following inequality (see Exercise 5.2),

$$\|w\|_{L^4} \le c\|w\|^{1/2}\|\nabla w\|^{1/2},$$

which gives

$$\frac{1}{2}\frac{d}{dt}\|w\|^2 + \|\nabla w\|^2 \le \|w\|_{L^4}^2\|\nabla u\| \le c\|w\|\|\nabla w\|\|\nabla u\| \le c\|\nabla u\|^2\|w\|^2 + \frac{1}{2}\|\nabla w\|^2.$$

Thus,

$$\frac{d}{dt}\|w\|^2 \le c\|\nabla u\|^2\|w\|^2.$$

Applying Gronwall's lemma (Lemma B.3), we conclude

$$\|w(t)\|^2 \le \|w(0)\|^2 \exp\left(c\int_0^t \|\nabla u(s)\|^2 ds\right).$$

Since $u \in L^2(0, T; V)$ and $w(0) = 0$, it follows that $w(t) = 0$ for all $t > 0$, which proves the uniqueness. □

For the existence theory of the Navier–Stokes equations, we refer to the additional notes available on the course platform.

5.4. Exercises

Exercise 5.1. Let $E(t)$ be the kinetic energy

$$E(t) := \int_\Omega \left(\frac{1}{2}\rho|u|^2\right) \mathrm{d}x = \frac{1}{2}\rho\|u\|^2_{L^2(\Omega)}, \tag{5.30}$$

for the Navier–Stokes equations ($\nu > 0$)

$$\frac{\partial u}{\partial t} + u \cdot \nabla u + \frac{1}{\rho}\nabla p = \nu\Delta u,$$
$$\nabla \cdot u = 0,$$

with zero the boundary condition $n \cdot u|_{\partial\Omega} = 0$. Prove that the kinetic energy is dissipative, that is,

$$\frac{\mathrm{d}}{\mathrm{d}t}E(t) \leq 0.$$

Exercise 5.2. (i) Let $u = u(x_1, x_2)$ be a smooth function of compact support in $\mathbb{R}^2$. Prove that

$$\max_{x_i} |u^m| \leq m \int_{-\infty}^{\infty} |u^{m-1}u_{x_i}| \mathrm{d}x_i, \quad (i = 1, 2),$$

for any $m \in \mathbb{N}, m \geq 1$.

(ii) Prove that for any smooth function u with compact support in some domain $\Omega \subset \mathbb{R}^2$, we have

$$\|u\|^4_{L^4(\Omega)} \leq 2\|u\|^2_{L^2(\Omega)}\|\nabla u\|^2_{L^2(\Omega)}.$$

Hint: Use the fact that $u^2u^2 \leq \max_{x_1} u^2 \max_{x_2} u^2$ and integrate both sides over $\mathbb{R}^2$, then use (i) and Cauchy-Schwarz inequality.

5.5. Summary and notes

This chapter highlights some fundamental aspects of fluid mechanics, including the derivation of Euler and Navier–Stokes equations (Section 5.1, which is taken from [**DG95**]). The weak formulation of the Navier–Stokes equations (Section 5.3) and the uniqueness of the equations in 2D (Section 5.3 follow from [**RRS16**]). The theory of Sobolev spaces and Sobolev embeddings, briefly introduced in Chapter 3, are crucial in the study of Navier–Stokes equations. For a more in-depth analysis of the Navier–Stokes equations, we refer to [**Gal11**].

APPENDIX A

Useful facts on Calculus

A.1. Divergence theorem

THEOREM A.1 (Gauss). *For a compact region $V \subset \mathbb{R}^n$ with a smooth boundary $S = \partial V$, oriented outward, and a continuously differentiable vector field $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$, we have*

$$\int_V \nabla \cdot \mathbf{F}\, dV = \int_{\partial V} \mathbf{F} \cdot d\mathbf{S},$$

where

- $\nabla \cdot \mathbf{F} = \sum_{i=1}^n \frac{\partial F_i}{\partial x_i}$ *is the divergence of* $\mathbf{F}$,
- $d\mathbf{S}$ *is the outward-oriented surface measure on* ∂V, *given by* $d\mathbf{S} = \mathbf{n}\, dS$, *where* $\mathbf{n}$ *is the unit outward normal.*

A.2. Green's identities

THEOREM A.2. *For scalar functions u and v that are twice continuously differentiable in a region $V \subset \mathbb{R}^n$ with boundary $S = \partial V$, we have*

(i) $\int_V \nabla u \cdot \nabla v\, dV + \int_V v \nabla^2 u\, dV = \int_{\partial V} v \frac{\partial u}{\partial n}\, dS$,
(ii) $\int_V (u \nabla^2 v - v \nabla^2 u)\, dV = \int_{\partial V} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n}\right) dS$.

A.3. Gaussian integral

The Gaussian integral is given by

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}.$$

More generally, for $a > 0$,

$$I := \int_{-\infty}^{\infty} e^{-ax^2}\, dx = \sqrt{\frac{\pi}{a}}.$$

PROOF. By squaring the integral and using Fubini's theorem (see Appendix A.4),

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-ax^2}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-ay^2}\, dy\right) = \int_{\mathbb{R}^2} e^{-a(x^2+y^2)}\, dx\, dy.$$

Using polar coordinates ($x = r\cos\theta, y = r\sin\theta$), the integral transforms as

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-ar^2} r\, dr\, d\theta = \int_0^{2\pi} \frac{1}{2a} d\theta = \frac{\pi}{a},$$

as required. □

A.4. Fubini's theorem

THEOREM A.3 (Fubini). *Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measure spaces, and let $f : X \times Y \to \mathbb{R}$ (or $\mathbb{C}$) be a measurable function. Suppose that f is absolutely integrable, i.e.,*

$$\int_{X\times Y} |f(x,y)|\, d(\mu\otimes\nu)(x,y) < \infty.$$

Then, the iterated integrals exist and satisfy

$$\begin{aligned}\int_{X\times Y} f(x,y)\, d(\mu\otimes\nu)(x,y) &= \int_X \left(\int_Y f(x,y)\, d\nu(y)\right) d\mu(x)\\ &= \int_Y \left(\int_X f(x,y)\, d\mu(x)\right) d\nu(y).\end{aligned}$$

REMARK A.4. If $f(x,y)$ is measurable and nonnegative then the absolute integrability condition is not required, and the iterated integrals still exist and are equal. This is known as Tonelli's theorem.

A.5. Polar coordinates

THEOREM A.5. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and integrable. Then we have*

$$\int_{\mathbb{R}^n} f \mathrm{d}x = \int_0^{\infty} \left(\int_{\partial B(x_0,r)} f \mathrm{d}S\right) \mathrm{d}r,$$

for each $x_0 \in \mathbb{R}^n$.

This change of variables formula is pretty useful when integrating over radially symmetric functions in higher dimensions. In particular, this gives rise to the following differentiation of integral formula

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\int_{B(x_0,r)} f \mathrm{d}x\right) = \left(\int_{\partial B(x_0,r)} f \mathrm{d}S\right),$$

for each $r > 0$. This is again a special case of the Reynolds transport theorem, see Appenxdix A.6.

A.6. Differentiation with moving regions

Let $U(t) \subset \mathbb{R}^n$ be a smooth and bounded region that depends smoothly on the parameter $t \in \mathbb{R}$. Let $\mathbf{v}$ be the velocity of the moving boundary $\partial U(t)$ and $\boldsymbol{\nu}$ be the outward pointing unit normal (you may think of $\mathbf{v}$ as $\frac{d}{dt}\partial U(t)$, interpreted as a vector field defined on $\partial U(t)$). The following theorem is also known as the *Leibniz integral rule* in high dimensions, or the *Reynolds transport theorem*.

THEOREM A.6 (Differentiation for moving regions). *For any smooth function $f = f(x,t)$, we have*

$$\frac{d}{dt}\int_{U(t)} f \mathrm{d}x = \int_{U(t)} \frac{\partial}{\partial t} f \mathrm{d}x + \int_{\partial U(t)} f \mathbf{v}\cdot\boldsymbol{\nu}\mathrm{d}S.$$

PROOF. We introduce a smooth flow map Φ that parametrizes the motion of the domain

$$\Phi(t,y) : U(0) \to U(t),$$

where for each fixed y, the function $\Phi(t,y)$ describes the trajectory of a moving particle

$$x = \Phi(t,y).$$

The velocity of the moving boundary is given by

$$\mathbf{v}(x,t) = \frac{\partial}{\partial t}\Phi(t,y)\Big|_{y=\Phi^{-1}(t,x)}.$$

We rewrite the integral over $U(t)$ using this change of variables

$$I(t) = \int_{U(t)} f(x,t)\,dx = \int_{U(0)} f(\Phi(t,y),t)J(t,y)\,dy,$$

where $J(t,y)$ is the Jacobian determinant of Φ. By differentiating under the integral sign,

$$\frac{d}{dt}I(t) = \int_{U(0)} \left(\frac{\partial f}{\partial t}(\Phi(t,y),t)J + f(\Phi(t,y),t)\frac{dJ}{dt}\right) dy.$$

By the determinant differentiation formula,

$$\frac{dJ}{dt} = J\nabla\cdot\mathbf{v},$$

so we obtain

$$\frac{d}{dt}I(t) = \int_{U(0)} \left(\frac{\partial f}{\partial t} + f\nabla\cdot\mathbf{v}\right) J\,dy.$$

Transforming back to $U(t)$,

$$\frac{d}{dt}\int_{U(t)} f\,dx = \int_{U(t)} \left(\dot{f} + f\nabla\cdot\mathbf{v}\right) dx,$$

where $\dot{f}$ is the material time derivative of f, that is

$$\dot{f} = \frac{\partial f}{\partial t} + \nabla f \cdot \mathbf{v}.$$

Now using the product rule,

$$\nabla \cdot (f\mathbf{v}) = \nabla f \cdot \mathbf{v} + f \nabla \cdot \mathbf{v},$$

we then have

$$\int_{U(t)} \left(\dot{f} + f \nabla \cdot \mathbf{v} \right) \, dx = \int_{U(t)} \left(\frac{\partial f}{\partial t} + \nabla \cdot (f\mathbf{v}) \right) \, dx.$$

By the divergence theorem A.1,

$$\int_{U(t)} \nabla \cdot (f\mathbf{v}) \, dx = \int_{\partial U(t)} f\mathbf{v} \cdot \boldsymbol{\nu} \, dS.$$

Substituting this into our earlier differentiation result,

$$\frac{d}{dt} \int_{U(t)} f \, dx = \int_{U(t)} \frac{\partial f}{\partial t} \, dx + \int_{\partial U(t)} f\mathbf{v} \cdot \boldsymbol{\nu} \, dS.$$

This completes the proof. □

APPENDIX B

Useful facts on Functional Analysis

B.1. Useful inequalities

THEOREM B.1 (Cauchy). *For any $a, b \in \mathbb{R}$, we have $ab \leq \frac{1}{2}(a^2+b^2)$.*

PROOF. The inequality is equivalent to $(a-b)^2 \geq 0$. □

THEOREM B.2 (Cauchy-Schwarz). *Let f, g be measurable functions on a measure space (X, μ). Then the following inequality holds*

$$\left| \int_X f(x)g(x)\, d\mu(x) \right| \leq \left(\int_X |f(x)|^2\, d\mu(x) \right)^{\frac{1}{2}} \left(\int_X |g(x)|^2\, d\mu(x) \right)^{\frac{1}{2}}.$$

In compact form

$$\|fg\|_{L^1_\mu(X)} \leq \|f\|_{L^2_\mu(X)} \|g\|_{L^2_\mu(X)}.$$

Equality holds if and only if f and g are linearly dependent, meaning there exists a scalar λ such that $f = \lambda g$, μ-almost everywhere.

PROOF. Consider the function $\varphi(t)$ defined by

$$\varphi(t) = \int_X |f(x) + tg(x)|^2\, d\mu(x).$$

Expanding the square inside the integral, we obtain

$$\varphi(t) = \int_X \left(|f(x)|^2 + 2tf(x)g(x) + t^2|g(x)|^2 \right) d\mu(x).$$

This simplifies to

$$\varphi(t) = \int_X |f(x)|^2\, d\mu(x) + 2t \int_X f(x)g(x)\, d\mu(x) + t^2 \int_X |g(x)|^2\, d\mu(x).$$

Since the integral of a square is always nonnegative, we have $\varphi(t) \geq 0$ for all $t \in \mathbb{R}$. The function $\varphi(t)$ is a quadratic polynomial of the form

$$at^2 + 2bt + c \geq 0, \quad \forall t \in \mathbb{R},$$

where

$$a = \int_X |g(x)|^2\, d\mu(x), \quad b = \int_X f(x)g(x)\, d\mu(x), \quad c = \int_X |f(x)|^2\, d\mu(x).$$

The discriminant of this quadratic equation must be nonpositive, that is

$$4b^2 - 4ac \leq 0.$$

Simplifying, we obtain

$$b^2 \leq ac.$$

Substituting back the values of a, b, c, we get:

$$\left(\int_X f(x)g(x)\, d\mu(x) \right)^2 \leq \left(\int_X |f(x)|^2\, d\mu(x) \right) \left(\int_X |g(x)|^2\, d\mu(x) \right).$$

Taking the square root on both sides yields the desired result. Equality holds if and only if the discriminant is zero, which happens when f and g are linearly dependent, i.e., $f = \lambda g$ almost everywhere for some scalar λ. □

Note that the same strategy can be employed to prove the following discrete form of the Cauchy-Schwarz inequality. For any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the following inequality holds

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|,$$

that is

$$\left(\sum_{i=1}^{n} u_i v_i \right)^2 \leq \left(\sum_{i=1}^{n} u_i^2 \right) \left(\sum_{i=1}^{n} v_i^2 \right).$$

THEOREM B.3 (Grönwall). *Let $u(t)$ be a non-negative, differentiable function on an interval $[t_0, T]$, and suppose there exist non-negative functions $\alpha(t)$ and $\beta(t)$ such that*

$$\frac{d}{dt}u(t) \leq \alpha(t)u(t) + \beta(t), \quad \text{for all } t \in [t_0, T].$$

Then, for all $t \in [t_0, T]$, we have

$$u(t) \leq u(t_0)e^{\int_{t_0}^{t} \alpha(s)ds} + \int_{t_0}^{t} \beta(s)e^{\int_{s}^{t} \alpha(\tau)d\tau} ds.$$

PROOF. Define the integrating factor

$$\mu(t) = e^{-\int_{t_0}^{t} \alpha(s)ds}.$$

Multiplying both sides of the inequality by $\mu(t)$, we obtain

$$\mu(t)\frac{d}{dt}u(t) \leq \mu(t)\alpha(t)u(t) + \mu(t)\beta(t).$$

Since

$$\frac{d}{dt}\mu(t) = -\alpha(t)\mu(t),$$

we rewrite the left-hand side as

$$\frac{d}{dt}(\mu(t)u(t)) \leq \mu(t)\beta(t).$$

Integrating both sides from t_0 to t, we get

$$\mu(t)u(t) - \mu(t_0)u(t_0) \leq \int_{t_0}^{t} \mu(s)\beta(s)ds.$$

Since $\mu(t_0) = e^{-\int_{t_0}^{t_0} \alpha(s)ds} = 1$, it follows that

$$\mu(t)u(t) \leq u(t_0) + \int_{t_0}^{t} \mu(s)\beta(s)ds.$$

Multiplying by $e^{\int_{t_0}^{t} \alpha(s)ds} = \frac{1}{\mu(t)}$, we obtain

$$u(t) \leq u(t_0)e^{\int_{t_0}^{t} \alpha(s)ds} + \int_{t_0}^{t} \beta(s)e^{\int_{s}^{t} \alpha(\tau)d\tau}ds.$$

This completes the proof. □

B.2. Schwartz space

Definition B.4 (Schwartz space). The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ is the space of all infinitely differentiable functions $f : \mathbb{R}^n \to \mathbb{R}$ (or $\mathbb{C}$) such that for every multi-index $\alpha, \beta \in \mathbb{N}^n$, the seminorm

$$p_{\alpha,\beta}(f) = \sup_{x \in \mathbb{R}^n} |x^\beta D^\alpha f(x)|$$

is finite. In other words, $\mathcal{S}(\mathbb{R}^n)$ consists of smooth functions that decay at infinity faster than any polynomial and whose derivatives satisfy the same property.

It comes directly from the definition that $\mathcal{S}(\mathbb{R}^n) \subset C^\infty(\mathbb{R}^n)$. Moreover, we have the following inclusions

$$C_0^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n).$$

In fact, these inclusions are dense.

Lemma B.5. *The space $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$. The Schwartz space $\mathcal{S}(\mathbb{R}^n)$, in turn, is dense in $L^p(\mathbb{R}^n)$, $1 \leq p < \infty$.*

B.3. More on properties of Fourier transform

THEOREM B.6. *The Fourier transform maps $\mathcal{S}$ into itself.*

SKETCH OF PROOF. Let $f \in \mathcal{S}(\mathbb{R}^n)$, we need to show that the Fourier transform $\mathcal{F}(f)$ is also in $\mathcal{S}(\mathbb{R}^n)$.

1. Smoothness of $\mathcal{F}(f)$. The Fourier transform of f is given by

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x\cdot\xi}\, dx.$$

Differentiating under the integral sign, we compute

$$D^\alpha \mathcal{F}(f)(\xi) = \int_{\mathbb{R}^n} (-2\pi i x)^\alpha f(x) e^{-2\pi i x\cdot\xi}\, dx.$$

Since $|x^\alpha f(x)|$ is rapidly decreasing, the integral defining $D^\alpha\mathcal{F}(f)(\xi)$ is absolutely convergent for all ξ, showing that $\mathcal{F}(f)$ is smooth.

2. Rapid decay of $\mathcal{F}(f)$. To show that $\mathcal{F}(f)$ decays rapidly, we consider

$$\sup_{\xi\in\mathbb{R}^n} |\xi^\beta D^\alpha \mathcal{F}(f)(\xi)|.$$

Using integration by parts, let L be the differential operator

$$L = (1+|x|^2)^{-1} \sum_{j=1}^n \frac{\partial}{\partial x_j} x_j.$$

Since $L^k e^{-2\pi i x\cdot\xi} = (2\pi i|x|^2)^{-k} e^{-2\pi i x\cdot\xi}$, applying L^k to $f(x)e^{-2\pi i x\cdot\xi}$ and integrating by parts shows that

$$|\xi^\beta D^\alpha \mathcal{F}(f)(\xi)| \le C_{k,\alpha,\beta} \sup_{x\in\mathbb{R}^n} |x^\gamma D^\delta f(x)|.$$

Since f is in the Schwartz space, the right-hand side is finite for all γ, δ, proving that $\mathcal{F}(f)$ decays faster than any polynomial.

Since $\mathcal{F}(f)$ is smooth and has rapid decay, it belongs to $\mathcal{S}(\mathbb{R}^n)$, completing the proof. □

The following theorem records one of the most important facts about the Fourier transform in L^2.

THEOREM B.7 (Plancherel). *Let $f \in L^2(\mathbb{R}^n)$, and let $\mathcal{F}(f)$ denote its Fourier transform, defined by*

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x\cdot\xi}\, dx.$$

Then, the Fourier transform extends to an isometry on $L^2(\mathbb{R}^n)$, meaning that we have the identity

$$\|f\|_{L^2} = \|\mathcal{F}(f)\|_{L^2},$$

or explicitly,

$$\int_{\mathbb{R}^n} |f(x)|^2 \, dx = \int_{\mathbb{R}^n} |\mathcal{F}(f)(\xi)|^2 \, d\xi.$$

Consequently, the Fourier transform defines a unitary operator on $L^2(\mathbb{R}^n)$.

PROOF. We first consider the case where $f \in \mathcal{S}(\mathbb{R}^n)$, the Schwartz space, since such functions are dense in $L^2(\mathbb{R}^n)$. Define the Fourier transform of f as

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} \, dx.$$

Consider the L^2-norm squared of f

$$\|f\|_{L^2}^2 = \int_{\mathbb{R}^n} |f(x)|^2 \, dx.$$

To relate this to the Fourier transform, we compute

$$\int_{\mathbb{R}^n} |\mathcal{F}(f)(\xi)|^2 \, d\xi.$$

Using the definition of the inverse Fourier transform, we express $f(x)$ as

$$f(x) = \int_{\mathbb{R}^n} \mathcal{F}(f)(\xi) e^{2\pi i x \cdot \xi} \, d\xi.$$

Multiplying both sides by $\overline{f(x)}$ and integrating over $\mathbb{R}^n$, we obtain

$$\int_{\mathbb{R}^n} |f(x)|^2 \, dx = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} \mathcal{F}(f)(\xi) e^{2\pi i x \cdot \xi} \, d\xi \right) \overline{f(x)} \, dx.$$

By Fubini's theorem, we interchange the integrals

$$\int_{\mathbb{R}^n} \mathcal{F}(f)(\xi) \left(\int_{\mathbb{R}^n} e^{2\pi i x \cdot \xi} \overline{f(x)} \, dx \right) d\xi.$$

Recognizing the inner integral as $\overline{\mathcal{F}(f)(\xi)}$, we obtain

$$\int_{\mathbb{R}^n} |\mathcal{F}(f)(\xi)|^2 \, d\xi = \int_{\mathbb{R}^n} |f(x)|^2 \, dx.$$

This shows that $\mathcal{F}$ preserves the L^2-norm. By a density argument, the result extends to all functions in $L^2(\mathbb{R}^n)$, and the Fourier transform defines a unitary operator on $L^2(\mathbb{R}^n)$. □

REMARK B.8. The Plancherel theorem is related to the following result, called *Parseval's identity*, which states that if (e_n) is an orthonormal basis of a separable Hilbert space H, then it holds that

$$\|f\|_H^2 = \sum_n |\langle f, e_n \rangle|^2, \quad \text{for any } f \in H.$$

APPENDIX C

Solutions for selected exercises

TBA

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